Volume-uniform electric-shell isolation, exact shared-link hopping, and a homological carrier in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

Master edition, with machine-checked provenance

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. In the normalised charge-odd fundamental-plaquette sector, an explicit order-two Weingarten calculation gives the exact coefficient of hopping between adjacent plaquettes that share one link,

$$t_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0, \qquad N \geq 3.$$

The oriented cellular boundary $B = \partial_2^\dagger$ fixes the corresponding local geometry. A Casimir and short-cycle argument proves the volume-uniform electric window $\operatorname{spec}(H_E|_{\text{trivial flux}}) \cap [0, \frac{5}{2}C_F) = \{0, 2C_F\}$, with the $2C_F$ eigenspace exactly the oriented one-plaquette shell and an external margin at least $C_F/2$. A link-centre classification then proves that the four shared-link fusion channels exhaust every second-order off-diagonal process. Consequently the global order-u^2 symbol is $t_N u^2 B(k)B(k)^\dagger$, with no retained-shell or process-completeness hypothesis. Its spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$ up to a common scalar, with $q = 4\sum_j \sin^2(k_j/2)$, and its flat carrier $\ker \partial_2$ has dimension L^3+2. A normalised Fourier sum of radius-one cube boundaries is the exact carrier at every nonzero allowed momentum; on nested volumes its order-u^2 splitting is L-independent, although one local cube seed has only $O(L^{-3})$ weight at a fixed momentum.

A supplied $SU(3)$ two-cube calculation — an exact-Casimir $\mathcal{B} = 6$ basis of 1,590,462 states, reached by path enumeration without ever assembling the magnetic matrix — reports the same coefficient channel by channel rather than only in the sum, with each of the six link-resolved channels separately proportional to the same incidence Gram on all 56 adjacent pairs, and localises the $\mathcal{B}$-cutoff dependence to one symmetry orbit; those finite-precision contractions are corroboration, not an independent derivation. By arithmetic on the weight ledger alone, a cutoff keeping only the singlet and $\Lambda^2 F$ routes projects the hopping to $-1/12$ — opposite in sign to t_3 — and omits exactly 14/153, the two summing to t_3. A later detached replay on the same open prism yields nested radius-two finite-u Ritz spectra and the leading incremental order-u^4 and order-u^5 effects of entering Q_2 and restoring its retained magnetic block; these are finite-volume nested-model differences, not complete fourth- or fifth-order coefficients. Exact integer geometry also gives the closed-cube shell $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$ with Gram eigenvalues $0, 4, 6$.

For comparison we solve the spectrum of a defined unsigned-incidence operator in closed form, the cubic $\mu(\mu-p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p+q = 12$, so one zone function runs both sectors. Identifying that operator with the complete charge-even linked Hamiltonian requires a further derivation,

because the vacuum-mediated route must be accounted for. The organized upstream corpus records a cold exact replay of the retained $SU(3)$ third-order chain, although that microscopic replay is not bundled here. Fourth-order physical-kernel diagnostics remain conditional; in particular, the retained checkpoints provably do not determine the off-axis coefficient, and that non-identifiability is exhibited by an explicit second witness.

The exact coefficient results are finite-volume and finite-order. In addition, for $SU(3)$ a published spectral-localisation theorem gives a volume-uniform, all-orders Riesz island for the complete neutral charge-odd shell at sufficiently small local coupling; the threshold is existential and the result remains at fixed lattice spacing. No continuum glueball or Yang–Mills mass-gap claim is made.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations [12, 13]. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology and signed or line-graph constructions [11, 21, 25, 29, 31], compact localised states and their failure to span singular bands on a torus [28, 30], and supersymmetric incidence factorisations [34] are all established. Closely related closed-membrane structures also occur in two-form spin liquids [35]. A recent classification of exactly flat bands protected by local graph topology [39] reaches extensive face-graph degeneracy from an *unoriented* face–vertex incidence, where rank–nullity alone gives $|F| - |V|$ flat bands. That is a different mechanism from the one here, and the difference is measurable: on the signed face–edge operator of Section 3 the corresponding index is exactly 0 and bounds nothing, while both kernels equal $L^3 + 2$ — homology, not counting. It is recorded as a distinction, not an identification. We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

The new dynamical content is local and exact. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, the shared-link matrix element in the charge-odd fundamental-plaquette sector factorises into a colour coefficient and an oriented edge–face Gram entry. Representation theory fixes the former and the cubical chain complex fixes the latter,

$$\text{colour dynamics} \longrightarrow t_N,$$
$$\text{cellular geometry} \longrightarrow BB^\dagger.$$

What is proved, and what remains conditional

The principal second-order statements are self-contained here; where a higher-order or charge-even operator's identification with a physical sector is conditional, that condition is stated separately. Earlier editions carried three load-bearing prose premises. Two are now theorems, and this edition proves more than the premises asked for: the free electric window of Theorem 1 and the process classification of Lemma 17. The third, the charge-even identification, remains a hypothesis and is argued in Section 13; the coverage appendix (Appendix G) makes its unchecked status visible by omission.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2, derived from the order-two

Weingarten values (Theorem 13). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.

2. For every $N \geq 3$ and $L \geq 3$, the trivial-flux electric spectrum below $\frac{5}{2}C_F$ is exactly $\{0, 2C_F\}$; the charge-odd part of the $2C_F$ eigenspace is exactly the $3L^3$-dimensional one-plaquette shell and its external free margin is at least $C_F/2$ (Theorem 1). Every off-diagonal second-order intermediate is then either a vanishing same-face route or one of the four adjacent shared-link channels (Lemma 17). Theorem 18 gives their explicit projector cross matrix elements and coefficient t_N, so the global symbol $t_N u^2 BB^\dagger$ follows without a retained-shell or process-completeness hypothesis (Corollary 19).

3. The Bloch spectrum of $BB^\dagger$ is $\{0, q, q\}$ (Theorem 6), and on the periodic L^3 complex the zero-mode count is $L^3 + 2$ (Theorem 7). The two are the same statement, by a sum rule over the momentum grid (Theorem 9). At fixed nonzero nested momentum the carrier and its splitting are volume-independent, with the exact locality–momentum tradeoff given in Proposition 11.

4. The truncation that keeps only the singlet and $\Lambda^2 F$ routes projects the hopping to $w_{\Lambda^2 F} - w_{\mathbf{1}} = -1/12$, and what it omits is $w_{\mathrm{Sym}^2 F} - w_{\mathrm{Adj}} = 14/153$; the two sum to t_3 (Corollary 24). This is arithmetic on (32) and needs no input beyond Theorem 13.

5. The defined unsigned-incidence comparison operator has the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p + q = 12$ (Theorem 32); its range is exactly $[-4, 12]$, the top attained only at Γ and the floor on the three planes $k_j = \pi$, triple only at R (Proposition 33). Conditional on identifying this comparison operator with the complete linked charge-even symbol, its Γ-point splitting is $12t_{r,+}$ (Proposition 35). That identification is not derived here; the vacuum-mediated all-pairs term must first be reorganised or cancelled.

6. On the closed cube surface the charge-odd shell is $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$, with G-eigenvalues $0, 4, 6$; the labels follow from the rotation action rather than being assigned, the A_1^{--} level is the fundamental class of S^2, and it sits at the scalar for every hopping coefficient (Proposition 27). This is integer linear algebra and takes no input from any delivery.

7. The retained Γ/axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 41), and an explicit second witness exhibits the non-identifiability by construction (Proposition 43).

Four families of statements depend on computational artifacts. The radius-two parents and detached verifier are bundled with this edition; the earlier raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not. Repeated assertions in synthesis documents are not treated as independent reproductions. The organized upstream authority corpus records a cold exact 251/251 replay for the retained third-order operator chain; that is stronger than an unchecked delivery, but it is still cited here as external computational evidence because this release does not replay the raw chain.

8. A supplied two-cube delivery reports six link-irrep channel coefficients at $N = 3$, as rational reconstructions of finite-precision contractions. Conditional on those six numbers, they are the four fusion-channel weights individually — channel by channel and not merely in the sum (Reported result 22) — and the $\mathcal{B} = 4$ truncation of that space retains exactly the adjacent shared-link channels the published link cutoff of item 4 retains, so the two land on the same projected hopping — the same channels, not the same scheme. Conditional on the delivered

diagonals, the symmetry analysis of Proposition 26 applies to them: the orbit structure itself is exact and takes no delivered input. The $\mathcal{B} = 6$ two-cube build was not reproduced here.

9. A later hash-sealed release on the same open prism reports nested charge-odd and charge-even radius-two Ritz operators. Its detached verifier reconstructs the spectra and the leading $K_{2,\star} - K_1$ and $K_{2,\mathrm{full}} - K_{2,\star}$ coefficients from the sealed parity-sector parents. This establishes numerical consistency relative to those inputs, not convergence in action radius or a physical glueball gap (Reported result 29).

10. For $SU(3)$, the frozen upstream corpus records that all candidate third-order lifter classes vanish and gives the scalar and shared-link coefficients of Section 7. The census is three lifter classes evaluated over 32 five-trace numerator patterns each, 96 contractions in total, within a retained chain that passed 251/251 cold exact gates upstream. This compact verifier checks the resulting rational assembly but not those microscopic contractions. If the census is exhaustive and the linked operator ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3.

11. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2-1)^3]$ and its equality with the complete axial coefficient for $N = 3, \ldots, 12$. Conditional on those calculations, the $SU(3)$ carrier acquires nonzero axial dispersion at order u^4. The complete off-axis operator remains open — and by Theorem 41 it cannot be closed by any Γ or axial datum.

These coefficient statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but they do not by themselves solve the external spectral problem. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2}. Theorem 3 later proves all-orders persistence of the complete three-component cluster in an existential small-$|u|$ domain; it does not prove convergence of the displayed coefficient series, persistence of one internal sheet, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \tfrac{1}{2} \sum_{\ell} C_2(\ell) + \beta_N \sum_{p} \Big(1 - \tfrac{1}{N} \operatorname{Re} \operatorname{Tr} U_p\Big). \tag{1}$$

Dropping the additive constant,

$$H = H_E - uV, \qquad V = \sum_p (\chi_p + \bar{\chi}_p), \qquad u = \frac{\beta_N}{2N}, \tag{2}$$

with $\chi_p = \operatorname{Tr} U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$-specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u. Reading it as a coordinate would demand dividing the order-r coefficient by 4^r, which misses the printed tables by exactly 16 at u^2 and 64 at u^3.

All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\mathrm{exc}}(u) = H(u) - E_{\mathrm{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}}\,|0\rangle, \tag{3}$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\Big(\tfrac{1}{2}C_F\Big) = 2C_F = \frac{N^2-1}{N}, \qquad C_F = \frac{N^2-1}{2N}. \tag{4}$$

This external energy is the input to the resolvent denominators of Section 4, where it is checked together with them. For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes. The analytic continuation of the unsigned-channel coefficient is nonzero there, which locates the $N^2 - 4$ zero in the channel difference rather than in the incidence. Because the charge-conjugation split degenerates for $SU(2)$, this continuation is not asserted to be an $SU(2)$ charge-even spectrum.

For each coordinate direction choose a dual noncontractible cut Σ_j, and fix one orientation for every geometric edge. For $z \in \mathbb{Z}_N$, let $Z_j(z)$ act by $U_e \mapsto z^{\varepsilon(e,\Sigma_j)}U_e$, where the signed intersection number $\varepsilon(e, \Sigma_j) \in \{-1, 0, +1\}$. These unitaries commute with H_E, while charge conjugation obeys $CZ_j(z)C^{-1} = Z_j(z^{-1})$. The simultaneous $+1$ eigenspace of every $Z_j(z)$ is therefore charge-conjugation invariant; call it the *trivial electric one-form sector*. A loop of N-ality k and winding vector w has eigenvalue z^{kw_j}, so contractible plaquettes lie in this sector while a once-winding fundamental loop does not.

Theorem 1 (Uniform first electric spectral window)**.** *Let $N \geq 3$ and $L \geq 3$. Let the pure-gauge physical Hilbert space be $\mathcal{H}_{\text{phys}} = L^2\big(SU(N)^{E(\Lambda_L)}\big)^{SU(N)^{V(\Lambda_L)}}$, and define the trivial electric one-form sector and its charge-odd part by*

$$\mathcal{H}^{(0)}_{\text{phys}} = \{\Psi \in \mathcal{H}_{\text{phys}} : Z_j(z)\Psi = \Psi\ \forall j, z\}, \qquad \mathcal{H}^{(0,-)}_{\text{phys}} = \{\Psi \in \mathcal{H}^{(0)}_{\text{phys}} : C\Psi = -\Psi\}.$$

Normalise the electric Hamiltonian on the Peter–Weyl spin-network basis by $H_E\Psi_{\{R_e\},\{\iota_v\}} = \frac{1}{2}\sum_e C_2(R_e)\Psi_{\{R_e\},\{\iota_v\}}$. Then, with $E_{F,N} = 2C_F$,

$$\operatorname{spec}(H_E|_{\mathcal{H}^{(0)}_{\text{phys}}}) \cap \Big[0, \tfrac{5}{2}C_F\Big) = \{0, 2C_F\}. \tag{5}$$

The zero eigenspace is the vacuum line, while the $2C_F$ eigenspace is the direct sum, over elementary plaquettes, of their two oriented fundamental Wilson-loop lines. Consequently

$$\operatorname{Ran}\Big(\mathbf{1}_{\{E_{F,N}\}}(H_E)|_{\mathcal{H}^{(0,-)}_{\text{phys}}}\Big) = \operatorname{span}\{\,|p,-\rangle : p \in F_2(\Lambda_L)\,\}, \tag{6}$$

and this charge-odd eigenspace has dimension $3L^3$. In particular the full electric shell is separated from the rest of the trivial-flux spectrum by

$$\operatorname{dist}\Big(2C_F,\ \operatorname{spec}(H_E|_{\mathcal{H}^{(0)}_{\text{phys}}}) \setminus \{2C_F\}\Big) \ \geq\ \tfrac{1}{2}C_F, \tag{7}$$

uniformly in L.

Proof. Normalise the invariant form so every root has squared length 2, and write a nonzero dominant highest weight as $\lambda=\sum_{i=1}^{N-1}a_i\omega_i$. If $a_i>0$, put $\lambda=\omega_i+\mu$ with μ dominant. Positivity of the inverse A_{N-1} Cartan matrix gives $C_2(\lambda)-C_2(\omega_i)=\frac{1}{2}\big((\mu,\mu)+2(\omega_i+\rho,\mu)\big)\geq 0$, with equality only for $\mu=0$. Moreover

$$C_2(\omega_i)=\frac{i(N-i)(N+1)}{2N},\qquad C_2(\omega_i)-C_F=\frac{N+1}{2N}(i-1)(N-i-1)\ \geq\ 0.$$

Thus every nontrivial irrep satisfies $C_2(R)\geq C_F$, with equality only for F and $\bar{F}$.

We need one further all-rank shelf. If a dominant weight λ is neither ω_1 nor ω_{N-1}, then one of three cases occurs: an interior Dynkin coefficient is nonzero; an endpoint coefficient is at least two; or both endpoint coefficients are nonzero. Monotonicity of the preceding Casimir difference therefore bounds $C_2(\lambda)$ below by, respectively, $C_2(\omega_2)$ (for $N\geq 4$), $C_2(2\omega_1)$ or its conjugate, or $C_2(\omega_1+\omega_{N-1})$. The three comparisons with the required shelf are

$$C_2(\omega_2)-\tfrac{5}{4}C_F=\frac{(N+1)(3N-11)}{8N}>0\quad(N\geq 4),\tag{8}$$

$$C_2(2\omega_1)-\tfrac{5}{4}C_F=\frac{(N-1)(3N+11)}{8N}>0,\tag{9}$$

$$C_2(\omega_1+\omega_{N-1})-\tfrac{5}{4}C_F=\frac{3N^2+5}{8N}>0.\tag{10}$$

For $N=3$ there is no interior coefficient, so the final two cases are exhaustive. Hence every irrep other than $\mathbf{1},F,\bar{F}$ satisfies

$$C_2(R)>\tfrac{5}{4}C_F.\tag{11}$$

We also need the centre-neutral refinement. Zero N-ality is equivalent to λ lying in the A_{N-1} root lattice. Write $\lambda=\sum_i m_i\alpha_i$, $m_i\in\mathbb{Z}$, and $a_i=(\lambda,\alpha_i)\geq 0$. Since $m=A^{-1}a$ and every entry of A^{-1} is strictly positive, a nonzero dominant λ has $m_i\geq 1$ for every i. Hence $(\lambda,\rho)=\sum_i m_i\geq N-1$. Every nonzero root-lattice vector has squared norm at least 2, so

$$C_2(\lambda)=\tfrac{1}{2}\big((\lambda,\lambda)+2(\lambda,\rho)\big)\ \geq\ N.\tag{12}$$

Consider a non-vacuum spin-network component with energy $E<\frac{5}{2}C_F$. If S is the set of links carrying nontrivial irreps, then $\sum_{e\in S}C_2(R_e)=2E<5C_F$, so $|S|\leq 4$. Gauge invariance forbids support degree one, because a single nontrivial irrep has no invariant vector. The torus graph is simple for $L\geq 3$; therefore a nonempty connected support has at least a three-cycle. Two components would require at least six links, and a connected simple graph of minimum degree two with at most four edges is a single C_3 or C_4.

A three-cycle occurs only for $L=3$ and is a straight loop winding once around one coordinate direction. Indeed, lift its three unit steps to $\mathbb{Z}^3$: their sum is Lw, and the length and parity constraints force $L=3$, $w=\pm e_j$. Degree-two intertwiners transport one irrep R around the loop. Trivial one-form flux forces its N-ality to vanish, and (12) gives $H_E=\frac{3}{2}C_2(R)\geq\frac{3}{2}N>\frac{5(N^2-1)}{4N}=\frac{5}{2}C_F$, a contradiction. The support is therefore a four-cycle. Degree-two intertwiners transport one irrep R (with the dual on oppositely oriented links) around it, so its energy is $2C_2(R)$. Lift its four unit steps to $\mathbb{Z}^3$. For $L\geq 5$ the winding is zero; at $L=3$ a nonzero winding violates step parity; and at $L=4$ the only nonzero possibility is four collinear steps, a once-winding loop. Trivial flux then forces R to have zero N-ality, so (12) gives $H_E=2C_2(R)\geq 2N>\frac{5}{2}C_F$, again a contradiction.

With zero winding, simplicity forces, up to cyclic permutation, reversal, coordinate relabelling, and independent sign choices, the cyclic step sequence $\varepsilon_ie_i,\ \varepsilon_je_j,\ -\varepsilon_ie_i,\ -\varepsilon_je_j$, with $i\neq j$: the boundary of one elementary plaquette. Equation (11) now forces $R \in \{F, \bar{F}\}$, and its energy is $2C_F$. This proves (5); the empty support gives the unique vacuum.

At each degree-two vertex Schur's lemma makes the intertwiner unique. The two orientations are $\chi_p|0\rangle$ and $\bar{\chi}_p|0\rangle$, exchanged by charge conjugation, so their negative eigenspace is the line $|p,-\rangle$. Conversely every such state is physical, contractible, and has energy $4(C_F/2)$. Peter–Weyl orthogonality separates distinct plaquettes, of which there are $3L^3$. Finally, the vacuum lies $2C_F$ below the shell and (5) places every higher eigenvalue at least $C_F/2$ above it, proving (7). $\square$

The preceding theorem is enough to identify the physical content of a small-coupling spectral patch. The volume-uniform persistence of that patch is not obtained by bounding the norm of the extensive perturbation. Instead we use the local spectral enclosure of Yarotsky [10].

Proposition 2 (Periodic three-link cell descent)**.** *Group the three positively oriented outgoing links at each vertex into one cell and rescale the Hamiltonian by* $3/2$*; the smallest nonzero one-link electric energy is* $2/3$*, so the rescaled one-cell Hamiltonian has gap one. Assign the three positively oriented plaquettes based at a vertex to that cell. Then each rescaled cell interaction has norm at most* $27|u|$*, the constants of the spectral localisation theorem of [10] may be taken independent of L, and with* $\lambda = 27c_2|u|$ *and* $27|u| < c_1$*,*

$$\operatorname{spec} \widetilde{H}_L(u) \subset \bigcup_{a \in \operatorname{spec} H_{E,L}} [a(1-\lambda), a(1+\lambda)]. \tag{13}$$

Proof. Since $\|\chi_p + \bar{\chi}_p\| \le 6$, the norm of each rescaled cell interaction is bounded by $\frac{3}{2}|u| \cdot 3 \cdot 6 = 27|u|$. The cell spaces may be infinite dimensional and the onsite Hamiltonians may be unbounded in the theorem of [10]. Its linked-set proof uses only finite range, the local norm, and an exponential bound on the number of connected sets through a marked cell. The same bound holds on every periodic finite quotient because its interaction graph has the same uniformly bounded degree. Thus the constants are independent of L and, translated back to the present energy units, (13) follows. $\square$

Theorem 3 ($SU(3)$ all-orders finite-volume Riesz island)**.** *For periodic* L^3 *tori with* $L \ge 3$*, let* $\widetilde{H}_L(u) = H_L(u) - E_{0,L}(u)I$ *and* $\lambda = 27c_2|u|$*, where* $c_1, c_2 > 0$ *are the range-dependent constants in the spectral localisation theorem of [10]. If*

$$27|u| < c_1, \qquad \lambda < \tfrac{1}{33}, \tag{14}$$

then the part of $\operatorname{spec} \widetilde{H}_L(u)$ *descending from the free energy* $E_F = 8/3$ *is separated from every other free-energy patch by*

$$\delta_-(u) \;\ge\; \frac{1-15\lambda}{3}, \qquad \delta_+(u) \;\ge\; \frac{1-33\lambda}{6}, \tag{15}$$

uniformly in L. The circle

$$\Gamma_F: \; |z - 8/3| = \rho, \qquad \rho = \frac{1-\lambda}{12}, \tag{16}$$

encloses that patch and obeys

$$\sup_{z \in \Gamma_F} \big\|(\widetilde{H}_L(u) - z)^{-1}\big\| \;\le\; \frac{12}{1-33\lambda}. \tag{17}$$

Restricted to $\mathcal{H}^{(0,-)}_{\mathrm{phys}}$*, its Riesz projection has rank exactly* $3L^3$*.*

Proof. Proposition 2 supplies the enclosure (13) with $\lambda = 27c_2|u|$ uniform in L.

For an $SU(3)$ irrep (p, q), its one-link electric energy in units of $1/6$ has numerator $p^2 + q^2 + pq + 3p + 3q$. The nonzero one-link numerators below 18 are $4, 9, 10, 16$, so their additive semigroup has consecutive attainable values $14, 16, 17$ around the target 16. Consequently the lower and upper separations between the target interval and its nearest competitors are exactly the right-hand sides of (15); the upper one is positive precisely under the second condition in (14). The target interval has half-width $8\lambda/3$. Direct subtraction from (16) gives the minimum contour clearance $(1 - 33\lambda)/12$, proving (17).

Gauge transformations, charge conjugation and the electric one-form symmetries commute with $H_L(u)$ and hence with the contour projection. The same contour works along $t \mapsto tu$, $0 \le t \le 1$. Finite-volume compact resolvent and bounded analytic perturbation make the restricted projection norm-continuous, so its finite rank is constant. At $u = 0$ Theorem 1 identifies that rank as $3L^3$. ☐

Remark 4 (Exact boundary of the all-orders result). Theorem 3 transports the complete three-component shell, not one internally isolated band. The constants c_1, c_2 are not numerical, the result is confined to fixed-spacing strong coupling, and the separate one-particle theorem of [10] is not invoked: its nondegenerate one-site, nonresonance hypothesis fails for this composite, degenerate plaquette shell. The theorem therefore supplies an interacting finite-volume Riesz cluster, not an infinite-volume particle or Wilson-source residue.

Remark 5 (The two small-torus coincidences). At $L = 4$ a once-winding fundamental loop has energy $2C_F$ for every $N \ge 3$; trivial one-form flux is therefore load-bearing at every rank, not only at $SU(4)$. The genuinely rank-four coincidence occurs instead at $L = 3$: an $SU(4)$ loop in $\Lambda^2 F$ has $C_2 = 5/2$ and energy $3C_2/2 = 15/4 = 2C_F$, but its N-ality is two, so the same flux condition excludes it (and its character is charge-even). These are volume effects, not a continuum statement.

Set P_- equal to the projector in (6) and $Q_- = 1_- - P_-$, where 1_- is the identity on $\mathcal{H}^{(0,-)}_{\text{phys}}$. Both H_E and V preserve this space: H_E commutes with C and every $Z_j(z)$, while every contractible plaquette term in V is charge-even and centre-neutral. No degenerate $E_{F,N}$ direction is omitted, and degenerate perturbation theory gives

$$H_{\text{eff},-} = P_- H_E P_- + uH_1 + u^2 H_2 + u^3 H_3 + \cdots \tag{18}$$

with every H_r linked, vacuum-subtracted, and carrying the appropriate Q-space resolvents.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2\partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1\partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4\sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \tag{19}$$

Theorem 6 (Incidence factorisation). *With* $w(k) = (\bar{d}_3, -\bar{d}_2, \bar{d}_1)^\mathsf{T}$,

$$BB^\dagger = qI - ww^\dagger, \qquad B^\dagger w = 0, \qquad \|w\|^2 = q,$$

and therefore $\operatorname{spec}(BB^\dagger) = \{0, q, q\}$. *With* $S = BB^\dagger - 4I$ *the signed shared-link adjacency,* $\operatorname{spec} S(k) = \{-4,\ q-4,\ q-4\}$.

Proof. Multiplying (19) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. □

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ, $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis.

Theorem 7 (Finite-torus carrier)**.** *For the periodic cubic cellulation of* T^3, $\ker \partial_2 = \operatorname{im} \partial_3 \oplus H_2$ *with* $\dim \operatorname{im} \partial_3 = L^3 - 1$ *and* $\dim H_2 = 3$, *so* $\dim \ker \partial_2 = L^3 + 2$.

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\operatorname{rank} \partial_3 = L^3 - 1$ by rank–nullity. Discrete Hodge decomposition gives $\ker \partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. □

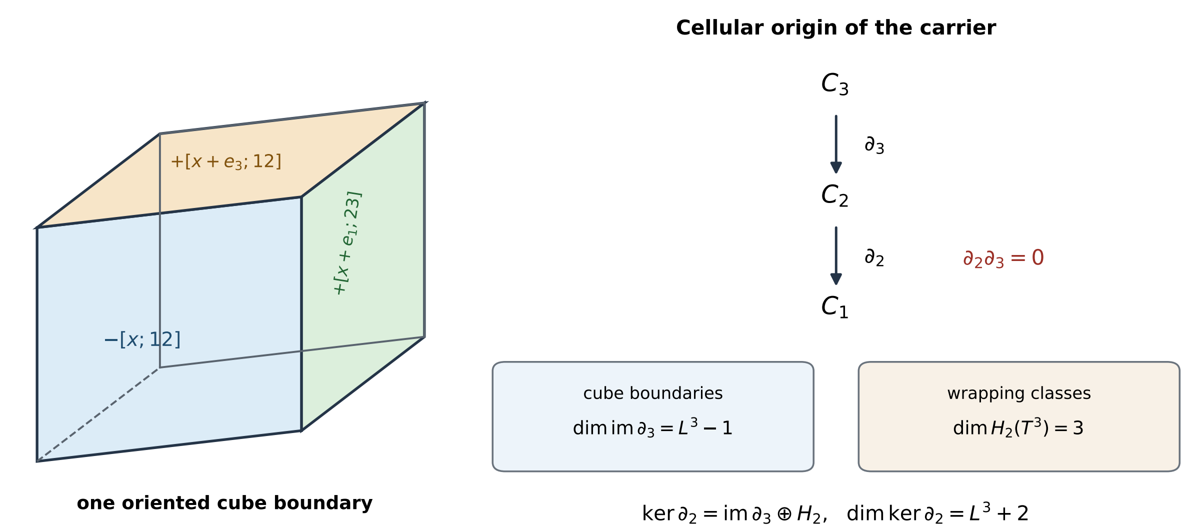

Figure 1: The carrier is a cellular cycle space. The left panel shows three visible terms in the outward-oriented boundary of one cube; the complete six-face formula is given in Appendix E. Cube boundaries span $\operatorname{im} \partial_3$, while three noncontractible wrapping classes complete $\ker \partial_2$ on the three-torus.

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix E; the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound the kernel from below, and rank over $\mathbb{F}_p$ bounds it from above, since rank can only drop under reduction. The two bounds meet at $L = 1, \dots, 5$.

Remark 8. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x + e$ and $x - e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

The following sum rule joins Theorems 6 and 7, which are otherwise two unrelated statements: a 3×3 spectrum and an $(L^3 + 2)$-dimensional space.

Theorem 9 (Bloch–chain bridge)**.** *On the periodic L^3 complex,*

$$\sum_k (3 - \operatorname{rank} B(k)) = L^3 + 2,$$

the sum running over the L^3 allowed momenta, with nullity profile 3 at Γ exactly once and 1 at each of the other $L^3 - 1$ momenta.

Proof. $B(0) = 0$ contributes 3. For $k \neq 0$ some $d_j \neq 0$, and Theorem 6 gives $\operatorname{rank} B(k) = 2$, contributing 1 each. Summing gives $3 + (L^3 - 1) = L^3 + 2$. □

Remark 10. Earlier editions of this paper read the agreement as a coincidence worth checking — "the 3 is the triple degeneracy of $B(0) = 0$, not $b_2(T^3)$". That was wrong, and the correction is the better statement. Under the discrete Fourier transform ∂_2 block-diagonalises and $\ker \partial_2 = \bigoplus_k \ker \partial_2(k)$; at $k \neq 0$ the kernel block is one-dimensional and is exactly $\operatorname{im} \partial_3(k)$, while $\partial_3(0) = 0$. So all of $\operatorname{im} \partial_3$ sits at nonzero momenta and the quotient is supported entirely at Γ: the Γ block *is* H_2, and the 3 *is* $b_2(T^3)$. The two routes do not agree by accident; they are the same decomposition read in two bases.

The same block decomposition says exactly what a local seed costs at sharp momentum.

Proposition 11 (Fixed-momentum carrier and the locality tradeoff)**.** *Let $b_x = \partial_3[x; 1, 2, 3]$ be the six-face boundary of the elementary cube at x, and use the unitary discrete Fourier transform $\hat{b}_L(k) = L^{-3/2} \sum_{x \in \Lambda_L} e^{-ik\cdot x} b_x$. For every allowed $k \neq 0$, $\hat{b}_L(k)$ has norm $\sqrt{q(k)}$ and, up to the Fourier phase convention, is the vector $w(k)$ of Theorem 6. Hence $\varphi_L(k) = \hat{b}_L(k)/\sqrt{q(k)}$ is exactly the unique unit carrier vector in that momentum fibre.*

If $k_0 = 2\pi(r_1, r_2, r_3)/L_0 \neq 0$, then k_0 is an allowed momentum on every nested volume $L = mL_0$, and the order-u^2 internal separation from each of the two incidence branches is

$$t_N u^2 q(k_0), \tag{20}$$

independent of m. More generally it is bounded below by $t_N u^2 q_*$ on any set of allowed momenta with $q(k) \geq q_* > 0$.

This exact sharp-momentum state is not local: it is the Fourier sum of all cube boundaries. Conversely, for the normalised radius-one seed $b_x/\sqrt{6}$,

$$|\langle \varphi_L(k),\, b_x/\sqrt{6} \rangle| = \sqrt{\frac{q(k)}{6L^3}}, \tag{21}$$

and $\hat{b}_L(0) = 0$.

Proof. Fourier transforming the six terms of the cubical boundary gives, in the $(12, 13, 23)$ face basis, $(d_3, -d_2, d_1)^T$, with conjugates if the opposite Fourier sign is chosen. Its squared norm is $q(k)$ and $\partial_2 \partial_3 = 0$ puts it in $\ker B(k)^\dagger$; uniqueness for $k \neq 0$ follows from Theorem 6. The nested-volume statement is the integer identity $k_0 = 2\pi(mr)/(mL_0)$, and (20) follows from Corollary 19. Inverting the unitary Fourier transform gives $b_x = L^{-3/2} \sum_k e^{ik\cdot x} \sqrt{q(k)}\, \varphi_L(k)$, which proves (21); the $k = 0$ coefficient vanishes. □

Remark 12 (What the L^{-2} collapse does and does not say)*.* The minimum of $q(k)$ over all nonzero finite-volume momenta is $4 \sin^2(\pi/L)$, so a contour isolating the single flat sheet from the other two sheets uniformly over the entire Brillouin zone cannot follow from the second-order symbol. This is an *internal* sheet separation. It does not erase the volume-uniform external electric-shell gap (7), and it does not affect the fixed-k coefficient (20). Neither fact by itself controls the interacting Riesz projection or supplies transfer-matrix overlap; those remain the substantive part of G18.

3.1 The wrapping sheets are cycles, not harmonic — and their harmonic representatives

Theorem 7 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x:\, x_m=0}[x; i, j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker \partial_2 / \operatorname{im} \partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s,\ \partial_3 \partial_3^\dagger\, s\rangle}{\|s\|^2} = 2 \qquad \text{exactly, for every } L \geq 2. \tag{22}$$

The distinction matters because H_2 is used in two senses: the quotient in Theorem 7, which the sheets span, and the harmonic subspace $\ker \partial_2 \cap \ker \partial_3^\dagger$, which they do not lie in.

Harmonic representatives do exist, and they are one line away. Averaging a sheet over its L translates in the complementary direction,

$$h_{(ij)} = \frac{1}{L} \sum_{r=0}^{L-1} s_{(ij)}(r), \qquad \partial_2 h_{(ij)} = 0, \qquad \partial_3^\dagger h_{(ij)} = 0, \tag{23}$$

gives three harmonic chains spanning $\ker \partial_2 \cap \ker \partial_3^\dagger$, which is exactly three-dimensional; and $s_{(ij)} - h_{(ij)} \in \operatorname{im} \partial_3$, so a sheet and its average are the same homology class. The averaging is the $k = 0$ Fourier projection, and what it removes is precisely the non-harmonic part measured above — which is Remark 10 again, in real space. Nothing about the finding is withdrawn — the generators Theorem 7 exhibits are still not harmonic — but a real-space statement phrased for the harmonic subspace now has representatives to point at, and Proposition 39 covers both, since $B^\dagger w = 0$ holds on all of $\ker \partial_2$.

4 Exact adjacent shared-link dynamics and global assembly

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \mathrm{Adj}, \qquad F \otimes F = \Lambda^2 F \oplus \mathrm{Sym}^2 F, \tag{24}$$

with dimensions and quadratic Casimirs

$$\begin{aligned}
d_{\mathbf{1}} &= 1, & C_{\mathbf{1}} &= 0,\\
d_{\mathrm{Adj}} &= N^2 - 1, & C_{\mathrm{Adj}} &= N,\\
d_{\Lambda^2 F} &= \tfrac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \tfrac{(N+1)(N-2)}{N},\\
d_{\mathrm{Sym}^2 F} &= \tfrac{N(N+1)}{2}, & C_{\mathrm{Sym}^2 F} &= \tfrac{(N-1)(N+2)}{N}.
\end{aligned} \tag{25}$$

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 13 (Shared-link channel weights)**.** *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2, for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\mathrm{Tr}(AU)\,\mathrm{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree-$(2,2)$ moment of U.

Second, two such moments settle both families. In the Weingarten calculus, whose asymptotic origin is [14] and whose finite-N closed forms are [27], the order-two values are $\mathrm{Wg}(e) = 1/(N^2-1)$ and $\mathrm{Wg}((12)) = -1/(N(N^2-1))$ — the inverse of the S_2 Gram matrix $\left(\begin{smallmatrix} N^2 & N \\ N & N^2 \end{smallmatrix}\right)$ — the index sums give

$$\begin{aligned} M_{\mathrm{direct}} &= \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \\ M_{\mathrm{cross}} &= \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N. \end{aligned} \tag{26}$$

The standard formula is often written for $U(N)$. A degree-$(2,2)$ integrand is invariant under $U \mapsto zU$ for $z \in U(1)$, so integrating it over $SU(N)$ and over $U(N) = (SU(N) \times U(1))/\mathbb{Z}_N$ gives the same number provided the $SU(N)$-invariants at this degree are exhausted by the permutation contractions — which holds exactly when the degree is below the rank, so that no determinant (Levi-Civita) contraction is available. At degree two this is the condition $N \geq 3$, which is where the construction already lives. For the like family, $\mathrm{Tr}(AU)\,\mathrm{Tr}(BU)$ decomposes into $\mathrm{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\mathrm{direct}} \pm M_{\mathrm{cross}})/2$. Normalising,

$$n_{\mathrm{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\mathrm{Sym}^2 F}}{N^2}, \qquad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij}\bar{U}_{lk}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1 against $M_{\mathrm{direct}} = N^2$, so $n_{\mathbf{1}} = 1/N^2 = d_{\mathbf{1}}/N^2$ and $n_{\mathrm{Adj}} = 1 - 1/N^2 = d_{\mathrm{Adj}}/N^2$. □

Remark 14. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (26) are explicit, so the reader can check it without leaving this page. That the pair is not two fitted constants is separately checked — the identity-permutation value times $N^2 - 1$ is 1 identically in N, so the $SU(3)$ numbers are a specialisation of a rank-generic formula. Together these make the *local shared-link coefficient* self-contained. By themselves they do not enumerate every second-order process; that separate step is supplied by Lemma 17. Appendix B records the index sums.

Remark 15 (External corroboration). The same numerator appears in the quantum-simulation literature from an unrelated direction. Ciavarella, Burbano and Bauer [36] publish a finite-rank plaquette matrix element $|M_\rho|^2 = \dim\rho/(\dim A \dim R)$; for two fundamental factors $\dim A = \dim R = N$ and this is d_ρ/N^2 identically in N. That is provenance for a chain already checked here, not a second derivation, and it is not a second derivation. The exhaustion of the shared-link routes is proved independently in Lemma 17.

Remark 16 (A falsified zero backend). The organized authority corpus contains an independent exact diagnostic at $N = 3$:

$$\mathrm{Wg}(e) = \frac{1}{8}, \qquad \mathrm{Wg}((12)) = -\frac{1}{24}, \qquad \int_{SU(3)} |U_{11}|^4 \, dU = \frac{1}{6} \neq 0.$$

The related fourth moment $\int |U_{11}|^4 \, dU = 2/[N(N+1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2,2)$ sector. It therefore falsifies the archived stranded-flux rule that discards this sector. This is useful negative evidence for retaining the contractions above; it is not a second derivation of t_N.

4.2 The off-diagonal process list is complete

Write $X_q = \chi_q + \bar{\chi}_q$, so $V = \sum_q X_q$, and let $Q_0 = 1 - |0\rangle\langle 0|$.

Lemma 17 (Second-order process completeness)**.** *For $N \geq 3$, $L \geq 3$, and distinct plaquettes p, p', define*

$$K_{2,-} = P_- V Q_- (E_{F,N} - H_E)^{-1} Q_- V P_-, \tag{27}$$

$$H^{\mathrm{exc}}_{2,-} = K_{2,-} - E^{(2)}_{\mathrm{vac}} P_-, \qquad E^{(2)}_{\mathrm{vac}} = \langle 0|V Q_0 (0 - H_E)^{-1} Q_0 V|0\rangle. \tag{28}$$

If p and p' share no link, then $\langle p', -|K_{2,-}|p, -\rangle = 0$. If they share the unique link e, every nonzero intermediate lies in exactly one of $F \otimes \bar{F} = \mathbf{1} \oplus \mathrm{Adj}$, $F \otimes F = \Lambda^2 F \oplus \mathrm{Sym}^2 F$ or the conjugate of the second decomposition. Thus these four shared-link channels exhaust every off-diagonal second-order process.

The conclusion is preserved by the stated link-Casimir budget cutoff, and more generally by cutoffs that are functions only of the link Casimirs and therefore act identically on all four orientation assignments. A general projection inside the degenerate $2E_{F,N}$ eigenspace need not preserve the non-adjacent cancellation.

Proof. Expand the external states and both insertions in oriented characters. A term $\chi_p^{\varepsilon}\chi_q^{\eta}$, with $\varepsilon, \eta \in \{\pm 1\}$ and $\chi^{-1} = \bar{\chi}$, has linkwise centre charge $\partial(\varepsilon p + \eta q)$ in $C_1(\Lambda_L; \mathbb{Z}_N)$. Since H_E commutes with the independent link-centre actions, a common electric intermediate between the p and p' sides requires

$$\partial(\varepsilon p + \eta q) = \partial(\varepsilon' p' + \eta' r) \pmod{N} \tag{29}$$

for action plaquettes q, r and signs $\varepsilon, \eta, \varepsilon', \eta'$.

We first record a small-support fact. A nonzero 2-cycle over $\mathbb{Z}_N$ cannot be supported on at most four elementary plaquettes. Choose a supported face f. Each of its four boundary links needs another supported face to cancel its nonzero coefficient, while two distinct cubical faces share at most one link. Four distinct additional faces would therefore be required.

Apply this fact to $z = \varepsilon p + \eta q - \varepsilon' p' - \eta' r$. Equation (29) makes z a 2-cycle over $\mathbb{Z}_N$, so $z = 0$ there. Because $p \neq p'$, the four occurrences must pair with opposite signs. The only possibilities are

$$q = p, \quad \eta = -\varepsilon, \quad r = p', \quad \eta' = -\varepsilon', \tag{30}$$

or

$$q = p', \quad r = p, \quad \eta = \varepsilon', \quad \eta' = \varepsilon. \tag{31}$$

This also closes the determinant exceptions. For $SU(3)$ a coefficient ± 3 would require three occurrences of one face and leave a distinct fourth face with coefficient ± 1; for $SU(4)$ a coefficient ± 4 would require all four occurrences to be the same face. Both contradict $p \neq p'$.

The first alternative is absent from the charge-odd vector:

$$X_p|p, -\rangle = \frac{(\chi_p + \bar{\chi}_p)(\chi_p - \bar{\chi}_p)}{\sqrt{2}}|0\rangle = \frac{\chi_p^2 - \bar{\chi}_p^2}{\sqrt{2}}|0\rangle,$$

so its same-face zero-N-ality component, including the vacuum, cancels. Only the exchange alternative remains.

If p, p' are link-disjoint, sharing a vertex being immaterial, linkwise factorisation gives $X_{p'}|p, -\rangle = \sqrt{2}\,|p, -\rangle|p', +\rangle$ and $X_p|p', -\rangle = \sqrt{2}\,|p, +\rangle|p', -\rangle$. Both vectors have electric energy $2E_{F,N}$, so the reduced resolvent is scalar on their span, while their inner product is zero by charge orthogonality.

If p, p' share e, the six nonshared links each carry one fundamental or antifundamental factor and the shared link carries exactly two. The two complete tensor-product decompositions above therefore give all common intermediates. Their energies and gaps are $E_\rho = 3C_F + \frac{1}{2}C_\rho$ and $E_\rho - E_{F,N} = C_F + \frac{1}{2}C_\rho > 0$. Spectral resolution yields only their four projectors. Vacuum subtraction is diagonal and leaves this cross matrix element unchanged. The two-face term is connected, so the linked Möbius projection leaves it unchanged as well. □

With Theorem 13 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \tag{32}$$

and summing within the two orientation families,

$$A_N = w_{\mathbf{1}} + w_{\mathrm{Adj}} = -\frac{2N^3}{(N^2-1)(2N^2-1)}, \tag{33}$$

$$B_N = w_{\Lambda^2} + w_{\mathrm{Sym}^2} = -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)}. \tag{34}$$

4.3 The shared-link law

Theorem 18 (All-rank adjacent shared-link law)**.** *Let $p \neq p'$ be oriented plaquettes sharing the single link e. For every integer $N \geq 3$, let $s = s(p,e)$, $s' = s(p',e)$, and let $\hat{\Pi}_\rho$ be the charge-conjugation-closed electric projector onto the shared-link channel ρ (including the conjugate irrep on the conjugate like-channel subspace). Define*

$$\eta_\rho = \begin{cases} -1, & \rho \in \{\mathbf{1}, \mathrm{Adj}\}, \\ +1, & \rho \in \{\Lambda^2 F, \mathrm{Sym}^2 F\}. \end{cases}$$

Then

$$\hat{\Pi}_\rho X_{p'}|p,-\rangle = \eta_\rho s s' \, \hat{\Pi}_\rho X_p |p',-\rangle, \qquad \big\|\hat{\Pi}_\rho X_{p'}|p,-\rangle\big\|^2 = \frac{d_\rho}{N^2}, \tag{35}$$

$$\langle p',-|X_p \, \hat{\Pi}_\rho \, X_{p'}|p,-\rangle = \eta_\rho \, s(p',e)\, s(p,e)\, \frac{d_\rho}{N^2}. \tag{36}$$

Consequently the linked order-u^2 contribution through all four channels is

$$\langle p',-|H_2|p,-\rangle = t_N \, s(p',e)s(p,e) = t_N (BB^\dagger)_{p'p},$$

where $s(p,e) = \pm 1$ is the cellular boundary incidence and

$$t_N = B_N - A_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0.$$

Proof. Because distinct cubical faces share at most one link, $(BB^\dagger)_{p'p} = ss'$. Expand $|p;\varepsilon\rangle_{\mathrm{or}} := \chi_p^\varepsilon|0\rangle$, with $\chi_p^{+1} := \chi_p$ and $\chi_p^{-1} := \bar{\chi}_p$, so that $|p,-\rangle = 2^{-1/2}\sum_{\varepsilon=\pm1}\varepsilon\,|p;\varepsilon\rangle_{\mathrm{or}}$. On the shared link the two oriented factors are εs and $\varepsilon' s'$. A like channel requires $\varepsilon\varepsilon' = ss'$, while a mixed channel requires $\varepsilon\varepsilon' = -ss'$. Thus the two projected route vectors differ by exactly $\eta_\rho ss'$. Their two charge-conjugate assignments are orthogonal for $N \geq 3$; the two factors $1/\sqrt{2}$ therefore sum to one channel weight, not twice one. Theorem 13 gives the norm d_ρ/N^2, proving (35) and (36).

On this channel, $(E_{F,N} - H_E)^{-1}\hat{\Pi}_\rho = -\hat{\Pi}_\rho/(C_F + C_\rho/2)$. Its contribution is therefore $\eta_\rho ss' w_\rho$. Summing gives $ss'(w_{\Lambda^2 F} + w_{\mathrm{Sym}^2 F} - w_{\mathbf{1}} - w_{\mathrm{Adj}}) = ss'(B_N - A_N)$. Substitution of (33)–(34) gives the displayed rational function. Every denominator factor and $N^2 - 4$ is positive for $N \geq 3$. □

The opening geometric ingredient of the proof is no longer prose: the verifier enumerates every distinct face pair of the periodic complex at $L = 3$ and $L = 4$ and confirms that no pair meets in more than one link. Two faces sharing a link span at most two sites per axis, so these windows already exhibit every local configuration and larger L adds translates only.

Corollary 19 (Global torus assembly)**.** *On the trivial-flux charge-odd electric shell of Theorem 1, translation and cubic symmetry make the remaining diagonal contribution a scalar, and*

$$H^{(N)}_{\mathrm{eff},-}(k, u) = E_{\mathrm{flat},N}(u)I + t_N u^2 B(k)B(k)^\dagger + O(u^3).$$

Consequently the three bands through second order are $E_{\mathrm{flat},N}(u)$ and $E_{\mathrm{flat},N}(u) + t_N u^2 q(k)$ twice, and the flat carrier is $\ker \partial_2$, of dimension $L^3 + 2$.

Proof. The centre-charge argument of Lemma 17, now with only three plaquette occurrences, gives $\langle p', -|P_- V P_-|p, -\rangle = 0$ for $p \neq p'$. A possible $SU(3)$ determinant balance would require all three occurrences to be the same face and is therefore excluded off diagonal. Translation and cubic symmetry make the remaining first-order diagonal a scalar, including any $SU(3)$ same-face determinant contribution, and it is absorbed into $E_{\mathrm{flat},N}(u)$.

At second order Lemma 17 and Theorem 18 assemble every off-diagonal entry. Translations act transitively on faces of a fixed orientation and the cubic group permutes the orientations, so the invariant second-order diagonal is also a multiple of the identity. Thus the adjacency part is $t_N(BB^\dagger - 4I)$; absorb $-4t_N$ into the scalar and apply Theorem 6. □

The status change deserves one sober sentence rather than celebration. What earlier editions carried as two prose premises is now proved relative to the standard degenerate perturbation framework of (18), the classical Casimir arithmetic of Theorem 1, and the centre-charge classification of Lemma 17. The finite-volume $L = 3, 4, 5$ enumerations in the verifier are regressions on those proofs, not substitutes for them. The charge-even identification of Section 6 remains a hypothesis, and nothing about convergence, higher orders, or the continuum has changed.

Two consequences,

$$t_3 = \frac{5}{612}, \qquad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \cdots \tag{37}$$

The expansion is governed by a closed form. Writing the planar-normalised coefficient as $4N^3 t_N$,

$$1 - 4N^3 t_N = \frac{2N^4 + 31N^2 - 9}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}, \tag{38}$$

an identity of rational functions, and the right side is positive and the left strictly decreasing for every real $N \geq 2$: substituting $N = M + 2$ makes every numerator coefficient of both the remainder and the derivative nonnegative, so $4N^3 t_N$ increases monotonically to 1. At fixed canonical coordinate u the conditional bandwidth of Corollary 19 is $12t_N u^2$, so the entire dispersive content of the projected sector switches off at exactly the rate N^{-3}, with computable subleading coefficient $-1/(4N^2)$: conditional on the assembly, the carrier's exact degeneracy is approached by the *whole* sector in the planar limit. This is consistent with the general suppression of glueball mixing at large N, but no 't Hooft-coupling identification is made here: the statement is about the coefficient in the canonical coordinate, and it is exact.

4.4 The vacuum-mediated route

Lemma 17 is specific to the charge-odd shell. The corresponding assertion is false in general: the charge-even sector admits a vacuum-mediated route. Its cancellation in the odd sector is the charge-conjugation selection rule below, and omitting it is a recorded error, not a hypothetical one.

Proposition 20 (Vacuum-mediated route)**.** *The reduced-resolvent route*

$$\frac{\langle p'|V|0\rangle\langle 0|V|p\rangle}{E_{F,N}-E_0}=\frac{\langle p'|V|0\rangle\langle 0|V|p\rangle}{E_{F,N}}$$

is available for any pair of plaquettes, sharing a link or not. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p,-\rangle=0$. Consequently the charge-odd hopping $t_N=B_N-A_N$ carries no such term, while the charge-even shared-link element acquires $1/C_F$: the normalised charge-even matrix-element product is 2 and $E_{F,N}=2C_F$.

$$\ell_N=A_N+B_N+\frac{1}{C_F}=-\frac{2N(3N^2-5)}{(N^2-1)(2N^2-1)(4N^2-9)}.$$

Remark 21. At $N=3$, $A_3+B_3=-481/612$ and $1/C_F=3/4$, giving $\ell_3=-11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

In the charge-odd shell, Lemma 17 and this proposition dispose of non-adjacent pairs: the exchange vectors are orthogonal and the vacuum route vanishes. In the charge-even sector, however, this proposition does not dispose of the non-adjacent vacuum route. Taken over all pairs it would be a rank-one operator supported at $k=0$, not the twelve-neighbour adjacency multiplied by ℓ_N. The check above establishes the all-rank shared-link value ℓ_N; the reorganisation or cancellation of the charge-even all-pairs term remains a separate requirement.

4.5 Finite-volume width

Let $q_{\max}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\max}(L)=\begin{cases}12, & L \text{ even},\\ 12\cos^2(\pi/2L), & L \text{ odd},\end{cases} \tag{39}$$

because only at even L does some k_j reach π exactly. The width of the full retained charge-odd manifold is $W^{(-)}_{N,L}=q_{\max}(L)\,t_N u^2+O(u^3)$, so for even L, and coefficientwise as $L\to\infty$ at fixed perturbative order, $[u^2]W^{(-)}=12t_N\sim 3/N^3$. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

The smallest nonzero incidence eigenvalue is $q_{\min}=4\sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

4.6 The orientation signs are essential

Removing the orientation signs — from the entries of (19) and from the phases, so each $e^{ik}-1$ becomes $e^{ik}+1$ — gives an unsigned incidence $\mathcal{N}(k)$. Define its comparison adjacency by $A_+(k):=\mathcal{N}(k)\mathcal{N}(k)^\dagger-4I$. Its adjacency spectra at Γ and R are $\{12,0,0\}$ and $\{-4,-4,-4\}$, which share no

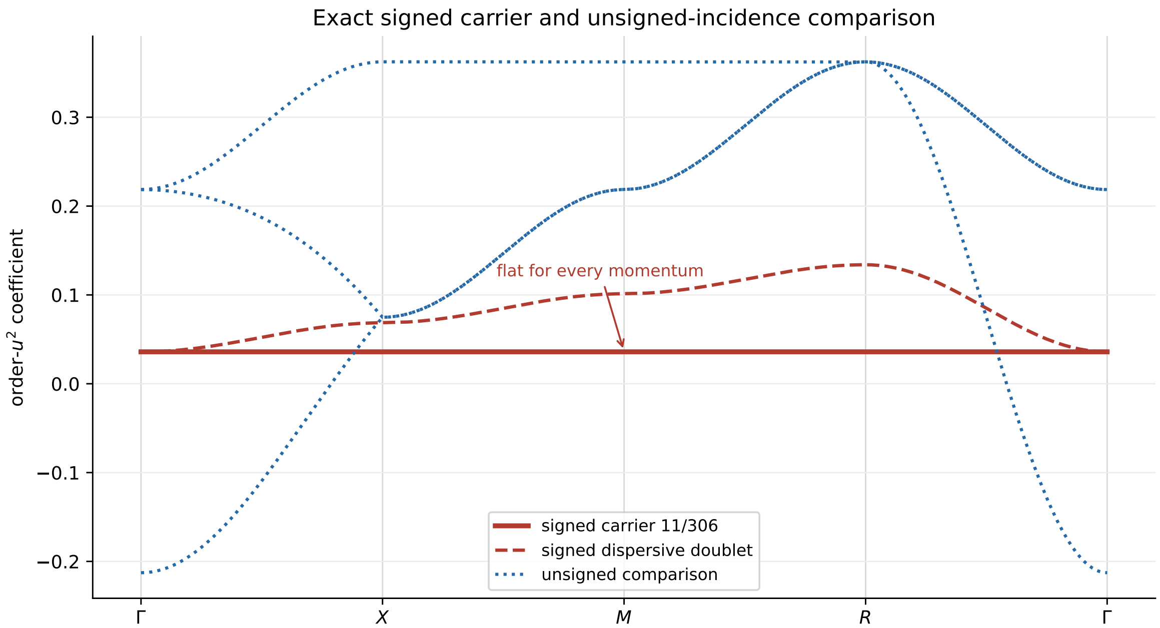

Figure 2: Order-u^2 coefficients along Γ–X–M–R–Γ. The red carrier and dispersive doublet are the signed incidence result; their conditional global assembly is Corollary 19. The blue curves are the exact unsigned-incidence comparison spectrum. Reading the blue curves as physical charge-even bands requires the separate symbol hypothesis of Section 6.

eigenvalue, so no level can be momentum independent. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

Section 6 computes the unsigned comparison spectrum in closed form. The signed span is $8-(-4)=12$, giving the charge-odd second-order width $12t_3 = 5/51$ at $N=3$. The unsigned continuous-zone span is $12-(-4)=16$. Interpreting $16|\ell_3| = 88/153$ as a charge-even bandwidth is conditional on deriving the complete linked charge-even symbol; on a finite torus the value -4 is sampled only when L is even.

5 The second-order coefficient on a two-cube space

Everything above derives t_N from a resolvent ledger indexed by the fusion channel of one shared link. That ledger is representation theory carried out on a single plaquette pair; it has never been tested against an operator calculation on a space large enough for two cubes to interact. This section reports such a test and, more importantly, states what it settles.

The construction is the open face-sharing $(3,2,2)$ prism with its eleven oriented plaquettes, in the exact-Casimir $\mathcal{B}=6$ $SU(3)$ truncation, of basis dimension 1,590,462. The delivery reports that it reaches the relevant sector through 794 paths and 398 reachable states; it does not assemble a 1,590,462-dimensional magnetic matrix. Its complete charge-odd one-plaquette shell at $E_* = 8/3$ has dimension eleven — a census result and not the plaquette count. The full E_* eigenspace has dimension 22, split $11_{C=+} \oplus 11_{C=-}$ by tensor-derived charge conjugation, and the delivery removes all 22 from the pseudoinverse; because the eleven columns exhaust the odd half, that agrees with the $Q = 1_- - P_-$ prescription of (18). Writing K_{LR} for the raw second-order operator and K_L, K_R, K_F for the left-cube, right-cube and shared-face operators in their own coordinates, the connected operator is defined by the literal Möbius fold

$$K_{\mathrm{conn}} = K_{LR} - J_L K_L J_L^{\dagger} - J_R K_R J_R^{\dagger} + J_F K_F J_F^{\dagger}, \tag{40}$$

and the same fold applied to the geometry gives G_{conn}. The fold is literal in the sense that no eigenvalue or fitted scalar is subtracted: each source operator is folded in its own coordinates and lifted. The delivery corroborates that internally by applying the same weights history by history over its 1,984 ordered two-step histories, of which 1,192 cancel at weight zero, and the two routes agree to 4.9×10^{-15}.

First order settles itself. On the shell $PVP = I_{11}$, so the first-order term is nonzero but scalar: it shifts all eleven branches equally and creates no branch ambiguity. Its connected fold then vanishes *identically*, and by counting rather than by cancellation — each of the eleven faces is covered exactly once by $\{L, R\}$ once the shared face is restored by F, so every Möbius weight in (40) applied to the identity is $1 - 1 = 0$. Second order is therefore the leading connected term and not a correction to a first-order one.

At second order the result is

$$K_{\mathrm{conn}}^{(2)} = \frac{5}{612}\, G_{\mathrm{conn}} + D, \tag{41}$$

with D diagonal and delivered. The geometry alone fixes the *splitting* inside each block and nothing else: G_{conn} turns out to be four disjoint 2×2 blocks plus three isolated directions (Appendix F), so each pair splits by $2 \times 5/612 = 5/306$ about its own diagonal entry. Given D, which Appendix F prints, the spectrum is then closed form: 0 once, $-\frac{15}{4}$ twice, $-\frac{34}{9}$ four times, and $-\frac{129}{34}$ four times — exact arithmetic on entries that are themselves reconstructions, a boundary made precise below. The delivery reports the coefficient recovered target-blind — supplied to no builder, graph fit, branching rule or validator — and reports that the payload hashes were produced before the recorded comparison, a chronology it itself calls record-backed and only partly auditable; the value is byte-present in the nested $\mathcal{B} = 4$ package that the $\mathcal{B} = 6$ build seals as an authority object, so the defensible statement is dependency-path nonuse and not corpus-wide absence. This paper re-thresholds those delivered gates rather than re-running the build, including a deliberately wrong-sign control that the held-out scaling rejects: on a 66-dimensional held-out word space the correct second-order residual falls with log-log slope 2.90 while the sign-reversed control falls with slope 2.00, a minimum error ratio of 325. What all this establishes is runtime nonuse of the comparator on the delivered dependency path, which is not a preregistration. The finite-u validations are asymmetric: the $\mathcal{B} = 4$ delivery includes a held-out diagonalisation of the full 4,180-dimensional charge-odd space with an $O(u^3)$ residual, whereas the $\mathcal{B} = 6$ check is restricted to the 66-dimensional $P + Q_1$ star.

Resolving (41) by the irreducible representation the shared link actually carries in the ranked intermediate state gives six coefficients rather than four, because a link distinguishes ρ from $\bar{\rho}$:

$$\begin{array}{c|cccccc} \rho & \mathbf{1} & \mathbf{3} & \bar{\mathbf{3}} & \mathbf{6} & \bar{\mathbf{6}} & \mathbf{8} \\ \hline c_\rho & \frac{1}{12} & -\frac{1}{12} & -\frac{1}{12} & -\frac{1}{9} & -\frac{1}{9} & \frac{16}{51} \end{array} \tag{42}$$

That these six sum to $5/612$ is weak evidence: six rationals reach one target in many ways, and a fitted set would pass it. The statement worth making is stronger, and it is what makes this more than a coincidence of one sum. First, though, the boundary.

The six coefficients of (42) are not symbolic: the delivery stores them as rational reconstructions of pinned finite-precision Clebsch–Gordan contractions, with a maximum reconstruction residual of 2.2×10^{-15} per channel — 2.1×10^{-14} across the delivery's reconstructed matrices as a whole — and its own certificate declaring `symbolic_exact_local_amplitudes: false`. What follows is therefore reported, conditional on those six numbers, and not a symbolic identity in the sense of Theorem 13.

Reported computational result 22 (The census is the Weingarten ledger)**.** At $N = 3$ the six reconstructed link-irrep coefficients of (42) are the four fusion-channel weights (32) individually:

$$c_{\mathbf{1}} = -w_{\mathbf{1}}, \qquad c_{\mathbf{8}} = -w_{\mathrm{Adj}},$$

$$c_{\mathbf{3}} = c_{\bar{\mathbf{3}}} = \tfrac{1}{2} w_{\Lambda^2 F}, \qquad c_{\mathbf{6}} = c_{\bar{\mathbf{6}}} = \tfrac{1}{2} w_{\mathrm{Sym}^2 F}.$$

Substituting (25) into (32) at $N = 3$ gives $w_{\mathbf{1}} = -1/12$, $w_{\mathrm{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$ and $w_{\mathrm{Sym}^2 F} = -2/9$; comparison with (42) is then arithmetic. Six reconstructed rationals landing on four predicted ones, with the two coincidences the correspondence requires, is a strong consistency statement and not a proof that the underlying contraction is exact.

One further feature of the census is worth separating out, because it is the paper's central architecture measured rather than assumed. The six link-resolved matrices are not merely six numbers that sum correctly: each is *separately* proportional to the same incidence Gram. The eleven oriented plaquettes of the prism have exactly 56 ordered adjacent pairs, each sharing one link with signed overlap ± 1; the delivery reports a constant channel-to-geometry ratio on all 56 of them, for every one of the six channels, with worst residual 2.2×10^{-15}. Colour dynamics fixes a number per channel and the cellular boundary operator fixes the matrix; if the two mixed, the ratio would be constant for the total and not for the parts.

The correspondence is scoped to what the Casimir closure exhausts: adjacent, one-action, shared-link intermediates at order u^2. Same-face diagonal routes, on-site terms and higher-action intermediates lie outside it — the $\mathcal{B} = 6$ scalar below is exactly a place where one of them is missing — and so does any untruncated-Hilbert-space statement.

It is scoped in one further way, which matters more. The two-cube build draws its local Wilson amplitudes from the same hash-pinned upstream Clebsch–Gordan archive as the single-cube reconstructions of the next subsection, and the squared amplitudes that archive produces are the published d_ρ/N^2 [36] — the numerator of (32). The census therefore does not confirm the weights from an independent source: it tests the *assembly* that carries them, which is the reachable-channel list, the resolvent denominators, the operator-level fold, the even split between conjugates, and the geometry that survives each channel separately. That is the statement worth making, and it is a different statement from confirmation.

Two features of the correspondence carry the content. The mixed family enters negated because $t_N = B_N - A_N$, which is the same relative sign the incidence orientation carries. And each like-family *fusion* weight is split evenly between the irrep and its conjugate — the two orientations of the shared link weighted equally — which the delivery resolves on the two-cube space rather than assuming. It is not measured *here*: what this repository checks is the identity the six delivered numbers satisfy.

Conjecture 23 (All-rank even split)**.** For every $N \geq 3$ each like-family fusion weight splits evenly between an irrep and its conjugate in the link-resolved census.

This is a conjecture and nothing here promotes it. Falsifying it means a link-resolved census at some $N \geq 4$ in a truncation that still retains every adjacent shared-link channel; at $N = 4$ the endpoint budgets make that $\mathcal{B} \geq 33/4$, and no such construction has been run.

5.1 One truncation, two constructions

The two-cube space also isolates why a cutoff can reverse the sign of t_3. An adjacent shared-link channel ρ is reachable only if the endpoint Casimir budget $C_2(\rho) + 2C_2(\mathbf{3})$ fits inside the truncation, meaning $C_2(\rho) + 2C_2(\mathbf{3}) \leq \mathcal{B}$: the sextet needs exactly 6 and the adjoint 17/3, so both survive $\mathcal{B} = 6$ and neither survives $\mathcal{B} = 4$.

The weak inequality is not a convenience. Both retentions this section uses are decided by an equality: $\Lambda^2 F = \bar{\mathbf{3}}$ has budget exactly 4 and so sits exactly on the $\mathcal{B} = 4$ cutoff, and the sextet sits exactly on the $\mathcal{B} = 6$ one. Under a strict rule $\mathcal{B} = 4$ would reach $\{\mathbf{1}\}$ alone and $\mathcal{B} = 6$ would lose the sextet, contradicting both delivered censuses — so the deliveries are what fix the convention, which is also why the budget arithmetic is not an independent confirmation of either census.

Corollary 24 (What a link cutoff drops)**.** *Retaining only the singlet and* $\Lambda^2 F$ *routes projects the second-order hopping to*

$$w_{\Lambda^2 F} - w_{\mathbf{1}} = -\tfrac{1}{12},$$

opposite in sign to t_3 *and* 10.2 *times larger in magnitude. What is dropped is*

$$w_{\mathrm{Sym}^2 F} - w_{\mathrm{Adj}} = \tfrac{14}{153},$$

and $-\frac{1}{12} + \frac{14}{153} = \frac{5}{612} = t_3$*. This is arithmetic on* (32) *alone.*

The published $p + q \leq 1$ link cutoff retains exactly those two routes — a per-link restriction, where $\mathcal{B} = 4$ is a total endpoint budget, so the two schemes agree on the retained channels at this order and are not the same rule — so $-1/12$ is its projected adjacent hopping and $14/153$ is the adjacent-hopping correction that completes it — not $5/612$, which would double-count what it already carries. On the two-cube space the first of those two numbers is delivered rather than inferred: a separately sealed $\mathcal{B} = 4$ run on the identical geometry gives $K^{(2),\,\mathcal{B}=4}_{\mathrm{conn}} = -\frac{1}{12} G_{\mathrm{conn}} + D_{\mathcal{B}=4}$ outright, and the Casimir budgets of the previous paragraph say without reference to any census that $\mathcal{B} = 4$ can reach only $\{\mathbf{1}, \mathbf{3}, \bar{\mathbf{3}}\}$. Conditional on (42), the same split of the census reproduces it, $c_{\mathbf{1}} + c_{\mathbf{3}} + c_{\bar{\mathbf{3}}} = -1/12$, and gives the omitted half as $c_{\mathbf{6}} + c_{\bar{\mathbf{6}}} + c_{\mathbf{8}} = 14/153$.

Remark 25. Two single-cube reconstructions bracket the same statement. A full $\mathcal{B} = 4$ open cube assembled from the pinned `ymcirc` $d = 3$, $\mathcal{B} = 4$ plaquette data [37, 38] reproduces $-1/12$ and the truncated shell ordering to 4.3×10^{-11}; a $\mathcal{B} = 6$ cube reproduces $+5/612$ and the inverse ordering to 8.4×10^{-15}. Both locate the missing weight at the same $14/153$.

Neither is an independent replication. Both read tabulated plaquette matrix elements of the same lineage — and the two-cube build draws its Clebsch–Gordan amplitudes from the same hash-pinned `pyclebsch` archive as the $\mathcal{B} = 6$ cube, so that agreement is a cross-check of assembly and not of the amplitudes. The published closed form [36] is d_ρ / N^2 — the very numerator of (32). This is a cross-check of assembly, not a third confirmation of t_3; the exact rational identity is Corollary 24, which needs no tolerance. The $p + q \leq 1$ nomenclature and the finite-rank matrix element are both from [36], and single-cube $SU(N)$ diagonalisation in this style goes back to [26].

The correction to the adjacent-hopping coefficient of a published T_1 Hamiltonian is therefore $14/153$ and not $5/612$, which would double-count the two routes T_1 already carries. One coincidence of value is worth naming: $-1/12$ is also the singlet weight $w_{\mathbf{1}}$, and the cutoff value is a *difference* of two weights, not a weight.

The $\mathcal{B} = 6$ cube is channel-complete for the hopping and *not* for the scalar: its scalar is $39/68$ against the bridge's $11/34$, differing by exactly $1/4$, the local same-face sextet route, which first enters at cutoff $20/3$ — integer label $\mathcal{B} = 7$, so any cutoff above $20/3$ already carries it.

That is a prediction with a cheap test, and the test was executed. A six-face $\mathcal{B} = 7$ reachable-state probe returns $\alpha_{\mathcal{B}=7} = 11/34$ and $t_{\mathcal{B}=7} = 5/612$ with scalar-plus-Gram residual 1.1×10^{-15}, so raising the cutoff past the same-face threshold lowers all three absolute coefficients by exactly $1/4$ and leaves the relative shell $\{0,\ 5/153,\ 5/102\}$ untouched — the scalar moves, the hopping does not. That probe's artifact did not travel, so it is reported here and not checked.

5.2 What the cutoff moves, and where

The connected diagonal D is eleven numbers in both deliveries and is treated as a remainder in both. It is not eleven numbers. The open prism has a symmetry group — the cell exchange $x \mapsto 2-x$, the two transverse reflections and the $y \leftrightarrow z$ swap, of order 16 — and D is constant on its orbits, which are $8+2+1$: the eight side faces, the two end caps, the shared face.

Proposition 26 (Where the truncation lives)**.** *The eight-element orbit is exactly the support of the four cross-cell pairs, and it is the only orbit whose diagonal value differs between the two truncations: the end caps carry* $-15/4$ *and the shared face* 0 *at both* $\mathcal{B}=4$ *and* $\mathcal{B}=6$*, while the side value moves from* $-7/4$ *to* $-2317/612$*. Together with the four cross-cell blocks carrying the whole off-diagonal change, the entire* $\mathcal{B}=4 \to \mathcal{B}=6$ *difference is confined to the faces that carry connected transport.*

This also says what the nonuniform D is. Where the faces of a complex form a *single* orbit — the closed cube of the next subsection, or the periodic torus of Corollary 19 — orbit-constancy forces the diagonal to be scalar, and the operator is exactly $\alpha I + tG$. The prism has three orbits because it has a boundary. D is the open boundary showing up in the diagonal, not a failure of the incidence form.

5.3 The one-cube shell, and why its flat level cannot move

Those two single-cube reconstructions are usually read as numerical cross-checks. They are more than that, because the object they diagonalise is the same chain complex. The six faces of a cube are a closed surface, so in the outward orientation $\ker \partial_2$ is one-dimensional and spanned by the sum of all six — the fundamental class, $b_2(S^2)=1$. That is Theorem 7 on a complex with no three-cells, where the L^3-1 boundaries are absent and only the class survives.

Proposition 27 (One-cube shell)**.** *On the closed cube surface* $G = \partial_2^\top \partial_2$ *commutes with all* 24 *proper rotations. The combinations* $b_{+i} - b_{-i}$ *carry the defining vector representation and the combinations* $b_{+i} + b_{-i}$ *the permutation representation of the three unordered axes, so with the charge-odd inversion* $Pb_{+i} = -b_{-i}$ *the shell is*

$$A_1^{--} \oplus T_1^{+-} \oplus E^{--},$$

with G*-eigenvalues* 0*,* 4*,* 6 *and multiplicities* 1*,* 3*,* 2*. Consequently* $\alpha I + tG$ *has levels* α*,* $\alpha + 4t$*,* $\alpha + 6t$*.*

Proof. The rotations permute the six outward faces, and G is built from the boundary map, so it commutes with the permutation action. On the antisymmetric combinations that action is the defining representation by inspection — $b_{+i} - b_{-i}$ transforms as e_i — and on the symmetric combinations it is the permutation representation of $\{\pm e_i\}$, which is $A_1 \oplus E$. Under $Pb_{+i} = -b_{-i}$ the antisymmetric sector is parity even and the symmetric one parity odd. In these sectors G is $4I$ and $4I - 2(J-I)$ respectively, of spectra $\{4,4,4\}$ and $\{0,6,6\}$; since the three isotypic components have distinct dimensions the eigenvalues attach to them uniquely. □

Three things follow. The cubic labels are derived here rather than assigned: both deliveries assert them and neither builds the symmetry matrices. The A_1^{--} level is Proposition 39's mechanism on a second complex — $Gw = 0$ there, so that level sits at α for every t, and no truncation, no channel and no coefficient in dispute anywhere in this paper can move it. And the reversal the two runs report is therefore a statement about the sign of one number rather than about the shell as a whole: at $\mathcal{B}=6$ the levels are $\{39/68,\ 371/612,\ 127/204\}$ and at $\mathcal{B}=4$ $\{37/12,\ 11/4,\ 31/12\}$, in the opposite order, with A_1^{--} in each case sitting exactly at the scalar. The relative shell $\{0, 4t, 6t\}$ is $\{0,\ 5/153,\ 5/102\}$ at $\mathcal{B} = 6$.

The finite-volume fingerprint of the 28 August finite-N bridge certificate — a different release, and named here because a reader tracing it to the two-cube one would find nothing — is worth recording because its most discriminating multiplicity is not a feature of the truncation: on the periodic 3^3 torus the incidence eigenvalues $0, 3, 6, 9$ have multiplicities $29, 12, 24, 16$, respectively, and $29 = L^3 + 2$ is Theorem 7.

Remark 28 (Scope of the second-order delivery). Everything above in this section is second order in u, finite volume, strong coupling and charge odd. It bears on t_N and decides nothing about the fourth-order coefficient of Section 9. The $\mathcal{B} = 6$ two-cube construction was not re-run here: its arithmetic is audited against geometry rebuilt independently, and its diagonal D is B-truncated and not proved cutoff-stable. The older held-out finite-u validation that rejects the wrong-sign control is a 66-dimensional $P + Q_1$ word space with the Q_1–Q_1 magnetic block set to zero; it is not the radius-two $K_{2,\star}$ operator defined below. The two branches are not equally covered: at $\mathcal{B} = 4$ the delivery reports a direct diagonalisation of the complete 4,180-dimensional charge-odd block at four small couplings, two of them held out, with the residual after $E_* + u + u^2 \operatorname{eig} K_{\mathrm{eff}}^{(2)}$ scaling as u^3; at $\mathcal{B} = 6$ no such diagonalisation exists and the 66-dimensional star stands in for it. That $\mathcal{B} = 4$ report carries no check here — its certificate did not travel — and is recorded as the delivery's rather than as this paper's. No external group has reproduced that release. The later radius-two release is a distinct construction. It neither upgrades the $\mathcal{B} = 6$ cutoff-stability claim nor supplies a radius-three test.

5.4 A detached-replayed radius-two Ritz extension

A later sealed release (29 August 2026) on the same open $(3, 2, 2)$ prism assembles the charge-odd and charge-even $SU(3)$ two-cube Ritz operators with no Casimir cutoff inside its declared frontiers: the electric seed P (eleven charge-odd faces), the complete one-action frontier Q_1, and the exact-energy-factorized two-action frontier Q_2, with cumulative charge-odd dimensions $11, 77, 773$ and charge-even dimensions $1, 2, 9$. The release calls this construction *cutoff-free*: here that means no $\mathcal{B}$ cutoff inside the declared one- and two-action frontiers. The radius-two projection itself remains a truncation of the gauge-theory Hilbert space, and the phrase is used in no wider sense below.

To keep action radius distinct from perturbative order, write the three nested matrices as K_1, $K_{2,\star}$ and $K_{2,\mathrm{full}}$. In $P + Q_1 + Q_2$ block order,

$$H_0 = \operatorname{diag}(E_s I, H_1, H_2), \qquad V = \begin{pmatrix} A_0 & B & 0 \\ B^\dagger & G & C \\ 0 & C^\dagger & D_2 \end{pmatrix}. \tag{43}$$

Here $E_s = 8/3$ in the odd seed and $E_s = 0$ in the even seed. The three finite-coupling operators use $K(u) = H_0 + uV$, with

$$\begin{aligned} K_1(u) &= (H_0 + uV)|_{P+Q_1}, \\ K_{2,\star}(u) &= H_0 + u\big[V \text{ with only } D_2 = Q_2 V Q_2 \text{ set to zero}\big], \\ K_{2,\mathrm{full}}(u) &= (H_0 + uV)|_{P+Q_1+Q_2}. \end{aligned} \tag{44}$$

Thus "full" means complete only within the retained Q_2 block. Every finite-coupling Cauchy interlacing inequality between the three is checked, with recorded violation exactly 0. All quoted quantities are invariant under diagonal basis rephasing, which is why no signed hopping entry is read as an observable there. The delivery is finite-precision throughout; everything in this subsection is reported at stated tolerances, never promoted.

Reported computational result 29 (The rational sub-block of the cutoff-free second-order spectrum)**.** Six of the eleven eigenvalues of the sealed cutoff-free second-order block $C_2 = BR_1B^\dagger$ on the odd seed are rationals to within 10^{-12} (the worst residual is 5×10^{-14}, about 30 ulps):

$$-\tfrac{2429}{306}\ (\times 2), \qquad -\tfrac{404}{51}\ (\times 3), \qquad -\tfrac{2419}{306}\ (\times 1),$$

and the outer levels straddle the central triple by exactly

$$\tfrac{2429}{306} - \tfrac{404}{51} = \tfrac{404}{51} - \tfrac{2419}{306} = \tfrac{5}{306} = 2t_3.$$

The remaining five eigenvalues are not rational at this precision.

The point is what this operator was *not* told. The census of Reported result 22 recovered the shared-link channel coefficients from a build that retained those channels; here the sealed radius-two operator retains every second-order route the two-action frontier supports, makes no shared-link truncation, and the shared-link coefficient $t_3 = 5/612$ of Theorem 18 appears anyway — as the exact splitting scale of a rational sub-block, embedded in a spectrum whose other levels are algebraic. A structural identification of that sub-block (which face combinations carry it, and why the central level is threefold) is not derived here and is recorded as a lead, not a claim; the open two-cube cluster has face-inequivalent environments, so a non-scalar diagonal is expected and no contradiction with Corollary 19, which is a torus statement, arises.

The nested-difference coefficients have a provable onset, which is what makes them readable at all.

Proposition 30 (Onset of the nested-radius differences)**.** *Let $R_i = (E_sI - H_i)^{-1}$. If $PVQ_2 = 0$, the leading seed-space operator present in $K_{2,\star} - K_1$ is*

$$I_4 = BR_1CR_2C^\dagger R_1B^\dagger, \qquad I_5 = BR_1CR_2D_2R_2C^\dagger R_1B^\dagger, \tag{45}$$

where I_5 is the leading term present in $K_{2,\mathrm{full}} - K_{2,\star}$.

Proof. Any contribution that enters Q_2 must follow the shortest closed block walk $P \to Q_1 \to Q_2 \to Q_1 \to P$, which contains four insertions of V. Distinguishing the full operator from the star operator additionally requires one insertion of D_2, hence five insertions. The blocks A_0 and G may dress these walks but cannot shorten them. Evaluating the intermediate resolvents at the seed energy gives (45). □

The proposition is exact conditional on the declared frontier and sealed block structure. The following values are direct floating-point evaluations of its closed products, not exact rationals and not eigenvalue extrapolations.

Reported computational result 31 (Detached radius-two replay)**.** The lowest eigenvalue of the odd second-order seed operator is isolated by 0.08438094122012973; the even seed is one-dimensional. Projection of (45) gives

	$C = -$	$C = +$	odd–even gap
I_4	-25.63495194947556	-24.807728943850304	-0.8272230056252567
I_5	-4.933069330883140	-9.116955631684514	$+4.183886300801373$

The last entry is the leading $K_{2,\text{full}} - K_{2,\star}$ contribution to the retained odd–even gap. It is *not* the complete fifth-order coefficient of either unablated sector Hamiltonian. The weak-u cubic intercept 4.183884987168061 differs by 1.31×10^{-6}; the direct product, not that fit, defines the reported coefficient.

The hash-pinned detached verifier starts from both sealed parity-sector parent operators and recomputes the two coupling grids, all three nested spectra, residuals, interlacing, direct products, rephasing and adaptive path integrals. It returned zero maximum recomputation error and maximum rephasing discrepancy 4.09×10^{-14}; all 24 hostile, algebra, numerical-safety and serialization tests passed. This is an independent downstream replay of the sealed parents, not an independent derivation of their microscopic gauge-basis contractions.

The boundary is quantitative. No odd Q_3 matrix is present, so no matched radius-two error enclosure or convergence claim exists. At $g = 1$ the odd and even ground seed weights have fallen to 0.274 and 0.277, while the even exterior residual is 3.784; the retained eigenvalues are still accurately computed, but their accuracy as approximations to a larger space is uncontrolled there.

One further feature bears on Section 9. The delivery's fifth-order object on the eleven-dimensional seed is recorded as a *matrix*: its traceless part is nonzero, with $\mathrm{tr}(\mathrm{shape}^2) = 0.359$ and eleven recorded shape eigenvalues. It is therefore the first delivered instrument in this corpus whose output is not a Γ-point scalar. That observation is about the instrument class only: an open prism carries no off-axis momentum, so I_4 and I_5 in (45) do not choose a value of C_{shp}, and nothing here weakens Theorem 41.

6 The unsigned-incidence comparison operator

The unsigned incidence $\mathcal{N}(k)$ of Section 4 defines an exact algebraic comparison operator $G_+(k) = \mathcal{N}(k)\mathcal{N}(k)^\dagger$. Its identification with the complete linked charge-even effective Hamiltonian is an additional hypothesis, not a result of the shared-link calculation: Proposition 20 exposes an all-pairs vacuum route whose cancellation or linked-cluster reorganisation has not been derived. With that boundary explicit, write

$$\hat{a}_m = 2 + 2\cos k_m = |1 + e^{ik_m}|^2, \qquad p = \sum_m \hat{a}_m.$$

The hat is not decoration: $\hat{a}_m$ is the *complement* of the $a_m = 4\sin^2(k_m/2)$ used in the charge-odd shape basis (47) below, $\hat{a}_m + a_m = 4$, and that complementarity is the sum rule below.

Theorem 32 (Unsigned-incidence Bloch cubic)**.** *$p + q = 12$ pointwise, and the characteristic polynomial of $\mathcal{N}\mathcal{N}^\dagger$ is*

$$\mu^3 - 2p\,\mu^2 + p^2\mu - 4\hat{a}_1\hat{a}_2\hat{a}_3 \;=\; \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3,$$

with the unsigned comparison-adjacency eigenvalues $\lambda = \mu - 4$. The signed counterpart of the same computation is $\mu(\mu - q)^2$.

The algebraic closed form is complete rather than merely consistent at sampled momenta: the characteristic polynomial of the 81×81 and 192×192 finite plaquette adjacencies factorises coefficient for coefficient over $\mathbb{Z}$ into this cubic over the momentum grid.

Proposition 33 (Range and attainment)**.** *$\lambda \in [-4, 12]$ exactly. The top $\lambda = 12$ is attained only at Γ ($\hat{a}_1 = \hat{a}_2 = \hat{a}_3 = 4$). The bottom $\lambda = -4$ is attained on the whole of the three planes $k_j = \pi$, where the constant term vanishes, and only at R is it a triple root.*

Proof. Write $f(\mu) = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3$. Then $f(0) = -4\hat{a}_1\hat{a}_2\hat{a}_3 \le 0$ and $\mu(\mu - p)^2 < 0$ for $\mu < 0$, so no root lies below 0; and $f'(\mu) = (3\mu - p)(\mu - p) > 0$ for $\mu > p$, with $\hat{a}_1\hat{a}_2\hat{a}_3 \le (p/3)^3$ by AM–GM and

$$f(16) \ge 16(16-p)^2 - \frac{4p^3}{27} = \frac{4}{27}(12-p)(48-p)^2 \ge 0.$$

Hence $\mu \in [0, 16]$ and $\lambda = \mu - 4 \in [-4, 12]$. Equality at the top forces $p = 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 = 64$, which is Γ. Equality at the bottom is $f(0) = 0$, i.e. $\hat{a}_1\hat{a}_2\hat{a}_3 = 0$, i.e. some $k_j = \pi$; there $f(\mu) = \mu(\mu - p)^2$ and the root $\mu = 0$ is simple unless $p = 0$, which is R alone. □

Remark 34. A flat unsigned-comparison branch would need $f(\mu_0) = 0$ identically in k, which the constant term already forbids. Its minimum is attained on three measure-zero planes, against the signed operator's flat band. This contrast is an exact statement about two incidence constructions, not by itself a derivation of two physical charge sectors.

6.1 Conditional charge-even assembly

The shared-link coefficient at $r = 2$ is all-rank and analytic. Turning it into a charge-even band structure assumes the symbol identification stated above; the displayed second-order scalar additionally uses the supplied $SU(3)$ leakage/tower identification. The $r = 3$ values — $t_{3,+} = -6335/249696$ and everything computed from it — come from the same supplied domino ledger as Section 7, and are conditional on it in the same way.

Conditional on that identification, the scalar ledger is not a list of separate facts. Writing λ for the adjacency eigenvalue and $\mathrm{tower}_{r,s}$ for the within-plaquette term in canonical u,

$$E_s(\lambda, r) = \mathrm{tower}_{r,s} + 12\,\mathrm{leak}_{r,s} + \lambda\, t_{r,s}, \tag{46}$$

with $\lambda \in \{-4, 8\}$ in the signed charge-odd sector (carrier and dispersive top), $\lambda \in \{12, -4\}$ in the unsigned comparison, and $\lambda = 0$ returning the on-site diagonals. This reproduces every registered diagonal and band value — nine of them — across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta_\pm(3u/2)$ on its own sector's series. The letter Δ is reused elsewhere with other subscripts — Δ_s for $s \in \{X, M, P, R\}$ are the fourth-order checkpoint differences of Section 9, Δ_C the gap between the two recorded off-axis coefficients, and Δ_L the finite-volume incidence separation of Section 13; only the sector subscript $\pm$ means this series. With that convention, (46) takes no unregistered input. The twelve is a count on this lattice, not a citation: the plaquette graph is 12-regular and two distinct faces share at most one link. The two towers are different objects and both are needed: the charge-odd one contributes $8/3 + u + u^2/2 + 7u^3/32$ and the charge-even one $8/3 - u + (13/20)u^2 + (101/200)u^3$, each of them $4\Delta(3u/2)$ on its own sector's series, so $\mathrm{tower}_{2,-} = 1/2$ against $\mathrm{tower}_{2,+} = 13/20$ and $\mathrm{tower}_{3,-} = 7/32$ against $\mathrm{tower}_{3,+} = 101/200$.

Because the supplied ledger has $\mathrm{leak}_{r,+} = t_{r,+}$ at both computed orders, the conditional charge-even assembly collapses to $\mathrm{tower}_{r,+} + (12 + \lambda)\, t_{r,+}$, a one-parameter family in the charge-even tower. That equality is verified and unexplained: at third order the charge-odd leakage separates from the charge-odd hopping while the charge-even identity survives. It is recorded as a unifying candidate with a falsifier — an order at which the two are computed independently and differ — not as a mechanism.

6.2 Where the rest frame is not blind

Conditional corollary 35 (Charge-even Γ splitting)**.** If the unsigned symbol is the complete linked charge-even symbol, then at Γ its comparison spectrum is $\{12, 0, 0\}$, so the A_1^{++} level and the E^{++} doublet are distinct, and

$$E(A_1^{++}) - E(E^{++}) = 12\, t_{r,+}$$

exactly: $-22/51$ at $r = 2$ and $-6335/20808$ at $r = 3$.

This is the structural counterpart of the charge-odd blindness recorded in Section 7. The signed spectrum at Γ is $\{-4, -4, -4\}$, one level, so no charge-odd rest-frame series can see the hopping; the unsigned comparison spectrum has two levels, so the charge-even rest frame could fix the coefficient under the symbol hypothesis. The A_1^{++} branch is isotropic at leading order without assuming it, with curvature $\frac{4}{3}|t_{r,+}|$ in all three directions — 22/459 at second order and 6335/187272 at third.

On a finite L^3 grid the conditional charge-even width is

$$W_{r,L}^{(+)} = |t_{r,+}|\,[12 - \lambda_{\min}^{(+)}(L)], \qquad \lambda_{\min}^{(+)}(L) = -4 \iff L \text{ is even.}$$

Thus it equals $16|t_{r,+}|$ for even L and approaches that coefficientwise as $L \to \infty$; the charge-odd manifold width is $12|t_{r,-}|$ for even L. At any order at which the correction stays in the incidence algebra — which Proposition 39 warns is not automatic — the second-order ratio is $4(3N^2 - 5)/[3(N^2 - 4)]$, finite and greater than one at every rank $N \geq 3$, decreasing from 88/15 at $N = 3$ towards 4. So at second order the charge-even band is the wider one at every rank under the charge-even symbol hypothesis, and that asymmetry is a channel statement rather than a geometric one.

7 $SU(3)$ through third order

The *third-order* content of this section is an externally verified computational input: the frozen upstream corpus reports a 251/251 cold exact replay for the retained chain, while this compact release checks only the assembled rationals. At second order the hopping coefficient t_3 is analytic, but the displayed flat scalar additionally uses the upstream $SU(3)$ identification $\mathrm{leak}_2 = \ell_3$ and its within-plaquette tower. That tower contributes $8/3 + u + u^2/2 + 7u^3/32$.

Reported computational result 36 (Third-order ledger)**.** The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \qquad \mathrm{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12\,\mathrm{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

Conditional corollary 37 (Third-order factorisation)**.** If the lifter census is exhaustive and the linked ledger complete, then

$$H_{\mathrm{eff},-}(k,u) = E_{\mathrm{flat}}(u)I + \tau(u)B(k)B(k)^\dagger + O(u^4),$$

$$E_{\mathrm{flat}}(u) = \tfrac{8}{3} + u + \tfrac{11}{306}u^2 - \tfrac{109151}{249696}u^3,$$
$$\tau(u) = \tfrac{5}{612}u^2 + \tfrac{1975}{124848}u^3.$$

The second-order diagonal is the $\lambda = 0$ value $\frac{1}{2} + 12\,\mathrm{leak}_2 = 7/102$ with $\mathrm{leak}_2 = -11/306$; subtracting $4t_3$ gives the flat scalar $11/306$ at $\lambda = -4$.

Remark 38 (What the external comparison can and cannot do). Hamer's 1989 rest-frame series [23] agrees with E_{flat} at every shared order. It cannot test τ. The hopping enters the spectrum only through $\tau(u)q(k)$ and $q(0) = 0$, so every *charge-odd* Γ-point datum is blind to it. Under the unsigned-symbol hypothesis the charge-even rest frame is not, by Conditional result 35; the one-parameter family $(b_3 + \varepsilon,\ \text{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\,\text{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (46). Conditional result 35 is the charge-*even* analogue and separates leak_+ from t_+; it says nothing about b_3 or leak_3.

8 What the homology protects

Proposition 39 (Boundary-factorised rigidity)**.** *If an order-r Bloch correction has the form $H_r(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $H_r(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$, both from Theorem 6. The middle term is annihilated whatever M_r is, which is the content: the protection belongs to the boundary operator and not to the dynamics. □

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\operatorname{End}(C_2)$, and a general higher-order correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \tag{47}$$

with $e_2 = \sum_{i<j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4\sin^2(k_i/2)$. The $1/q$ is the carrier projection: $\|w\|^2 = q$ by Theorem 6, so the carrier-projected correction is $\varepsilon_4 = (w^\dagger H_4 w)/q$ and the numerator, not ε_4, is the polynomial.

9 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient C_{shp}. This edition does not adjudicate them, and no fourth-order band or bandwidth is used above. What it does is state precisely where the disagreement lives, because that geography is now checked and it is what a decisive calculation must target.

After subtracting the rest value, the recorded carrier ansatz is

$$\Delta_4(k) := \varepsilon_4(k) - \varepsilon_4(0) = Aq + Be_2 + C_{\text{shp}}\frac{4e_2}{q} + D\frac{e_3}{q}. \tag{48}$$

The continuous $k \to 0$ limit is used at $q = 0$. Denote the two recorded mixed-gradient values by C_{old} and C_{new} and set $\Delta_C = C_{\text{new}} - C_{\text{old}}$. A supplied all-rank cube ledger also reports

$$\alpha_N = \frac{640}{N(N^2-1)^3}, \tag{49}$$

but this formula and its identification with the complete axial coefficient remain conditional on that ledger. Sealed packages containing other fourth-order scalars use incompatible observable labels; none is promoted here to a canonical rest coefficient.

9.1 An exact theorem for the saved historical kernel

The upstream authority corpus supplies the hash-pinned 189-record historical kernel. Pulling its centered fourth-order operator back to cube amplitudes with $\mathsf{C} = \partial_3$, define

$$Q_4 = \mathsf{C}^\dagger H_4 \mathsf{C}, \qquad G = \mathsf{C}^\dagger \mathsf{C}, \qquad \mathcal{Q}_4 = Q_4 - s_4 G,$$

where $s_4 = \operatorname{tr} H_4(\Gamma)/3$. If $L_i = 2I - T_i - T_i^{-1}$, then $G = \sum_i L_i$ and the centered coefficient is the generalized eigenvalue $\mathcal{Q}_4\phi = \lambda_4 G\phi$. The representative change $(Q_4, G) \sim (Q_4 + \delta G, G)$ is a scalar gauge only when the scalar anchor changes simultaneously; it cannot change an off-axis shape.

Reported computational result 40 (Exact saved-kernel Hodge pencil)**.** For the supplied historical kernel,

$$q^{(4)}_{\mathrm{old}} = -\frac{20721577909065127111}{7250590288602460800}, \qquad \alpha_{\mathrm{old}} = \frac{5}{12}, \qquad \beta_{\mathrm{old}} = \frac{17607806155349}{275331901291200},$$

and

$$\mathcal{Q}_{4,\mathrm{old}} = \frac{5}{48}\sum_i L_i^2 + \frac{17607806155349}{1101327605164800}\sum_{i<j} L_i L_j \succeq 0. \tag{50}$$

Its continuous-zone and even-L width is

$$W_{4,\mathrm{old}} = \alpha_{\mathrm{old}} + \beta_{\mathrm{old}} = \frac{132329431693349}{275331901291200}.$$

These identities are exact for the saved kernel and its cold fixed-kernel checks. They do not establish that this kernel is the unique physical linked fourth-order operator.

The corresponding 25-point numerator stencil has

$$w_0 = \tfrac{9}{2}\alpha + 3\beta, \qquad w_1 = -(\alpha+\beta), \qquad w_2 = \tfrac{1}{4}\alpha, \qquad w_d = \tfrac{1}{4}\beta,$$

and obeys $w_0 + 6w_1 + 6w_2 + 12w_d = 0$ exactly. For $\alpha, \beta > 0$, the generalized dispersion is bounded by $0 \le \lambda_4 \le \alpha + \beta$, with the continuous-zone extrema at Γ and R respectively; an odd-L grid does not contain R.

The two records do agree on the axial datum, but the corpus's own quoted tolerance does not cover that agreement: the α gap is 2.43×10^{-13} against a stated bound of 2.3×10^{-13}. The sealed core holds internally — α_{new} and $4A_{\mathrm{new}}$ agree to 2×10^{-15} — which is consistent with the quoted bound being too tight, but two float comparisons do not discriminate that from a shared upstream slip. Recorded as a discrepancy rather than resolved, and the bound is not widened.

9.2 The obstruction is polynomial

Theorem 41 (Obstruction certificate)**.** *The two recorded off-axis coefficients enter* (47) *only through the shape $4e_2/q$, so the two records' carrier numerators differ by $4\Delta_C e_2$ and their bands by $\Delta_C\,(4e_2/q)$. On any one-dimensional axial cut two of the three a_i vanish, so e_2 is the* zero polynomial *there and both differences vanish identically. Consequently no Γ-point or axial datum distinguishes the two records, at any precision. Off axis they do differ: $e_2(M) = 16$ and $e_2(R) = 48$, so the numerators differ by $64\Delta_C$ and $192\Delta_C$ and the bands by $8\Delta_C$ and $16\Delta_C$.*

Remark 42. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

On the axial cut itself the invariants collapse to $q = q_{\mathrm{ax}}(k) = 4\sin^2(k/2)$ with $e_2 = e_3 = 0$. Dividing the raw cube-boundary numerator $(\alpha_N/4)q_{\mathrm{ax}}^2$ by $\|w\|^2 = q = q_{\mathrm{ax}}$ leaves one power of q_{ax} with coefficient $\alpha_3/4 = 5/48$.

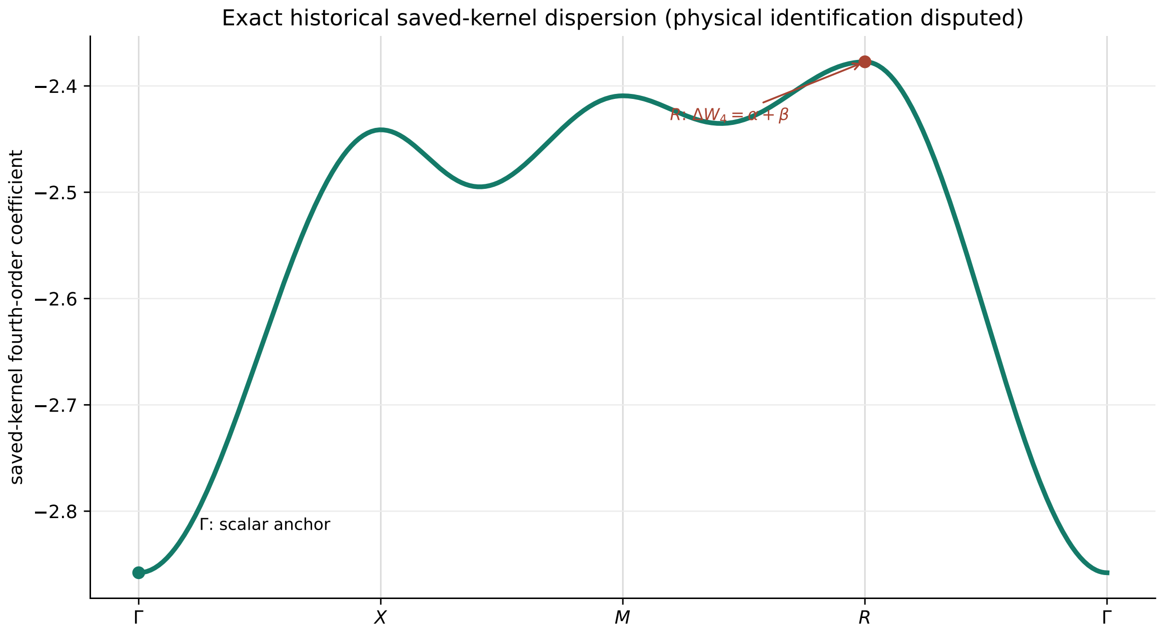

Figure 3: Exact dispersion of the saved historical kernel in Reported result 40. The plot is a theorem about that hash-pinned kernel. Its identification with the complete physical linked $SU(3)$ fourth-order operator is disputed, so the curve is not used as a settled physical band.

9.3 A second witness

Non-identifiability is better exhibited than argued from a difference between two records, since a difference could always be a computational error in one of them. The witness below is not itself a candidate for the physical coefficient. It is the balanced continuation obtained in the supplied source by setting an auxiliary bookkeeping parameter ϵ to zero; that source label carries no physical meaning here. Its sole role is to make the $4\Delta_C e_2$ family concrete with an exactly known shift.

Proposition 43 (Explicit second witness)**.** *There is a value C_{alt} with $C_{\text{alt}} - C_{\text{old}} = 25/1024$ exactly, appearing verbatim in the pinned source edition, whose induced band separations are 0 at X, $25/128$ at M and $25/64$ at R. It is identical to C_{old} on every retained datum and distinct from it off-axis.*

9.4 Where the coefficient lives

The four checkpoint deltas are the Γ-subtracted carrier corrections $\Delta_s = \varepsilon_4(k_s) - \varepsilon_4(0)$ at $k_X = (\pi, 0, 0)$, $k_M = (\pi, \pi, 0)$, $k_P = (\pi, \pi/2, 0)$ and $k_R = (\pi, \pi, \pi)$ — P is the off-axis, off-corner point that makes the system invertible, with $a = (4, 2, 0)$. Its algebraic role is explicit here; no additional provenance is inferred for that checkpoint. In the shape basis they are

$$\begin{aligned}
\Delta_X &= 4A, \\
\Delta_M &= 8A + 16B + 8C_{\text{shp}}, \\
\Delta_P &= 6A + 8B + \tfrac{16}{3}C_{\text{shp}}, \\
\Delta_R &= 12A + 48B + 16C_{\text{shp}} + \tfrac{16}{3}D,
\end{aligned} \tag{51}$$

and they invert exactly, so the four shape coefficients are recoverable from four checkpoints. Δ_X contains A alone, which is why the axial cuts agree whatever the dispute does.

Four further measurements locate the coefficient and bound where a disagreement could live. The distinction matters and is easy to lose: showing where a coefficient's *value* is concentrated is not by itself showing where a *discrepancy* can sit. Only the third item below does the second thing.

1. **The shape fit's C row sums to zero**, so no error in the Γ anchoring can move C_{shp} at all; the recorded run's own re-anchoring moved it by 4.6×10^{-15}, as the zero row sum requires. A scalar shift is translation-local and changes nothing observable, which is why the two anchored scalars $q^{(4)}_{\text{band}}$ and $m^{(4)}_{\Gamma}$ are differently anchored coordinates and not rival estimates.

2. **The coefficient's value is concentrated, not spread.** Six of the 189 kernel records — the normal $(0, 0, 1)$ shell — carry 81% of the total weight, while the 120 rotation records carry 6% and are 255 times less sensitive per record. This says where the coefficient lives, not yet where the two records can differ.

3. **The agreed axial coefficient pins the normal sector.** All six normal records share one amplitude h, and it sets both shapes rigidly: $A_{\text{normal}} = -h$ while the raw normal amplitude is $2h$, which in the shape normalisation of (47) is $C_{\text{normal}} = h/2$, so

   $$C_{\text{normal}} = -\tfrac{1}{2} A_{\text{normal}} \quad \text{exactly.}$$

   (The factor of four between $2h$ and $h/2$ is the raw-to-shape conversion, not a discrepancy.) Granting $A = 5/48$ — which both records share — together with the block decomposition, which puts $5/48 + 4x$ on the normal block and $-4x$ on the mixed one, therefore fixes h and with it 81% of the coefficient; and everything free to move without disturbing A totals less than half the disputed gap. A rival kernel must differ *structurally* in the A-carrying sector; no redistribution reaches the gap.

4. **The two kernels agree almost everywhere.** Their supports are identical and 144 of 189 records ride one common scale; the divergent cross sector is a single uniform ratio. The dispute is three amplitudes.

Read together these four sharpen the target and prefer no side of it. They say that a rival kernel must differ structurally in the A-carrying sector rather than by redistribution. That is a constraint on candidates built on the recorded 189-record support — which is where the measurement was made; a kernel on a different support is not constrained by it at all. Neither record is promoted here and neither is withdrawn: the interval of *values* between them is not narrowed. What is narrowed is the class of *kernels* that can reach either end, and that constraint bears on both records alike.

9.5 What cannot decide it

A large sealed sweep over 609 marked clusters has been the standing candidate for settling this. It cannot: its engine assembles coefficients through a Γ-point scalar and emits no kernel block at all, and by Theorem 41 Γ-point data are structurally incapable of constraining C_{shp}. This is a statement about the instrument, made by reading it, not a report of a failed run.

The instrument class that could eventually decide it now demonstrably exists: the sealed radius-two delivery of Section 5.4 emits a seed-space *matrix* at fifth order, not a scalar. What it lacks is off-axis momentum — a two-cube cluster has none — so it too cannot adjudicate. The requirement is now sharp on both counts: a decisive instrument must emit more than a Γ scalar *and* resolve an off-axis point such as P; the marked-cluster engine fails the first condition, the radius-two delivery the second, and a target-blind calculation satisfying both is exactly the next step named in the conclusion.

9.6 The kernel is six amplitudes

Every non-axial number in the dispute has been a float, and for one reason: the shape coefficients were obtained by fitting ε_4 at four points of the zone, with condition number 45.8. That is avoidable. Clearing the $1/q$ turns the ansatz into a linear identity between Laurent polynomials in $z_j = e^{ik_j}$,

$$T(k) \;=\; \psi^\dagger S(k)\psi \;=\; c_0 e_1 + Ae_1^2 + B\,e_1e_2 + 4C\,e_2 + D\,e_3,$$

with $s_j = 4\sin^2(k_j/2) = 2 - z_j - z_j^{-1}$ and e_r the elementary symmetric functions of the s_j. Both sides are polynomials with rational coefficients, so the system either has an exact solution valid on the whole zone or none at all. For the saved historical kernel it has one, with identically vanishing residual, and it returns $A = 5/48$, $B = D = 0$ and the registered $C_{\rm shp}$ as exact rationals.

The decomposition makes a structure visible that the fit could not. The 189 records carry only *six* distinct weight magnitudes, and each magnitude is one orbit of the cubic symmetry: a 132-record skeleton at amplitude u; a 12-record same-plane $(0,1,1)$ in-plane orbit at u_2; a 24-record cross-plane orbit at ρ; a 12-record same-plane nearest-neighbour in-plane orbit at π; a 6-record normal orbit at ν; and the 3 on-site records at σ. Each orbit's amplitude, read off its own shape contribution, equals the weight its records carry up to sign — which is what makes it an amplitude rather than a cluster.

Five of the six carrier projections then follow from the orbit's displacement set in two lines. Writing n for a plane's normal and $\{i,j\}$ for its own axes, and using $|\psi_P|^2 = s_{n(P)}$,

$$\begin{aligned}
\text{on-site}: &\quad S_P = \sigma && \longrightarrow\ \sigma\, e_1,\\
\text{normal}: &\quad S_P = \nu\,(2 - s_n) && \longrightarrow\ \nu\,(2e_1 - e_1^2 + 2e_2),\\
\text{in-plane}: &\quad S_P = \pi\big((2 - s_i) + (2 - s_j)\big) && \longrightarrow\ \pi\,(4e_1 - 2e_2),\\
\text{doubled}: &\quad S_P = u_2(2 - s_i)(2 - s_j) && \longrightarrow\ u_2(4e_1 - 4e_2 + 3e_3),\\
\text{rotation}: &\quad S_{PQ} = -\epsilon_P\epsilon_Q\,\rho\,\bar d_{n(Q)}\overline{\bar d_{n(P)}} && \longrightarrow\ -2\rho\, e_2,
\end{aligned}$$

the last because the carrier annihilates the phases outright: $T = -\rho\sum_{m\neq n} s_m s_n$. The skeleton's displacement set is larger and its row is solved rather than read off, giving $u\,(12e_1 - 4e_1^2 + 12e_2 - 6e_3)$, again with zero residual. Each of the six is an exact Laurent identity on that orbit's own records.

Collecting the six gives the whole shape table in closed form:

$$A = -\nu - 4u, \qquad B = 0, \qquad D = 3u_2 - 6u, \qquad 4C = 2\nu - 2\pi - 4u_2 - 2\rho + 12u.$$

Two consequences are immediate. Only two orbits carry A, so the agreed axial value fixes the normal amplitude outright, $\nu = -(5/48 + 4u)$: the normal channel is not independently disputed, it is a consequence of A. And substituting that together with $u_2 = 2u$ collapses the fourth-order off-axis coefficient to three signed numbers,

$$\boxed{C_{\rm shp} \;=\; -\tfrac{5}{96} \;-\; u \;-\; \tfrac{1}{2}(\rho + \pi)}$$

which reproduces the registered value exactly. The seven-row displacement-block ledger is a presentation of three amplitudes.

9.7 The tier collapse, in the orbit basis

The two shape coefficients B and D vanish in the recorded kernel even though the two-hop enumeration generates both, and the degree count does *not* forbid them. The carrier numerator reaches a-degree three, because the projection contributes one further power of a: for any vector whose components have squared modulus a_i — the carrier is one, in whichever of the two equivalent labellings — the quadratic form $\mathrm{diag}(a_i^2)$ evaluates to $\sum_i a_i^3 = q^3 - 3qe_2 + 3e_3$, an explicit witness carrying a nonzero e_3. An earlier degree-bound mechanism proposed for the collapse was therefore retracted.

At the level of displacement shells the vanishing looks like two integer cancellations: every degree-three block amplitude is an integer multiple of one raw record weight x, and $B = 0$ reads $(+1, +1, -2) \cdot x$ while $D = 0$ reads $(-3, +6, -3) \cdot x$. That grouping is the wrong one, and the orbit basis shows why. B is the coefficient of $e_1 e_2$, and *no orbit's closed form contains that monomial*. There is nothing to cancel: B is unpopulated, and needs no mechanism at all. Inside the skeleton the four B-carrying displacement blocks do cancel in $\pm$ pairs, same-plane against off-plane at the same shell and at unit weight, which is the $(+1, +1, -2)$ above with the two cross-plane blocks summed — but that is an artefact of slicing one amplitude by displacement.

D is carried by exactly two orbits, the skeleton at $-6u$ and the doubled orbit at $+3u_2$, and every other orbit has $D = 0$. So the entire e_3 collapse is the single equality

$$u_2 = 2u,$$

after which $D = -6u + 3u_2$ vanishes for *any* value of u. It is an identity in one amplitude, not a numerical coincidence between sectors, and it holds in the independently computed cold kernel too, with that kernel's own u. What a mechanism has to explain is no longer two weight vectors: it is why the same-plane $(0, 1, 1)$ in-plane orbit carries exactly twice the generic weight.

That, in turn, is not an independent coincidence. Fix a plane with own axes $\{i, j\}$ and normal n, and write $X = 2\cos k_i$, $Y = 2\cos k_j$, $Z = 2\cos k_n$. Everything the plane sends to itself beyond nearest neighbour is sixteen records: the $(0, 0, 2)$ shell at amplitude u, contributing $(X^2-2)+(Y^2-2)$; the mixed $(0, 1, 1)$ records at u, contributing $(X + Y)Z$; and the in-plane $(0, 1, 1)$ records at u_2, contributing $(u_2/u)XY$. Their sum is

$$u\big[(X + Y)(X + Y + Z) - 4\big]$$

if and only if the cross term is $2XY$ — that is, if and only if $u_2 = 2u$. The doubling is the condition for the same-plane long-range symbol to be a product rather than an inhomogeneous quadratic. With the product form, $D = 0$ holds separately in the same-plane sector and in the 96-record cross-plane skeleton, so neither vanishing is a cancellation between sectors at all — the cross-plane one because it is $-2u\, e_2(e_1 - 8)$ exactly, pure e_2 times a linear factor, which is also why it carries no A. What remains of the question is one level up: why the fourth-order dynamics produces a product there.

9.8 Both branches in one basis

Two things follow for the dispute itself, and neither promotes a side.

First, a caution about the ϵ-free branch, recorded here because the present authors first got it wrong. Evaluating the pinned structured B_N expression at $N = 3$ and comparing with β recomputed from the raw 189 records gives a difference of exactly $25/64$, and it is tempting to read $C_{\text{hist}} + 25/1024$ as an exact balanced value of C_{shp}. That reading is wrong. At $N = 3$ the structured expression returns exactly $P_{17}(9)/(3R_{20}(9))$, and the source formula is stated only for $N \geq 4$ with the explicit caution that it is not to be substituted at $N = 3$; the continuation route to that rank is separately closed, D_{34} carrying $(z - 9)^3$ with $Q_{32}(9) \neq 0$. The relation is real — it is what $\Delta\beta_3$ records — but no direct balanced contraction at $N = 3$ is held here, so it is a relation among recorded quantities and not an established branch.

What the exceptional ranks do show is a pattern about the determinant sector, and it is checked rather than asserted. At $SU(5)$ the shipped stage-1 scan holds 895,524 candidate support/output pairs with *zero* determinant sectors. At $SU(6)$ the sole determinant orbit shifts the scalar by exactly $6/343$ at q, X, M and R while A, B and the bandwidth are unchanged. At $SU(4)$ the recorded offsets are $\Delta A_4 = \Delta B_4 = 0$. So at every exceptional rank where the determinant sector has been computed independently it does not reach B, and $N = 3$ would be the one rank where it does. That is a pattern worth explaining. It is not a licence for the substitution.

The orbit basis makes that pattern sharper than a list, because it gives "momentum-independent" an exact meaning. The on-site orbit's carrier projection is exactly σe_1, so $\varepsilon = T/q = \sigma$ is constant over the zone and carries no A, B, C or D; conversely a correction recorded as a pure scalar shift *is* an on-site-orbit shift and nothing else. Read that way the exceptional record is uniform — at $SU(6)$ the determinant sector shifts the on-site orbit and stops there, at $SU(5)$ there is no such sector to shift, at $SU(4)$ the offsets vanish — and $N = 3$ is the single rank at which it leaves that orbit.

Where it goes instead is then forced, and the consequence is exact. Both sides of the dispute agree $\alpha = 5/12$, so $\Delta A_3 = 0$; with $A = -\nu - 4u$ that gives $\Delta\nu = -4\Delta u$. The primitive cube-completion channel is link-balanced — one U and one $U^\dagger$ on every link, so $\nu_\ell = 0$ — hence ϵ-blind, and that channel is the normal orbit, so $\Delta\nu = 0$ and therefore $\Delta u = 0$; and $\Delta u_2 = 0$ because $u_2 = 2u$ holds exactly in both recorded kernels. Since $C_{\rm shp} = -\frac{5}{96} - u - \frac{1}{2}(\rho + \pi)$, the sector has exactly one place left to act. With $\Delta\beta_3 = 25/64$,

$$\Delta(\rho + \pi) \;=\; -\tfrac{25}{512}, \qquad \Delta u = \Delta u_2 = \Delta\nu = \Delta\sigma = 0.$$

A direct balanced contraction at $N = 3$ — which this repository does not hold — must therefore return $\rho + \pi$ shifted from the historical $-12778990891087/1595026186790400$ by exactly that amount. The falsifier is correspondingly sharp: any such contraction differing from the historical amplitudes anywhere but in $\rho + \pi$, or shifting $\rho + \pi$ by anything else, refutes the reading. It prefers neither side of the fourth-order dispute, and it does not use the forbidden substitution — only $\Delta\beta_3$, which is recorded independently.

Second, the two rival kernels are the same computation. The cold $v10a.26$ record dump has the same six orbit sizes, satisfies $u_2 = 2u$ and $\nu = -(5/48 + 4u)$ to one part in 10^{13}, and gives the same normalised shape table row for row. Its skeleton unit is 4.1327437 times the historical one, and its ρ and π are *opposite in sign.* So the free disagreement is two sign-flipped amplitudes and one scale — not a re-weighting, and not a diffuse discrepancy. Six of the eighteen same-plane nearest-neighbour records are the normal orbit, which $A = 5/48$ pins identically on both sides; neither kernel is self-refuting on its own A, so the comparison localises the disagreement without deciding it.

9.9 A near-Γ statement that does not adjudicate

One quantitative fourth-order statement can be made without adjudicating anything, because it depends on the kernel only through $\sqrt{W_4}$. Here $W_4 = \sup_k(\varepsilon_4(k) - \varepsilon_4(0))$ is the zone spread of the carrier-projected fourth-order correction about Γ, and $\theta \in (0, 1)$ is the fraction of the second-order isolation gap the fourth-order spread is allowed to occupy; we take $\theta = 1/2$ throughout.

What the criterion needs is a lower bound on q over the region $|k| \ge r$, and there it is sharp: writing $q = \sum_m h(k_m^2)$ with $h(t) = 2(1 - \cos\sqrt{t})$, the identity $(s\cos s - \sin s)' = -s \sin s \le 0$ makes h concave and increasing on $[0, \pi^2]$, so the minimum over the simplex sits at a vertex and

$$\min_{|k|\ge r} q(k) = 4\sin^2(r/2), \tag{52}$$

attained on a coordinate axis. With $\tau(u) \geq t_3 u^2$ for $u > 0$, the fourth-order shape terms are therefore dominated by the second-order dispersion outside a ball of radius $2\arcsin\bigl(u\sqrt{W_4/(\theta t_3)}\bigr)$, which is Ku to leading order with $K = \sqrt{W_4/(\theta t_3)}$: the two records give $K = 10.85$ and $K = 15.06$ — floating point, since one of the two bandwidths exists only as a float — and the larger bounds both, so the exclusion radius can be stated for either record without choosing between them.

Jordan's inequality $q(k) \geq (4/\pi^2)|k|^2$ also holds on the whole zone and is tight at $k = 0$ and at the corner, but the corner is not where this bound is used; on the axis it is loose by $\pi^2/4$, and the constants it gives, 17.04 and 23.66, are exactly $\pi/2$ times the sharp ones.

It is not a bound on the open coefficient: a value of C_{shp} outside the two records carries its own W_4, and nothing here confines the answer to the interval the records span. The statement is non-vacuous only for $u < 0.133$ at $\theta = 1/2$, and that threshold is the same for both constants — the sharp radius reaches π when $u\sqrt{W_4/(\theta t_3)} = 1$, which is exactly where $Ku = \pi$ for the Jordan constant, because the two bounds agree at the zone corner. The excluded zone fraction grows as u^3.

10 Infinite volume at fixed lattice spacing

Theorem 3 is a finite-volume statement: it transports the complete three-component shell through a contour whose constants are uniform in L, but it says nothing about the thermodynamic limit, and its closing remark is explicit that it supplies an interacting Riesz cluster rather than a particle. This section closes that step at fixed spacing. The thermodynamic ground state and infinite-volume Hamiltonian come from the quantum-perturbation theorems of [1] applied to the same three-link cell decomposition as Proposition 2; the close-projection and GNS-transport chain then carries the coefficient shell into the physical sector, and the second-order coefficient of the exact orthonormal symbol isolates a rank-one internal sheet on every momentum ball compactly separated from Γ.

Two boundaries survive and are stated where they arise. No rank-one separation is claimed *at* Γ, where the cubic T_1 triplet meets; and the identification of the resulting semigroup with an isotropic Wilson-action transfer matrix, together with every continuum statement, is a separate problem taken up in Section 11.

10.1 Thermodynamic carrier reduction at fixed lattice spacing

The finite-volume Riesz island first has a thermodynamic consequence that is strictly weaker than a particle theorem but stronger than a finite-volume rank statement. We isolate that projection step before proving the stronger band, source-frame and patchwise carrier theorems below; no survival of a finite-volume eigenvalue is silently assumed.

Let $\mathcal{A}_{\mathrm{loc}}$ be the local observable algebra of the unconstrained link system on $\mathbb{Z}^3$. Write $(\mathcal{H}_u, \pi_u, \Omega_u)$ for the GNS triple of its interacting ground state, and let $H_\infty(u)$ be the GNS Hamiltonian normalized by $H_\infty(u)\Omega_u = 0$. Put

$$E_F = \frac{8}{3}, \qquad I_F(u) = \left[\frac{8}{3}(1-\lambda), \frac{8}{3}(1+\lambda)\right], \qquad \lambda = 27c_2|u|.$$

Proposition 44 (Infinite-volume external Riesz projection). *Assume*

$$27|u| < c_1, \qquad \lambda < \frac{1}{33}.$$

Then Theorems 2 and 3 of [1], applied to the same three-link cell decomposition as Proposition 2, give a translation-invariant thermodynamic ground state and a self-adjoint $H_\infty(u)$ satisfying

$$\operatorname{spec} H_\infty(u) \subset \bigcup_{a \in \operatorname{spec} H^{\mathrm{kin}}_{E,\infty}} [a(1-\lambda), a(1+\lambda)]. \tag{53}$$

Consequently the contour Γ_F in (16) lies in the resolvent set of $H_\infty(u)$ and defines

$$P_{F,\infty}(u) = -\frac{1}{2\pi i} \oint_{\Gamma_F} (H_\infty(u) - z)^{-1}\, dz. \tag{54}$$

The target interval is externally separated from the rest of the spectrum by

$$\delta_-(u) \geq \frac{1-15\lambda}{3}, \qquad \delta_+(u) \geq \frac{1-33\lambda}{6}.$$

For local $A, B \in \mathcal{A}_{\mathrm{loc}}$, along Yarotsky's open-box thermodynamic sequence (or a symmetry-preserving boundary sequence for which the same common enclosure and weak-resolvent limit have separately been proved),

$$\lim_{L\to\infty} \langle A\Omega_L, P_{F,L}(u) B\Omega_L \rangle = \langle \pi_u(A)\Omega_u, P_{F,\infty}(u)\pi_u(B)\Omega_u \rangle. \tag{55}$$

The local gauge-invariant, charge-odd subspace is reducing, so all of these statements remain true after restriction to the physical charge-odd local GNS sector.

Proof. The rescaling and local norm estimate in Proposition 2 verify the hypotheses of [1, Theorems 2 and 3]. Those theorems give the weak-star ground-state limit, the weak-resolvent limit, and the same multiplicative spectral enclosure with the finite-volume free spectrum replaced by its finite-support additive semigroup. The Casimir arithmetic in the proof of Theorem 3 is unchanged for that semigroup: the nearest free energies to $16/6$ are $14/6$ and $17/6$. This proves the displayed gaps and places Γ_F in a common resolvent set.

Yarotsky's weak-resolvent convergence is initially stated for $z \in \mathbb{C} \setminus \mathbb{R}$. Choose one connected tubular neighbourhood of Γ_F disjoint from every finite- and infinite-volume spectral enclosure. Matrix elements of the resolvents are uniformly bounded there by the common spectral clearance. The normal-family theorem and the identity theorem therefore extend their convergence to the contour. Integrating the matrix elements around Γ_F proves (55).

Invariance of the thermodynamic state under finite-support gauge transformations and charge conjugation gives their canonical GNS implementing unitaries. The corresponding automorphisms commute with the dynamics, so the implementers commute with $H_\infty(u)$. Define the physical odd sector to be the closed cyclic subspace generated by local gauge-invariant, charge-odd observables. It is reducing for the Hamiltonian, its resolvent and the contour projection, which proves the last assertion. $\square$

Remark 45 (What the enclosure alone does not say)*.* The enclosure is one-sided: it says where spectrum may occur, not that every free patch survives. Thus Proposition 44 by itself does not prove $P_{F,\infty}(u) \neq 0$. Nor does the finite-volume rank $3L^3$ imply a nonzero matrix element against any fixed local source. The next two results supply the missing close-projection, totality and source estimates; they are not consequences of the enclosure by itself.

10.1.1 Coefficient close projection and transport to the physical GNS sector

We use the three-outgoing-link cell decomposition and the gap-one normalization of Proposition 2. If $J \Subset \mathbb{Z}^3$ is a nonempty exact cell support, write $\mathcal{H}'_J$ for the corresponding excited tensor factor and $H_{J,0}$ for the restriction of the free cell Hamiltonian. Thus

$$H_{J,0} \geq |J|\, 1_{\mathcal{H}'_J}. \tag{56}$$

The superscript minus below means both local gauge invariance and odd charge conjugation parity. Define

$$\mathcal{X}_0^- = \left\{ v = (v_J) : v_J \in \mathcal{H}'_J,\ \|v\|_0 := \sum_J \|v_J\| < \infty \right\}^-, \tag{57}$$

$$\mathcal{X}_1^- = \left\{ v \in \mathcal{X}_0^- : v_J \in \operatorname{Dom} H_{J,0},\ \|v\|_1 := \sum_J \|H_{J,0} v_J\| < \infty \right\}. \tag{58}$$

The empty coefficient is absent in the odd sector. Consequently (56) makes $\|\cdot\|_1$ a norm, and the finite coefficient families with entries in $\operatorname{Dom} H_{J,0}$ form a core, denoted $\rfloor_{00}^-$, for

$$D(v_J)_J = (H_{J,0} v_J)_J, \qquad \operatorname{Dom} D = \mathcal{X}_1^-. \tag{59}$$

Lemma 46 (Exact-support symmetry)**.** *The exact-support projections commute with every local gauge transformation and with charge conjugation. Hence D, Yarotsky's coefficient perturbation F_u, the free shell projection R^-, and the coefficient Riesz projection $\mathcal{P}_u^-$ preserve $\mathcal{X}_j^-$, $j = 0, 1$.*

Proof. Let $p_x = |\Omega_x\rangle\langle\Omega_x|$ and $q_x = 1 - p_x$ on a grouped cell. In a finite box the exact-support projector is a tensor product of q_x for $x \in J$ and p_x for $x \notin J$. A gauge transformation at a vertex acts on incident link factors by left or right regular representations. After the links are regrouped into cells this action still factorizes over the cell tensor factors, and every factor fixes the constant cell vacuum Ω_x. It therefore commutes with p_x, q_x, and every exact-support projector. Charge conjugation is the linkwise pullback by the group automorphism $U \mapsto \overline{U}$; it also factorizes and fixes Ω_x, so the same conclusion holds.

The free electric Hamiltonian and the plaquette interaction commute with these symmetries. The symmetries fix the product vacuum and preserve the normalization $\langle \widetilde{\Omega}_\Lambda(u), \Omega_{\Lambda,0}\rangle = 1$. Since the interacting ground line is unique in the small-coupling domain, this normalization fixes its representative and makes that representative invariant. Uniqueness of the finite-volume coefficient expansion then implies that D and F_u commute with the induced coefficient actions. Passing to Yarotsky's thermodynamic coefficient limits preserves these commutators. Functional calculus for D and the contour definition of $\mathcal{P}_u^-$ prove the last assertion. □

Lemma 47 (Unweighted coefficient close projection)**.** *Let $g = \sup_x \|\widehat{\phi}_x(u)\| = 27|u|$. There is a function $a(g) \to 0$ such that*

$$\|F_u v\|_0 \leq a(g)\|v\|_1, \qquad v \in \mathcal{X}_1^-. \tag{60}$$

Assume in addition the spectral-enclosure conditions

$$27|u| < c_1, \qquad \lambda := 27 c_2 |u| < \frac{1}{33}. \tag{61}$$

On the circle

$$\gamma_u = \{z : |z - 4| = r_u\}, \qquad r_u = \frac{1-\lambda}{8}, \tag{62}$$

put

$$R^- = -\frac{1}{2\pi i}\oint_{\gamma_u}(D-z)^{-1}\,dz, \qquad \mathcal{P}_u^- = -\frac{1}{2\pi i}\oint_{\gamma_u}(D+F_u-z)^{-1}\,dz. \tag{63}$$

If $q(g) := 35a(g) < 1$, *then*

$$\boxed{\|\mathcal{P}_u^- - R^-\|_{\mathcal{X}_0^- \to \mathcal{X}_0^-} \le \kappa(g) := \frac{q(g)}{1-q(g)}.} \tag{64}$$

In particular, $\kappa(g) \to 0$*; after decreasing the small-coupling interval,* $\kappa(g) < 1$*.*

Proof. Fix any $0 < \rho < 1$. Yarotsky's estimate (14), for an input of exact support J, gives

$$\sum_K \|(F_u v_J)_K\| \rho^{-(d(K,J)+1)} \le C_{14}(\rho) g |J| \|v_J\|.$$

Since the weight on the left is at least ρ^{-1},

$$\sum_K \|(F_u v_J)_K\| \le \rho C_{14}(\rho) g |J| \|v_J\| \le \rho C_{14}(\rho) g \|H_{J,0} v_J\|.$$

Summing the columns proves (60) with $a(g) = \rho C_{14}(\rho) g$. This also proves that $D + F_u$, with domain $\mathcal{X}_1^-$, is closed when $a(g) < 1$.

The retained-shell gaps put γ_u in the resolvent set of D and give

$$K_0 := \sup_{z\in\gamma_u} \|(D-z)^{-1}\|_{0\to 0} \le \frac{8}{1-\lambda}.$$

Moreover,

$$D(D-z)^{-1} = 1 + z(D-z)^{-1}, \qquad K_1 := \sup_{z\in\gamma_u} \|D(D-z)^{-1}\|_{0\to 0} \le 2 + \frac{32}{1-\lambda} < 35,$$

because $|z| \le 4 + r_u$ on γ_u. Therefore

$$\|F_u(D-z)^{-1}\|_{0\to 0} \le a(g) K_1 \le q(g).$$

For $q(g) < 1$, Yarotsky's Neumann series (15) converges in $\mathcal{X}_0^-$ and yields

$$\|(D+F_u-z)^{-1} - (D-z)^{-1}\|_{0\to 0} \le K_0 \frac{q(g)}{1-q(g)}.$$

The length of γ_u is $2\pi r_u$, and $r_u K_0 \le 1$. Contour integration therefore proves (64). ☐

Lemma 48 (Finite-support shell bridge)**.** *In the physical charge-odd coefficient space,*

$$\operatorname{spec}(D|_{\mathcal{X}_0^-}) \cap (-\infty, 5) = \{4\}, \qquad \operatorname{Ran} R^- = \overline{\operatorname{span}}^{\|\cdot\|_0}\{w_{x,\alpha} : x \in \mathbb{Z}^3,\ \alpha = 1,2,3\}. \tag{65}$$

Proof. Let v_J be a finite-support gauge-invariant, charge-odd eigen-coefficient of rescaled electric energy below 5. Choose a periodic side length L larger than the occupied-edge bounding box and embed the support injectively in the L^3 torus, extending by the cell vacuum. No occupied path winds around the torus, so the embedded coefficient belongs to the trivial electric one-form sector. The retained-shell theorem then forces its energy to be 4 and its energy-4 component to be a linear combination of odd elementary plaquette vectors. The cell Laplacians have discrete Casimir spectrum, and energy below 5 bounds the number of occupied cells. Thus this diagonal direct sum has no additional approximate spectrum below 5. Finite coefficient families are a core for D, and spectral truncation in each finite tensor factor followed by truncation in J is dense in $\mathcal{X}_0^-$. Applying these truncations to the spectral projection of D proves (65). ☐

Lemma 49 (Coefficient totality)**.** *Assume $\kappa(g) < 1$. Then*

$$\operatorname{Ran} \mathcal{P}_u^- = \overline{\mathcal{P}_u^- \operatorname{Ran} R^-}^{\|\cdot\|_0}. \tag{66}$$

Using the finite-support retained-shell classification,

$$\operatorname{Ran} \mathcal{P}_u^- = \overline{\operatorname{span}}^{\|\cdot\|_0}\{\mathcal{P}_u^- w_{x,\alpha} : x \in \mathbb{Z}^3,\ \alpha = 1, 2, 3\}. \tag{67}$$

Proof. Write $P = \mathcal{P}_u^-$, $R = R^-$, and $Q = 1 - R$. Both R and Q are blockwise orthogonal projections and hence contractions in the direct-sum ℓ^1 norm. For $v \in \mathcal{X}_0^-$ set

$$y_0 = Qv, \qquad y_{n+1} = QPy_n.$$

Every y_n lies in $\operatorname{Ran} Q$, and hence

$$\|y_{n+1}\|_0 = \|Q(P - R)y_n\|_0 \le \kappa(g)\|y_n\|_0. \tag{68}$$

Idempotence of P gives the exact identity

$$Py_n = P^2 y_n = PRPy_n + PQPy_n = PRPy_n + Py_{n+1}. \tag{69}$$

Consequently, for every $N \ge 0$,

$$Pv = PRv + \sum_{n=0}^{N} PRPy_n + Py_{N+1}. \tag{70}$$

The last term tends to zero by (68). Every preceding term belongs to $P \operatorname{Ran} R$, which proves (66). The reverse inclusion is immediate because P is a projection.

Lemma 48 says that $\operatorname{Ran} R^-$ is the ℓ^1-closed span of the $w_{x,\alpha}$. Applying the bounded operator P and using (66) proves (67). This is the complete version of the iteration compressed into equations (21)–(24) of [1]; no one-site nondegeneracy assumption is used. □

Lemma 50 (Coefficient-to-GNS transport)**.** *Let $\mathcal{H}_{u,\mathrm{phys}}^-$ be the closed local gauge-invariant, charge-odd cyclic subspace in the GNS representation of the thermodynamic ground state. Put*

$$\widehat{H}_\infty^-(u) = \frac{3}{2} H_\infty^-(u), \tag{71}$$

and let $P_{F,\infty}^-(u)$ be its shell Riesz projection on γ_u. This is the same projection as the shell Riesz projection of $H_\infty^-(u)$, because multiplication of the Hamiltonian and contour by the same positive scalar does not change a spectral projection. Then the coefficient map extends to a contraction

$$\Gamma_u : \mathcal{X}_0^- \longrightarrow \mathcal{H}_{u,\mathrm{phys}}^-, \qquad \Gamma_u v = \pi_u\left(\sum_J \widehat{v}_J\right)\Omega_u, \tag{72}$$

and satisfies

$$P_{F,\infty}^-(u)\Gamma_u = \Gamma_u \mathcal{P}_u^-. \tag{73}$$

Moreover,

$$\operatorname{Ran} P_{F,\infty}^-(u) = \overline{\operatorname{span}}\{P_{F,\infty}^-(u)\pi_u(\widehat{w}_{x,\alpha})\Omega_u\}. \tag{74}$$

No injectivity assumption on Γ_u is required.

Proof. For a finite coefficient family,

$$\|\Gamma_u v\| \le \sum_J \|\pi_u(\widehat{v}_J)\Omega_u\| \le \sum_J \|\widehat{v}_J\| = \sum_J \|v_J\| = \|v\|_0.$$

Thus the series in (72) converges in operator norm and Γ_u extends by continuity to $\mathcal{X}_0^-$. Exact-support symmetry, Lemma 46, puts its range in the physical odd sector.

Yarotsky's equation (18), in the present gap-one normalization, gives on the finite coefficient core

$$\widehat{H}_\infty^-(u)\Gamma_u v = \Gamma_u(D + F_u)v. \tag{75}$$

Finite exact-support and local spectral truncations approximate every $v \in \mathcal{X}_1^-$ in the D-graph norm. The relative bound (60), boundedness of Γ_u, and closedness of $\widehat{H}_\infty^-(u)$ therefore extend (75) to all of $\mathcal{X}_1^-$. If z lies on γ_u and $a \in \mathcal{X}_0^-$, apply this equality to $v = (D + F_u - z)^{-1}a$ to obtain

$$(\widehat{H}_\infty^-(u) - z)^{-1}\Gamma_u a = \Gamma_u(D + F_u - z)^{-1}a. \tag{76}$$

Contour integration proves (73).

It remains only to justify density in the correct sector. The paragraph preceding equation (18) of [1] states that finite linear combinations of creator vectors $\pi_u(\widehat{v}_J)\Omega_u$ are dense in the GNS Hilbert space. To restrict this dense set to the physical odd sector, average a finite creator family over the compact product of gauge groups at the finitely many vertices incident to its support, and then apply $(1 - C)/2$. By Lemma 46, this operation neither enlarges exact support nor leaves the finite coefficient core. Since the corresponding Hilbert-space average is an orthogonal projection that fixes every physical odd vector, the averaged creator families are dense in $\mathcal{H}_{u,\mathrm{phys}}^-$. Hence

$$\operatorname{Ran} P_{F,\infty}^-(u) = \overline{P_{F,\infty}^-(u)\Gamma_u\rfloor_{00}^-} = \overline{\Gamma_u\mathcal{P}_u^-\rfloor_{00}^-}.$$

Now apply Lemma 49 and the boundedness of Γ_u to obtain (74). □

Corollary 51 (Uniform nonvanishing of each canonical seed)**.** *For all sufficiently small g, every vector*

$$\psi_{x,\alpha} = P_{F,\infty}^-(u)\pi_u(\widehat{w}_{x,\alpha})\Omega_u$$

is nonzero, uniformly in x and α.

Proof. The normalized creator has three-cell support S_s and obeys $\widehat{w}_s^*\widehat{w}_s = p_{S_s} := |\Omega_{S_s,0}\rangle\langle\Omega_{S_s,0}| \otimes I$. Here is the needed local product-vacuum estimate. In a periodic volume, the product vacuum is a trial state, while the sum of interaction terms is bounded below by $-|\Lambda|g$. Hence

$$E_\Lambda(u) \le |\Lambda|g, \qquad E_\Lambda(u) \ge \sum_{x\in\Lambda}\langle\widehat{h}_x\rangle_{\Lambda,u} - |\Lambda|g.$$

Translation invariance gives $\langle\widehat{h}_x\rangle_{\Lambda,u} \le 2g$. Since $1 - p_x \le \widehat{h}_x$ and $1 - p_{S_s} \le \sum_{x\in S_s}(1 - p_x)$, every periodic subsequential limit obeys

$$1 - \omega_u(p_{S_s}) \le 2|S_s|g = 6g.$$

Yarotsky uniqueness identifies that limit with ω_u. Since $R^-w_s = w_s$, Lemma 47 and the contraction property of Γ_u imply

$$\begin{aligned}\|\psi_s\| &= \|\Gamma_u\mathcal{P}_u^- w_s\| \\ &\ge \|\Gamma_u w_s\| - \|\Gamma_u(\mathcal{P}_u^- - R^-)w_s\| \\ &\ge \sqrt{1 - 6g} - \kappa(g).\end{aligned}$$

The last quantity is positive after decreasing the small-coupling interval. □

Remark 52 (Scope). Lemmas 47– 50 close CMP(1) for the complete three-component shell at fixed lattice spacing. They do not identify an individual internal dispersion sheet, an isotropic Wilson transfer-matrix particle, or a continuum Yang–Mills particle.

Lemma 53 (Exponentially rooted shell dressing)**.** *Under the hypotheses of Lemma 47, fix $0 < \eta < 1$ small enough for Yarotsky's weighted estimate (14). For every seed $s = (x, \alpha)$,*

$$\mathcal{P}_u^- w_s = w_s + c_s, \qquad \sum_J \eta^{-d(J,S_s)} \|H_{J,0} c_{s,J}\| \le A_\eta(g), \qquad A_\eta(g) \xrightarrow[g\to 0]{} 0, \tag{77}$$

uniformly in s and in finite-volume approximants.

Proof. For a root set S, put

$$\|v\|_{S,0}^{(\eta)} = \sum_J \eta^{-d(J,S)} \|v_J\|, \qquad \|v\|_{S,1}^{(\eta)} = \sum_J \eta^{-d(J,S)} \|H_{J,0} v_J\|.$$

Yarotsky's estimate (14), the triangle inequality $d(K, S) \le d(K, J) + d(J, S)$, and $H_{J,0} \ge |J|$ give

$$\|F_u v\|_{S,0}^{(\eta)} \le a_\eta(g) \|v\|_{S,1}^{(\eta)}, \qquad a_\eta(g) = \eta C_{14}(\eta) g.$$

On the dynamic contour γ_u of (62), put

$$K_1 = \sup_{z \in \gamma_u} \|D(D - z)^{-1}\|_{S,0\to S,0} < 35, \qquad q_\eta = a_\eta(g) K_1.$$

For $q_\eta < 1$, the Neumann resolvent series converges in the rooted norm. Since $R^- w_s = w_s$, applying D to the resolvent difference and integrating gives

$$A_\eta(g) \le r_u K_1 \frac{q_\eta}{1 - q_\eta}.$$

This proves (77). The argument accepts an arbitrary finite input support and does not use the one-site nondegeneracy assumption of [1, Theorem 4]. □

Theorem 54 (Fixed-spacing three-component physical band and Wilson-source frame)**.** *Under the hypotheses of Proposition 44, shrink the existential small-$|u|$ interval so that Lemmas 47 and 53 hold. In the physical charge-odd GNS sector,*

$$\operatorname{Ran} P_{F,\infty}^-(u) \simeq \ell^2(\mathbb{Z}^3) \otimes \mathbb{C}^3 \tag{78}$$

by a translation-intertwining unitary. In this representation the Hamiltonian is convolution by an exponentially summable Hermitian 3×3 kernel, so its Bloch matrix is analytic in a complex momentum strip. The complete band retains the external gaps in Proposition 44.

More precisely, let

$$\psi_{x,\alpha} = P_{F,\infty}^-(u) \pi_u(\widehat{w}_{x,\alpha}) \Omega_u, \qquad G_{st} = \langle \psi_s, \psi_t \rangle,$$

where $\widehat{w}_{x,\alpha}$ is the bounded rank-one plaquette creator. For some $\mu > 0$,

$$\|G - I\|_{\mu,\sharp} = g_0(u) < 1, \qquad G(k) \succeq (1 - g_0(u)) I_3 \quad \textit{for every real } k. \tag{79}$$

The same conclusion holds for the literal three-orientation Wilson multiplication source

$$O_{x,\alpha}^W = \operatorname{Im}\operatorname{Tr} U_{p(x,\alpha)}.$$

After the fixed normalization $c_W = i\sqrt{2}$, its projected synthesis map T_W is an isomorphism onto the band and its Gram symbol satisfies

$$G_W(k) \succeq (1 - \|G - I\|_{\mu,\sharp})(1 - \|E\|_{\mu,\sharp})^2 I_3 > 0 \quad \textit{for every real } k, \tag{80}$$

where $\|E\|_{\mu,\sharp} \to 0$ as $u \to 0$.

Proof. Lemma 50 gives

$$\operatorname{Ran} P^-_{F,\infty} = \overline{\operatorname{span}}\{\psi_{x,\alpha}\}.$$

It remains to prove that these generators form a frame rather than merely a dense family.

Yarotsky's cluster estimate (16) has the form

$$|\omega_u(A^*B) - \omega_u(A^*)\omega_u(B)| \le C_0^{|I|+|J|}\tau(g)^{\operatorname{dist}(I,J)}\|A\|\|B\|, \qquad \tau(g) \to 0.$$

The unit onsite gap and the variational energy-density bound also give, for every fixed cell set T,

$$1 - \omega_u(Q_T) \le 2|T|g, \qquad |\omega_u(A) - \omega_0(A)| \le 2\|A\|\sqrt{2|T|g} + 4|T|g\|A\|.$$

Use orthogonality of $P^-_{F,\infty}$ to expand only one Riesz mark:

$$G_{st} = \omega_u(\widehat{w}_s^*\widehat{w}_t) + \sum_J \omega_u(\widehat{w}_s^*\widehat{c}_{t,J}).$$

Charge oddness kills all one-point functions and the empty coefficient. For separated roots, the cluster estimate and (77) give

$$|G_{st} - \delta_{st}| \le C\{\tau(g)^{R_{st}} + A_\eta(g)\theta^{R_{st}}\}, \qquad \theta = \max\{C_0\eta, \tau(g)\} < 1.$$

For the finitely many contacting roots, local vacuum continuity gives $|G_{st} - \delta_{st}| \le C_{\mathrm{loc}}\sqrt{g} + A_\eta(g)$. Choose η first and then g so that $e^\mu C_0\eta < 1$ and $e^\mu\tau(g) < 1$. The number of roots at distance n grows only quadratically, hence the exponentially weighted row and column sums tend to zero with g. This proves (79).

The synthesis J_P therefore obeys

$$(1 - g_0)\|f\|_2^2 \le \|J_P f\|^2 \le (1 + g_0)\|f\|_2^2.$$

Its range is closed and, by totality, is the whole Riesz range. Thus $W = J_P G^{-1/2}$ proves (78).

For completeness, the energy-kernel estimate uses no unrecorded second Riesz mark. In the gap-one coefficient normalization put $\mathbb{D}_u = D + F_u$ and write $c_t = \mathcal{P}_u^- w_t - w_t$. Since $Dw_t = 4w_t$ and $\mathbb{D}_u$ commutes with $\mathcal{P}_u^-$,

$$\mathbb{D}_u\mathcal{P}_u^- w_t = 4w_t + k_t, \qquad k_t = Dc_t + F_u w_t + F_u c_t.$$

The rooted relative bound and (77) give

$$\sum_J \eta^{-d(J,S_t)}\|(k_t)_J\| \le B_\eta(g) := A_\eta(g) + 4a_\eta(g) + a_\eta(g)A_\eta(g) \longrightarrow 0.$$

The GNS intertwining (75) and orthogonality of the Riesz projection give the one-mark identity

$$\langle\psi_s, \widehat{H}^-_\infty(u)\psi_t\rangle = \omega_u\Big(\widehat{w}_s^*\,\widehat{\mathbb{D}_u\mathcal{P}_u^- w_t}\Big), \qquad \widehat{H}^-_\infty(u) = \frac{3}{2}H^-_\infty(u).$$

Applying the same finite-contact/cluster-tail split to $4w_t + k_t$ now shows that this energy kernel belongs to the same exponentially weighted convolution algebra. The convergent binomial series for $G^{-1/2}$ in that algebra then puts $W^* H_\infty W$ there as well, proving analyticity of its Bloch matrix.

For the literal source put

$$D_s = c_W O_s^W - \widehat{w}_s, \qquad M_D = 3\sqrt{2} + 1.$$

Character orthogonality gives $c_W = i\sqrt{2}$ and $D_s\Omega_0 = 0$. Hence

$$\|\pi_u(D_s)\Omega_u\|^2 \le 6gM_D^2.$$

Define the three synthesis maps by

$$T_c f = \sum_s f_s \psi_s, \qquad T_D f = \sum_s f_s P_{F,\infty}^-(u)\pi_u(D_s)\Omega_u, \qquad T_W = T_c + T_D.$$

Thus T_W is exactly the projected, c_W-normalized literal Wilson source. Its cross kernel is $C = T_c^* T_D$. For contacting roots, the displayed $O(\sqrt{g})$ norm bound and Cauchy–Schwarz control C_{st}. For separated roots, use $\psi_s = \Gamma_u(w_s + c_s)$: charge oddness removes both one-point terms, while the cluster bound controls w_s and the rooted estimate controls c_s. The same contact/tail summation therefore gives $\|C\|_{\mu,\sharp} \to 0$.

Since $G = T_c^* T_c$ is invertible, put $E = G^{-1}C$. Both T_c and T_D take values in the target band, and T_c is onto it. Hence the exact identities

$$T_c G^{-1} T_c^* = I_{\mathrm{Ran}\, P_{F,\infty}^-}, \qquad T_D = T_c G^{-1} T_c^* T_D = T_c E, \qquad T_W = T_c(I + E)$$

show that T_W is onto and bounded below once $\|E\|_{\mu,\sharp} < 1$. Taking Gram matrices, $G_W = (I+E)^* G (I+E)$, proves (80). $\square$

Remark 55 (Exact meaning of the closed result). Theorem 54 closes the complete three-component plaquette-shell band and the literal Wilson *source* residue at sufficiently small strong coupling and fixed lattice spacing. It does not isolate the one-dimensional $BB^\dagger$ kernel inside each nonzero-momentum fiber, and it does not identify the Hamiltonian semigroup with the isotropic Wilson-action transfer matrix.

There is also a sharp scalar consequence for the compact carrier seed. For an oriented plaquette p, define the normalized self-adjoint charge-odd source

$$A_p = \frac{\chi_p - \bar{\chi}_p}{i\sqrt{2}}.$$

Let c_x be the elementary cube based at x, with outward incidence signs $s(c_x, p)$, and put

$$C_x = \frac{1}{\sqrt{6}} \sum_{p \subset \partial c_x} s(c_x, p) A_p, \qquad \eta_x(u) = P_{F,\infty}(u)\pi_u(C_x)\Omega_u. \tag{81}$$

Its projected Gram kernel is

$$G_u(x) = \langle \eta_0(u), \eta_x(u) \rangle. \tag{82}$$

At $u = 0$, plaquette orthogonality and Proposition 11 give the exact symbol

$$\widehat{G}_0(k) = \sum_x e^{-ik\cdot x} G_0(x) = \frac{q(k)}{6}. \tag{83}$$

Corollary 56 (Compact carrier-source residue)**.** *Let $G_W(k)$ be the three-orientation literal-Wilson-source Gram symbol in (80). Then the cube-boundary source in (81) obeys*

$$\widehat{G}_u(k) \geq (1 - g_0(u))(1 - \|E\|_{\mu,\sharp})^2 \frac{q(k)}{6}. \tag{84}$$

In particular, on every $U \Subset \mathbb{T}^3 \setminus \{0\}$ with $q(k) \geq q_* > 0$, the compact carrier source has residue at least $(1 - g_0(u))(1 - \|E\|_{\mu,\sharp})^2 q_*/6$, uniformly for $k \in U$.

Proof. In the three-orientation plaquette fiber, the normalized cube boundary has coefficient vector $w(k)/\sqrt{6}$. Theorem 6 gives $\|w(k)\|^2 = q(k)$, while the literal source in (81) has the Wilson Gram, not the canonical-creator Gram. Equation (80) gives $G_W(k) \succeq (1 - g_0(u))(1 - \|E\|_{\mu,\sharp})^2 I_3$. Therefore

$$\widehat{G}_u(k) = \frac{1}{6} w(k)^* G_W(k) w(k) \geq (1 - g_0(u))(1 - \|E\|_{\mu,\sharp})^2 \frac{q(k)}{6}.$$

□

The full three-component band and its compact carrier-source residue are therefore closed at fixed spacing. The following continuation identifies the second-order coefficient of the exact orthonormal symbol and closes its rank-one internal sheet on every momentum ball compactly separated from Γ.

10.2 Exact isolation of the internal carrier sheet away from Γ

This subsection works at fixed lattice spacing in the $SU(3)$ Hamiltonian theory and in the unrescaled energy convention of Eq. (2). It uses the repaired close-projection theorem and Theorem 54; no Wilson-action transfer matrix is used. The result closes the internal rank-one sheet on every fixed momentum patch whose closure avoids Γ. It does not give a gap uniform as $k \to 0$.

For $\mu > 0$, let $\mathcal{A}_\mu$ denote the unital Banach $*$-algebra of 3×3 convolution kernels $K = (K_{\alpha\beta}(x))$ on $\mathbb{Z}^3$ with norm

$$\|K\|_{\mu,\sharp} := \max\left\{ \max_\alpha \sum_{x,\beta} e^{\mu|x|_1} |K_{\alpha\beta}(x)|, \max_\beta \sum_{x,\alpha} e^{\mu|x|_1} |K_{\alpha\beta}(x)| \right\}. \tag{85}$$

Its Fourier transform is analytic for $|\operatorname{Im} k_j| < \mu$ and $\sup_{k\in\mathbb{T}^3} \|K(k)\| \leq \|K\|_{\mu,\sharp}$.

Lemma 57 (Coupling-analytic exponentially local symbol)**.** *There are $\rho_0 > 0$ and $\mu > 0$ such that the orthonormal-frame Hamiltonian of Theorem 54 is represented by a map*

$$u \longmapsto h_u \in \mathcal{A}_\mu \tag{86}$$

which is holomorphic for complex $|u| < \rho_0$. For real u, $h_u(k)$ is Hermitian. The same statement holds for the canonical Gram kernel G_u and the physical one-mark energy kernel $\mathcal{K}_u$, and

$$h_u = G_u^{-1/2} \mathcal{K}_u G_u^{-1/2}. \tag{87}$$

Here the inverse square root is the branch satisfying $G_0^{-1/2} = I$.

Proof. Choose ρ_0 strictly inside both the repaired CMP(1) smallness domain and the complex ground-state contraction domain. With $\lambda_0 = 27c_2\rho_0 < 1/33$, one may use the fixed contour

$$\gamma_{\rho_0} : \quad |z - 4| = (1 - \lambda_0)/8$$

in the gap-one rescaling. It encloses the target cluster for every $|u| \le \rho_0$ and stays in the coefficient resolvent set. Indeed, the target disc has radius at most $4\lambda_0$, while the nearest other free centres are $7/2$ and $17/4$. The inequalities separating all three discs from this contour reduce on the upper side to $33\lambda_0 < 1$ (and on the lower side to the weaker $\lambda_0 < 1/9$). For complex u this statement uses the coefficient Neumann-resolvent bound, whose absolute majorant depends on $|u|$, rather than a self-adjoint spectral-ordering argument. This fixed contour, rather than a contour depending on the evaluation point u, is what makes the following holomorphic calculation literal.

The product-ground-state fixed-point map used in [1] is analytic in the bounded interaction. On a closed complex u-disc contained in the contraction domain its contraction constant and all coefficient majorants depend only on $|u|$. The analytic contraction theorem therefore makes its fixed point, the dressed coefficient perturbation F_u, and the Neumann resolvent series holomorphic in their weighted coefficient norms. Contour integration on γ_{ρ_0} makes the coefficient Riesz projection $\mathcal{P}_u^-$ holomorphic in every slightly weaker rooted norm.

We record why the spatial estimate remains valid on a complex disc. In the proof of [1, Theorem 1], Lemmas 3–4 give holomorphic polymer activities whose absolute values, uniformly on a closed parameter disc, obey a majorant of the form

$$|a_u(\chi)| \le C_\rho \varepsilon_\rho^{|\operatorname{supp}\chi|}, \qquad \varepsilon_\rho < 1,$$

after the coupling disc is made small enough. The number of polymers of size n through a specified point grows at most exponentially in n. Consequently the connected-cluster series is absolutely and locally uniformly convergent for complex u, which is the argument used there to prove weak-star analyticity. With two marked local insertions, every connected cluster joining supports I and J has size bounded below by a constant times $\operatorname{dist}(I, J)$. Summing the same activity majorant therefore gives the complex-disc estimate

$$\sup_{|u|\le\rho} |\omega_u^{\mathbb{C}}(A^*B) - \omega_u^{\mathbb{C}}(A^*)\omega_u^{\mathbb{C}}(B)| \le C_\rho^{|I|+|J|} \tau_\rho^{\operatorname{dist}(I,J)} \|A\|\|B\|, \qquad \tau_\rho < 1. \tag{88}$$

Here $\omega_u^{\mathbb{C}}$ denotes the holomorphic continuation; for real u it is the physical ground state. This is an absolute cluster bound, not an inference from real-u positivity or from pointwise weak-star analyticity.

Put $\mathbb{D}_u := D + F_u$ for the gap-one coefficient Hamiltonian and use the one-mark identities

$$(G_u)_{st} = \omega_u\left(\widehat{w}_s^* \,\widehat{\mathcal{P}_u^- w_t}\right), \tag{89}$$

$$(\widehat{\mathcal{K}}_u)_{st} = \omega_u\left(\widehat{w}_s^* \,\widehat{\mathbb{D}_u \mathcal{P}_u^- w_t}\right), \qquad \mathcal{K}_u := \frac{2}{3}\widehat{\mathcal{K}}_u. \tag{90}$$

They agree with the Hilbert-space Gram and energy kernels for real u but, unlike a two-mark norm formula, also give their holomorphic continuation to complex u. To make the summation explicit, write $\mathcal{P}_u^- w_t = w_t + c_t(u)$. The rooted Neumann majorant gives, uniformly for $|u| \le \rho < \rho_0$,

$$\sum_J \eta^{-d(J,S_t)} \|D_J c_{t,J}(u)\| \le A_{\eta,\rho},$$

and the same calculation applied to

$$\mathbb{D}_u \mathcal{P}_u^- w_t = 4w_t + Dc_t + F_u w_t + F_u c_t$$

gives a rooted majorant $A'_{\eta,\rho}$. Both coefficient families are holomorphic. Exact-support projections commute with charge conjugation, so w_t, $c_{t,J}$ and every displayed energy correction are charge odd.

The analytically continued ground-state functional is charge even; hence both one-point terms in (88) vanish identically, for complex as well as real u. For separated roots apply (88) term by term and use $\mathrm{dist}(S_s, S_t) \le \mathrm{dist}(S_s, J) + d(J, S_t)$, exactly as in the real CMP(3) Schur sum. Contacting roots form a fixed finite list and are bounded directly by the absolutely convergent marked-cluster series. Choose η and then the disc so that $\theta_\rho := \max\{C_\rho \eta, \tau_\rho\} < 1$. Since the number of roots at distance n grows quadratically, any $0 < \mu_0 < -\log \theta_\rho$ gives constants $C, \mu_0 > 0$ such that

$$\sum_{x\in\mathbb{Z}^3} e^{\mu_0 |x|_1} \sup_{|u|\le\rho} \left(|G_{\alpha\beta}(x;u)| + |\mathcal{K}_{\alpha\beta}(x;u)|\right) \le C, \qquad \rho < \rho_0.$$

Finite spatial truncations are holomorphic, and this summable supremum majorant makes them converge locally uniformly in $\mathcal{A}_\mu$ for any $0 < \mu < \mu_0$. Thus G_u and $\mathcal{K}_u$ are Banach-valued holomorphic maps.

After shrinking ρ_0 if necessary, $\|G_u - I\|_{\mu,\sharp} < 1$. The binomial series for $G_u^{-1/2}$ converges locally uniformly in the unital Banach algebra $\mathcal{A}_\mu$ and is holomorphic. Equation (87) proves the claim. The coefficient calculation in the gap-one Hamiltonian $\widehat{H} = (3/2)H$ would give the non-scalar coefficient $(3/2)t_3 = 5/408$. Multiplying the completed symbol by $2/3$ returns exactly $t_3 = 5/612$ in the unrescaled normalization of Eq. (2). □

Lemma 58 (Finite-to-infinite second-order coefficient)**.** *In the orthonormal frame of Theorem 54,*

$$[u^0]h_u(k) = \frac{8}{3} I_3, \qquad [u^1]h_u(k) = I_3, \qquad [u^2]h_u(k) = \sigma_2 I_3 + \frac{5}{612} B(k)B(k)^\dagger, \tag{91}$$

where σ_2 is a real scalar. If the separate $SU(3)$ diagonal ledger quoted in Section 7 is adopted, then $\sigma_2 = 11/306$; the internal-sheet conclusion below does not use this value.

Proof. First work in a sufficiently large finite periodic box ($L \ge 5$), in the physical charge-odd trivial-flux shell, and use a fixed contour valid on a complex u-disc. Kato parallel transport gives a holomorphic intertwiner from the free plaquette shell to the interacting Riesz island; on the real axis it is unitary. Differentiating its reduced operator at $u = 0$ gives the ordinary degenerate reduced-resolvent coefficient

$$P_-(-V)Q_-(E_{F,3} - H_E)^{-1}Q_-(-V)P_- - [u^2]E_{\mathrm{vac,L}}(u)I,$$

with the conventional scalar terms absorbed into the diagonal ledger. This is precisely the $Q_- = 1_- - P_-$ prescription of Eq. (18), rather than an eigenvalue-only comparison. The retained-shell theorem, the first-order charge/determinant census, and Corollary 19 therefore give

$$k_{L,0} = \frac{8}{3} I, \qquad k_{L,1} = I, \qquad k_{L,2} = \sigma_{2,L} I + \frac{5}{612} BB^\dagger.$$

The canonical Gram orthonormalization used to define $h_{L,u}$ is another holomorphic intertwiner which equals the free plaquette frame at $u = 0$ and is unitary for real u. The two identifications differ by a holomorphic near-identity similarity $Z(u)$, unitary on the real axis. If $Z_0 = I$ and

$$a(u) = a_0 I + u a_1 I + u^2 a_2 + \sum_{n\ge 3} u^n a_n,$$

then direct expansion gives

$$[u^2]\, Z(u)^{-1} a(u) Z(u) = a_2;$$

all possible frame terms are commutators with $a_0 I$ or $a_1 I$. Thus the full second-order matrix, not merely its eigenvalues, is unchanged by the orthonormal-frame choice. This is the point at which scalarity of both the zeroth- and first-order coefficients is essential.

It remains to justify passage to infinite volume without confusing torus convolution with convolution on $\mathbb{Z}^3$. Let $\mathcal{A}_{\mu,L}$ be the corresponding weighted convolution algebra on $\mathbb{Z}_L^3$, using the torus distance, and let

$$(\Pi_L K)([x]) = \sum_{n\in\mathbb{Z}^3} K(x+nL)$$

be periodisation. Then $\Pi_L : \mathcal{A}_\mu \to \mathcal{A}_{\mu,L}$ is a contractive algebra homomorphism. Let $G_{L,u}$ and $\mathcal{K}_{L,u}$ be the periodic finite-volume one-mark kernels. The coefficient fixed-point and marked-cluster series converge term by term to their infinite-volume series for every fixed centred displacement, locally uniformly on a smaller complex u-disc. The periodic ground-state limit is the same state as the open-box limit because the small-perturbation ground state is unique [2]. This first identifies the two limits for real u; local uniform holomorphy and the identity theorem then identify their complex continuations on the common disc.

The proof of Lemma 57 gives the stronger bound, uniformly in L,

$$\sum_{[x]} e^{\mu_0 d_L([x],0)} \sup_{|u|\le\rho} \left(|G_{L,\alpha\beta}(x;u)| + |\mathcal{K}_{L,\alpha\beta}(x;u)|\right) \le C_\rho.$$

Fix $0<\mu<\mu_0$. Split the torus norm at $d_L([x],0)=R$, with $2R<L$. The finite part of $G_{L,u}-\Pi_L G_u$ (and of the energy kernel) converges locally uniformly by the preceding termwise limit. Both tails, including every periodised image, are at most $C_\rho e^{-(\mu_0-\mu)R}$. Letting first $L\to\infty$ and then $R\to\infty$ proves

$$\|G_{L,u}-\Pi_L G_u\|_{\mu,\sharp,L} \longrightarrow 0, \qquad \|\mathcal{K}_{L,u}-\Pi_L\mathcal{K}_u\|_{\mu,\sharp,L} \longrightarrow 0$$

locally uniformly in u. On a smaller disc all Gram kernels obey a common $\|G_{L,u}-I\|_{\mu,\sharp,L} \le r_G < 1$. The binomial series is therefore uniform in L, and the homomorphism property of Π_L gives

$$\|h_{L,u}-\Pi_L h_u\|_{\mu,\sharp,L} \longrightarrow 0$$

locally uniformly. This also shows explicitly that wrap-around products in the finite-torus inverse square root are accounted for by periodisation, not silently identified with infinite convolution. Cauchy's formula on any circle inside that disc now gives

$$[u^2]h_u(x) = \lim_{L\to\infty}[u^2]h_{L,u}(x) \tag{92}$$

for each fixed displacement x. This is stronger than convergence at a few real couplings: local uniformity on a complex contour is what permits interchanging the thermodynamic limit with Taylor-coefficient extraction.

At order two every connected process is supported on one face or on two faces sharing one link. Hence, once the roots and this finite process neighbourhood fit without wrapping, the finite-volume coefficient is exactly the infinite-lattice coefficient. Lemma 17 excludes all other off-diagonal supports, and Theorem 18 assigns the surviving oriented shared-link element $t_3 = 5/612$. Translation and cubic symmetry make the remaining diagonal a scalar. The coefficient is finite range, so the entrywise identity also holds in every $\mathcal{A}_\mu$. This proves (91). □

Define

$$s_{\le 2}(u) := \frac{8}{3} + u + \sigma_2 u^2, \qquad t_3 := \frac{5}{612}. \tag{93}$$

Theorem 59 (Exact fixed-spacing carrier sheet on a nonzero-momentum patch)**.** *Assume the repaired CMP(1) lemma and Theorem 54. Let U be a relatively compact open momentum ball whose closure lies in $\mathbb{T}^3 \setminus \{0\}$, and put*

$$q_* := \inf_{k \in U} q(k) > 0.$$

Then there is $u_U > 0$ such that for every real $0 < |u| < u_U$ the exact infinite-volume target band over U has a rank-one analytic Riesz subbundle, descended from $\ker B(k)^\dagger$*, separated from the other two target branches by*

$$\mathrm{gap}_{\mathrm{int}}(k, u) \geq \frac{1}{2} t_3 u^2 q_*, \qquad k \in U. \tag{94}$$

Its direct integral is a reducing, translation-invariant subspace unitarily equivalent to $L^2(U)$, on which translations and the Hamiltonian act as

$$(T_x f)(k) = e^{ik \cdot x} f(k), \qquad (H_{\mathrm{car}}(u) f)(k) = E_{\mathrm{car}}(k, u) f(k). \tag{95}$$

The function $E_{\mathrm{car}}(k, u)$ is real analytic in k on U.

Proof. Choose $0 < r < \rho_0$ in Lemma 57 and set

$$M_r := \sup_{|\zeta| = r} \|h_\zeta\|_{\mu, \sharp} < \infty, \qquad C_r := \frac{2M_r}{r^3}.$$

Cauchy's estimate in $\mathcal{A}_\mu$, followed by summing the Taylor tail, gives for $|u| \leq r/2$

$$h_u(k) = s_{\leq 2}(u) I_3 + t_3 u^2 B(k) B(k)^\dagger + R_u(k), \qquad \sup_{k \in \mathbb{T}^3} \|R_u(k)\| \leq C_r |u|^3. \tag{96}$$

Indeed, the nth Banach-valued Taylor coefficient has norm at most $M_r r^{-n}$, and $\sum_{n \geq 3} (|u|/r)^n \leq 2|u|^3/r^3$ on this disc.

One may therefore take, in addition to the smallness thresholds required by CMP(1)–CMP(3),

$$u_U := \min\left\{\frac{r}{2}, \frac{t_3 q_*}{4C_r}\right\} > 0. \tag{97}$$

For $0 < |u| < u_U$,

$$\sup_{k \in U} \|R_u(k)\| \leq \frac{1}{4} t_3 u^2 q_*.$$

The comparison matrix in (96) has a simple carrier eigenvalue $s_{\leq 2}(u)$ and a double eigenvalue $s_{\leq 2}(u) + t_3 u^2 q(k)$. Weyl's inequality separates the corresponding exact spectral clusters by at least

$$t_3 u^2 q(k) - 2\|R_u(k)\| \geq \frac{1}{2} t_3 u^2 q_*.$$

The small contour surrounding the lower cluster therefore defines a rank-one Riesz projection analytic in k. Put $P_0(k) = v_0(k) v_0(k)^\dagger$ with $v_0(k) = w(k)/\sqrt{q(k)}$; this is the limiting comparison projection chosen by the $BB^\dagger$ coefficient, not a spectral projection of the triply degenerate $u = 0$ fiber. The same Riesz estimate and $\|R_u\|/(t_3 u^2 q_*) = O(|u|/q_*)$ show that, after decreasing u_U if necessary, $\sup_{k \in U} \|P_{\mathrm{car}}(k, u) - P_0(k)\| < 1/2$. Hence

$$v_u(k) := \frac{P_{\mathrm{car}}(k, u) v_0(k)}{\|P_{\mathrm{car}}(k, u) v_0(k)\|}$$

is a globally defined real-analytic unit section on the whole momentum ball. Fourier direct integration gives (95); analyticity of the simple eigenvalue follows from analyticity of the matrix symbol. $\square$

Corollary 60 (No dark carrier on a fixed momentum patch)**.** *After possibly decreasing u_U, the rank-one sheet in Theorem 59 is seen uniformly by a dressed cube-boundary Wilson source. More precisely, put*

$$v_0(k) := \frac{w(k)}{\sqrt{q(k)}}, \qquad k \in U,$$

and let $S_W(k,u)$ be the normalized literal-Wilson projected synthesis map written in the orthonormal band frame, with the fixed CMP(4) normalization chosen so that $S_W(k,0) = I_3$. If $P_{\mathrm{car}}(k,u)$ is the exact rank-one projection, then u_U may be chosen so that

$$\inf_{k\in U} \left\|P_{\mathrm{car}}(k,u)S_W(k,u)v_0(k)\right\|^2 \geq \frac{1}{4}, \qquad 0 < |u| < u_U. \tag{98}$$

The number $1/4$ is an absolute projected-norm bound for this normalized coefficient vector, not a spectral fraction normalized by the full raw-source covariance. Consequently every $f \in C_c^\infty(U)$ gives a rapidly decaying spatial smearing of the literal Wilson plaquettes, with cube-boundary incidence, whose projection onto the exact carrier sheet is nonzero and volume independent.

Proof. At $u = 0$ the three target levels coincide, so there is no spectrally defined rank-one carrier projection. Instead let $P_0(k) = v_0(k)v_0(k)^\dagger$ be the limiting comparison projection selected by the $BB^\dagger$ coefficient. Then $P_0(k)S_W(k,0)v_0(k) = v_0(k)$. The remainder estimate (96) and the order-u^2 gap imply, by the finite-dimensional Riesz-projection estimate, that

$$\sup_{k\in U} \|P_{\mathrm{car}}(k,u) - P_0(k)\| \longrightarrow 0 \quad (u \to 0).$$

CMP(3)–CMP(4) make this explicit. In the notation $W = T_c G^{-1/2}$ and $T_W = T_c(I+E)$,

$$S_W = W^* T_W = G^{1/2}(I+E),$$

so $\|S_W - I\|_{\mu,\sharp} \to 0$. Therefore

$$\|P_{\mathrm{car}} S_W v_0\| \geq 1 - \|P_{\mathrm{car}} - P_0\| - \|S_W - I\|.$$

Since $\overline{U}$ is compact, the right-hand side is at least $1/2$ after shrinking u_U. Squaring gives (98). Fourier transform then turns $f \in C_c^\infty(U)$ into a rapidly decaying spatial smearing. The identity $f(k)v_0(k) = w(k)f(k)/\sqrt{q(k)}$ identifies its coefficients as the cube-boundary incidence combination of the three literal Wilson plaquettes. □

Remark 61 (What has and has not been closed)*.* Theorem 59 removes the two hypotheses in the former conditional fixed-k statement: it proves both the exact orthonormal-symbol coefficient $[u^2]h_u = \sigma_2 I + (5/612)BB^\dagger$ and a remainder uniform in momentum on a nonzero-momentum patch. The threshold u_U is existential because the Yarotsky contraction majorants are existential, and it necessarily deteriorates as $q_* \downarrow 0$. No rank-one separation at Γ is claimed; the cubic T_1 triplet meets there.

For every $a_t > 0$, the exact Hamiltonian semigroup restricts to this subspace as multiplication by $e^{-a_t E_{\mathrm{car}}(k,u)}$. This does *not* identify $e^{-a_t H_\infty(u)}$ with the finite-temporal-spacing isotropic Wilson-action transfer matrix. The anisotropic-time/operator equivalence and every continuum-limit statement remain separate open problems.

11 The continuum bridge, and where it is blocked

The previous section works at one fixed lattice spacing. The remaining step is the continuum limit, and the first result below is that it cannot be reached by extending the domain of Theorem 3: in the Hamiltonian coordinate $u = g_H^{-4}$ an asymptotically free trajectory has $u(a) \to \infty$, while the proved island requires $u < u_*$. The hypotheses are eventually *disjoint*, so this is a theorem about the two domains rather than a caution about extrapolating between them, and computing the existential constants would not change it. The corollary that follows sharpens it: no nonzero finite strong-coupling Taylor truncation yields a finite continuum mass at all.

What the section does supply is the shape of what would be needed — the necessary scaling law an electric-unit branch must obey, an unconditional atom-passage proposition showing that a uniformly source-visible bottom atom survives vague convergence without any isolation or Haag–Ruelle hypothesis, and a reflection-positivity-preserving blocking corollary for the raw deterministic Balaban average with reflection-adapted block and path data. A two-point counterexample closes off the stochastic variant. The section ends by naming the remaining object explicitly, which is a Hamiltonian–Wilson matching package rather than a further estimate.

Proposition 62 (Exact strong/weak-coupling separation)**.** *For $SU(3)$ the Hamiltonian coordinate of this paper is $u = g_H^{-4}$. Put*

$$u_* := \min\left\{\frac{c_1}{27}, \frac{1}{891 c_2}\right\},$$

the coupling radius in Theorem 3. If $a \downarrow 0$ is any asymptotically-free Hamiltonian trajectory with $g_H(a) \to 0$, then $u(a) \to \infty$. In particular, there is $a_0 > 0$ such that $u(a) > u_*$ for $0 < a < a_0$. No tail of a continuum trajectory therefore lies in the proved Riesz-island domain.

Proof. The function $x \mapsto x^{-4}$ tends to infinity as $x \downarrow 0$. The eventual inequality is the definition of this limit. □

This is stronger than a warning about extrapolation: the hypotheses of the fixed-cutoff theorem and of the continuum trajectory are eventually disjoint. Numerical evaluation of the existential constants c_1, c_2 would not change that fact.

Conditional corollary 63 (Necessary continuum scaling of an electric-unit branch)**.** Use the universal pure-gauge beta-function coefficients and the standard two-loop lattice scaling law [5, 6, 7]. Let

$$b_0 = \frac{11N}{48\pi^2}, \qquad b_1 = \frac{34N^2}{3(16\pi^2)^2}, \qquad p = \frac{b_1}{2b_0^2} = \frac{51}{121}.$$

Assume that the Hamiltonian coupling has been matched to an asymptotically-free Wilson scheme, including the anisotropy/time normalisation, so that for a finite $\Lambda_H > 0$

$$a\Lambda_H = \exp\left[-\frac{1}{2b_0 g_H(a)^2}\right] \left(b_0 g_H(a)^2\right)^{-p}[1 + o(1)]. \tag{99}$$

Assume also that a vacuum-subtracted electric-unit excitation $E(u)$ obeys

$$aM(a) = c_H g_H(a)^2 E(u(a))[1 + o(1)], \qquad c_H > 0, \tag{100}$$

and $M(a) \to M \in (0, \infty)$. With $u = g_H^{-4}$, necessarily

$$E(u(a)) = \frac{M}{c_H \Lambda_H} b_0^{-p} u(a)^{(1+p)/2} \exp\left[-\frac{\sqrt{u(a)}}{2b_0}\right] [1 + o(1)]. \tag{101}$$

For $SU(3)$ the essential exponential is $\exp[-(8\pi^2/11)\sqrt{u}]$ and $(1 + p)/2 = 86/121$.

The Hamer crosswalk printed earlier gives $aM_{\mathrm{phys}} = (g^2/2)M_A(x)$ and $E(u) = M_A(2u)/2$. Hence $x = 2u$ and $g = g_H$ would fix $c_H = 1$, not $1/2$: the half is already present in $E = M_A/2$. We leave c_H explicit because the upstream hostile audit still records the operator normalisation $H = W/2$, $u = x/2$ as an asserted bridge and because the Hamiltonian–Wilson anisotropy/time map needed at weak coupling has not been derived. Proving that map would set $c_H = 1$ in the present convention.

Proof. Set $x = g_H^{-2} = \sqrt{u}$. Equation (99) gives $a = \Lambda_H^{-1} e^{-x/(2b_0)}(x/b_0)^p[1 + o(1)]$, while (100) gives $E = aMx/c_H[1 + o(1)]$. Substitution proves (101). □

Corollary 64 (Finite Taylor data cannot supply the continuum mass)**.** *Under the matching assumptions of Conditional result 63, substituting any nonzero polynomial $P(u)$ for $E(u)$ makes*

$$\left| \frac{c_H g_H(a)^2 P(u(a))}{a} \right| \longrightarrow \infty.$$

Thus no nonzero finite strong-coupling Taylor truncation yields a finite continuum mass.

Proof. If P has degree d and leading coefficient $q_d \neq 0$, then with $x = \sqrt{u}$ the expression is

$$c_H \Lambda_H q_d b_0^p e^{x/(2b_0)} x^{2d-1-p}[1 + o(1)].$$

Its absolute value diverges because the exponential dominates every real power. The conclusion is about finite truncations, not about a nonperturbatively continued branch. □

Conditional corollary 65 (Bare absolute-weight power counting)**.** Work first in a periodic box of fixed physical volume and let

$$\widetilde{O}_{a,n} = L(a)^{-3/2} \sum_x \operatorname{Im}\operatorname{Tr} U_{p_n(x)}$$

be the volume-normalized zero-momentum source made from plaquettes normal to direction n. Suppose a continuum operator matching exists in the form

$$\operatorname{Im}\operatorname{Tr} U_{p_n(x)} = a^d Z_d(a)\mathcal{O}_n(x) + o(a^d Z_d(a)) \tag{102}$$

in matrix elements against a fixed normalized state, and suppose the corresponding continuum matrix element is finite. Then

$$\left| \langle \Psi, \widetilde{O}_{a,n}\Omega \rangle \right|^2 = O\Big(a^{2d-3}|Z_d(a)|^2\Big). \tag{103}$$

For the charge-odd plaquette source, tracelessness and

$$\operatorname{Im}\operatorname{Tr} e^{iX} = -\frac{1}{6}\operatorname{Tr} X^3 + O(\|X\|^5) \tag{104}$$

give $d = 6$ and $\mathcal{O}_n \propto d^{abc}B_n^a B_n^b B_n^c$. Thus this state's absolute spectral-atom weight for the unrenormalized, volume-normalized source has the power $a^9|Z_6(a)|^2$. If Z_6 contains only the usual logarithmic matching factors, that absolute weight vanishes as $a \downarrow 0$.

Proof. At fixed physical volume, $L(a)^3 \asymp a^{-3}$. Replacing the lattice sum by its Riemann sum in (102) gives

$$L(a)^{-3/2} \sum_x a^d Z_d(a)\mathcal{O}_n(x) = a^{d-3/2} Z_d(a), \frac{1}{\sqrt{V}} \int_V \mathcal{O}_n(y)\, d^3y + o(a^{d-3/2} Z_d(a)).$$

Squaring proves (103). For a plaquette, $X = a^2 g_H F_{ij} + O(a^3)$; the linear term in $\operatorname{Tr} e^{iX}$ vanishes and the first imaginary term is cubic, which proves (104) and $d = 6$. □

This conditional result is an absolute-weight power count for the volume-normalized operator $\widetilde{O}_{a,n}$ above; it is not a statement about the Hilbert-normalized spectral fraction. At fixed physical Euclidean time the same overall operator normalization multiplies the bottom atom and the full correlator, so it cancels from their ratio. Consequently the factor $a^9|Z_6(a)|^2$ alone proves neither collapse nor survival of a normalized residue. Smearing over a radius fixed in physical units changes the operator and its matching factors; a uniform lower bound for the renormalized source fraction still requires independent spectral-measure and matching control. Moreover, the three diagonal tensors $d^{abc}B_n^aB_n^bB_n^c$ transform as T_1 under the cubic group but belong to a symmetric rank-three continuum tensor; the label T_1^{+-} alone does not distinguish continuum spin one from spin three. Auxiliary A_2^{+-} and T_2^{+-} operators are required for that degeneracy test. This is the standard cubic-subduction partner method used in continuum-spin assignments [8, 9].

Proposition 66 (Passage of a source-visible spectral atom)**.** *Let μ_n be positive locally finite Borel measures on $[0,\infty)$, and suppose*

$$M_n = \inf \operatorname{supp} \mu_n \in [m_-, m_+], \qquad \mu_n(\{M_n\}) \geq z_* > 0.$$

Assume that the source spectral measures converge vaguely, $\mu_n \Rightarrow \mu$. After passage to any subsequence on which $M_n \to M$, one has

$$\operatorname{supp} \mu \subset [M, \infty), \qquad \mu(\{M\}) \geq z_*. \tag{105}$$

Thus a uniformly source-visible bottom atom survives as a nonzero atom of the limiting spectral measure; no isolation annulus, spin assignment, or Haag–Ruelle hypothesis is needed for this weaker conclusion.

Proof. Compactness of $[m_-, m_+]$ supplies the stated subsequence. If $K \Subset [0, M)$, then $K \cap \operatorname{supp} \mu_n = \varnothing$ for all sufficiently large n, and vague convergence gives $\mu(K) = 0$. Hence the limiting support is contained in $[M, \infty)$.

For $\varepsilon > 0$, choose $f_\varepsilon \in C_c([0,\infty))$ with $0 \leq f_\varepsilon \leq 1$, equal to one on $[M - \varepsilon/2, M + \varepsilon/2] \cap [0,\infty)$ and supported in $[M-\varepsilon, M+\varepsilon] \cap [0,\infty)$. Eventually $f_\varepsilon(M_n) = 1$, so positivity and vague convergence give

$$\int f_\varepsilon \, d\mu = \lim_n \int f_\varepsilon \, d\mu_n \geq z_*.$$

Consequently $\mu([M-\varepsilon, M+\varepsilon] \cap [0,\infty)) \geq z_*$. Continuity from above as $\varepsilon \downarrow 0$ proves $\mu(\{M\}) \geq z_*$. ☐

For a positive transfer Hamiltonian and a charge-odd source, write

$$C_n(t) = \int e^{-tE} \, d\mu_n(E), \qquad Z_n = \mu_n(\{M_n\}).$$

Osterwalder–Schrader convergence [4] strong enough to imply vague convergence of these source measures therefore turns a uniform lower bound on Z_n and a two-sided physical bound on M_n into the nonzero limiting atom in Proposition 66. This is an exact reduction, not a proof of either uniform bound or of the required OS convergence.

Lemma 67 (What a finite-time correlator can certify)**.** *Let a finite positive measure have the form*

$$\mu = Z\delta_M + \nu, \qquad \operatorname{supp} \nu \subset [M + \Delta, \infty), \qquad \Delta > 0,$$

and put $C(t) = \int e^{-tE} \, d\mu(E)$ and $F(t) = e^{Mt} C(t)$. Then

$$Z \leq F(t), \qquad Z \geq \frac{F(t) - e^{-\Delta t} C(0)}{1 - e^{-\Delta t}}. \tag{106}$$

The second inequality may be replaced by zero when its right-hand side is negative.

Proof. Positivity gives $F(t) \geq Z$. The spectral gap gives

$$F(t) = Z + \int e^{-t(E-M)}\, d\nu(E) \leq Z + e^{-\Delta t}(C(0) - Z),$$

and rearrangement proves the lower bound. $\square$

Without a gap above the atom, every finite-t value of $F(t)$ is only an upper bound on Z; an absolutely continuous threshold can imitate a mass edge at finite time while having $Z = 0$. Lemma 67 therefore explains exactly where a uniform external gap can enter a nonperturbative residue proof, while Proposition 66 shows that isolation is not part of the final atom-passage statement itself.

Proposition 68 (Conditional OS gap passage)**.** *Let q_n be nonnegative closed forms of physical transfer Hamiltonians H_n, with vacuum projections P_n, after isometric embedding in a common Hilbert space. Suppose q_n Mosco-converges to the form q of an OS-reconstructed Hamiltonian H, $P_n \to P$ strongly,* $\operatorname{ran} P = \ker H$, *and for one $m > 0$ independent of n,*

$$q_n[\psi] \geq m\|(1 - P_n)\psi\|^2 \quad (\psi \in D(q_n)).$$

Then

$$q[\psi] \geq m\|(1 - P)\psi\|^2, \qquad \operatorname{spec}(H) \subset \{0\} \cup [m, \infty).$$

Proof. For $\psi \in D(q)$ choose a Mosco recovery sequence $\psi_n \to \psi$. The recovery and liminf properties give $q_n[\psi_n] \to q[\psi]$. Strong convergence of the projections gives $(1 - P_n)\psi_n \to (1 - P)\psi$. Passing to the limit proves the form bound; the spectral inclusion follows from the spectral theorem. $\square$

Proposition 69 (Deterministic reflection-positive pushforward)**.** *Let $(X, \mu, \theta; \mathcal{A}_+)$ be a reflected probability space with*

$$(\Theta F)(U) = \overline{F(\theta U)}, \qquad \int_X (\Theta F) F \, d\mu \geq 0 \quad (F \in \mathcal{A}_+).$$

Let $(X', \theta'; \mathcal{A}'_+)$ be a coarse reflected configuration space, and let $P : X \to X'$ be measurable. If

$$P\theta = \theta' P \quad \mu\text{-a.e.}, \qquad P^*\mathcal{A}'_+ \subset \mathcal{A}_+, \tag{107}$$

then the pushforward $P_\# \mu$ is reflection positive on $\mathcal{A}'_+$. On the gauge-invariant algebra the first equality may hold only modulo a coarse gauge transformation.

Proof. For $F' \in \mathcal{A}'_+$ put $F = F' \circ P$. Then

$$\int_{X'} (\Theta' F') F' \, d(P_\# \mu) = \int_X \overline{F'(\theta' P(U))} F'(P(U)) \, d\mu(U) = \int_X (\Theta F) F \, d\mu \geq 0.$$

If equivariance holds modulo a coarse gauge transformation, gauge invariance of F' gives the same middle equality. $\square$

Corollary 70 (Reflection-adapted Balaban-form blocking)**.** *Let the Wilson measure live on a periodic lattice of even temporal extent and be reflection positive on the gauge-invariant positive-time algebra. Let an odd block factor L divide that extent. Use centered L^d blocks, put the reflection plane between two adjacent layers of block centres, and choose the reference trees and Balaban paths in reflection pairs, with reversed coarse bonds defined by group inverse. Then the raw deterministic Balaban-form average of Eq. (15) in Ref. [3] pushes the Wilson measure to a reflection-positive coarse measure.*

Proof. Choose paths so that

$$\theta\Gamma_{c,x} = \Gamma_{\theta' c,\theta x}, \qquad \Gamma_{c^{-1},x'} = \Gamma_{c,x}^{-1}.$$

Holonomy is functorial under reflection, while logarithm and exponential commute with unitary conjugation. Reindexing the equal-weight average by $x \mapsto \theta x$ proves $P(\theta U) = \theta' P(U)$ off the principal-log branch set, which is Haar-null; Balaban's Eq. (11) supplies gauge covariance. For a positive coarse bond, both endpoint blocks lie wholly in the fine positive half. The locality statement following Eq. (43) of Ref. [3] therefore gives $P^*\mathcal{A}'_+ \subset \mathcal{A}_+$. Proposition 69 applies, using Wilson reflection positivity from Ref. [4]. □

The corollary proves preservation for the raw deterministic average with reflection-adapted block and path data. It does not cover Balaban's literal lower-corner convention, field-dependent gauge fixing, an asymmetric large/small-field partition, a softened stochastic constraint, or a complete multistep effective action. For the literal convention the finite-volume geometry gate is now exactly

$$P_{\mathrm{CMP98}}(\theta U) = g_\theta(U) \cdot \theta' P_{\mathrm{CMP98}}(U), \qquad \operatorname{supp} P_c \subset \Lambda_+ \quad (c \in E'_+), \tag{108}$$

where g_θ is the coarse gauge change induced by moving the reflected upper-corner basepoint back to the chosen lower corner. Block alignment by itself does not prove Eq. (108).

Remark 71 (Equivariant Markov blocking is insufficient). Let the fine space be one point. On the coarse space $X' = \{(s_-, s_+) : s_\pm = \pm 1\}$ with reflection $(s_-, s_+) \mapsto (s_+, s_-)$, take the reflection-invariant law

$$\nu = \tfrac{1}{2}\delta_{(+1,-1)} + \tfrac{1}{2}\delta_{(-1,+1)}.$$

The resulting Markov kernel is reflection equivariant and maps every positive-time observable to a fine constant. Nevertheless, for $F(s_-, s_+) = s_+$,

$$\int (\Theta' F) F \, d\nu = \mathbb{E}_\nu[s_- s_+] = -1.$$

Thus equivariance and one-sided mapping do not suffice for a stochastic kernel. A sufficient replacement is a reflection factorisation: there is a one-sided $T : \mathcal{A}'_+ \to \mathcal{A}_+$ such that

$$K\big((\Theta' F)G\big) = (\Theta T F)(TG) \qquad (F, G \in \mathcal{A}'_+).$$

The hypotheses of Proposition 68 are not supplied here. Theorem 3 controls a finite-volume cluster in a fixed-cutoff small-u domain, not the whole physical transfer Hamiltonian along $a \downarrow 0$. The separate local weak-well expansion in the upstream corpus also has no proved operator identity relating its β_{loc} to u. The exact next G19 object is therefore a Hamiltonian–Wilson matching package: the physical energy prefactor, anisotropy/speed-of-light renormalisation, g_H–g_W scheme map, and a renormalised spectral measure that can be followed to large u. On a Balaban route this package must also close Eq. (108) for the literal path convention and preserve reflection positivity through every later gauge-fixing and large/small-field operation.

12 External comparisons

External results count here because they are *independent* — produced without knowledge of this program — not because they are published. We use them only within the scope stated below; all normalisation conversions and coefficient matches are calculations of this paper.

Hamer 1989, both sectors. Let a_n denote the coefficient of x^n in Hamer's tabulated dimensionless mass series. The 1^{+-} rest-frame series agrees with E_{flat} at every shared order: $16/3 \div 2 = 8/3$ exactly, $a_1 = +1$, and $2a_2$, $4a_3$ matching $11/306$ and d_3 to 9.9×10^{-14} and 1.1×10^{-12}. Conditional on the unsigned-symbol identification, the 0^{++} series also matches the charge-even A_1^{++} Γ-point coefficients at x^2 and x^3, with the sign structure ($a_1 = -1$ in the scalar channel where the carrier's is $+1$) reproduced; the E^{++} doublet is not constrained by it, which is why Conditional result 35 uses the splitting rather than the level. The conversion $m_n = 2^{n-1}a_n$ is $x = 2u$ together with Hamer's normalisation $ma = (g^2/2)M$ — the 2^n from the substitution, the half from the normalisation — exactly, with no 4^r ambiguity anywhere. Hamer states in print that his x^3 and x^4 terms disagree with the earlier series of [13]. We report Hamer's comparison and do not independently adjudicate the earlier coefficient table.

The string tension. Let t_n denote the coefficient sequence in the transcribed Kogut–Pearson–Shigemitsu $SU(3)$ table [16] — read here from the 1980 preprint, the journal pages themselves being unread — and let σ_n denote the corresponding ledger coefficient after the displayed convention conversion. The transcription satisfies exactly $2^{n-1}t_n = \sigma_n$ for $n = 2, \ldots, 5$, in the physical sign convention $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$. The same series reproduces E_{flat} term by term through u^3, including the discriminating u^3 coefficient.

Euclidean series. Historical Euclidean strong-coupling series by Münster, Seo and Smit [15, 17, 18, 19, 20] provide context only. Two printings of the same Münster/Seo series — the second erratum to [15], which prints the definitive corrected table and credits Ukawa and Seo [17] with locating the error, and Smit's Table 1 scalar column [18], which credits them — agree rational for rational through u^8. The accompanying corpus also contains an apparent eighth-order discrepancy between the 1982 erratum and the 1985 table. Because every original table page has not been independently re-transcribed for this paper, neither the apparent equality nor the discrepancy is used as evidence. No Hamiltonian coefficient is inferred from these Euclidean records.

The overlap obstruction. Schierholz [22] measured in 1988 that at $\beta = 5.9$ — in the Euclidean scaling region, not at strong coupling — a bare local operator carries only a couple of percent of the state it interpolates, and gave the scaling a^5 for a local dimension-four operator. Together with Brandstaeter, Kronfeld and Schierholz [24] these document the degradation of local-operator projection and its critical signal-to-noise scaling as the lattice spacing is reduced. Nothing quantitative is transferred to the present strong-coupling finite-order setting. Read only as a warning, these results motivate posing `G18` in a smeared basis; they supply no bound and do not identify the carrier with a physical glueball.

Cross-regime discipline. Two indexed papers declare an explicit scope firewall and neither supplies a value here: Davoudi–Raychowdhury–Shaw [32], whose Hilbert-space cost comparison is locked to 1+1 dimensions with matter, and Kadam–Raychowdhury–Stryker [33]. The rule is enforced by the literature validator rather than recorded in prose.

13 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. Two of those exclusions are now sharper than a disclaimer. Section 10 *does* reach infinite volume — at one fixed lattice spacing — so what the first item excludes is the continuum, not the thermodynamic limit. And the continuum exclusion is a theorem rather than a caution: by Proposition 62 the hypotheses of Theorem 3 and those of an asymptotically free trajectory are eventually *disjoint* in the coupling coordinate $u = g_H^{-4}$, so no tail of a continuum trajectory lies in the proved domain and evaluating the existential constants c_1, c_2 would not change that. Corollary 64 adds that no nonzero finite strong-coupling truncation yields a finite continuum mass at all. Section 11 names what would be required instead. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-}; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L. Under the third-order factorisation input, the minimum internal separation of one sheet from the incidence sheets is $\Delta_L(u) = 4\tau(u)\sin^2(\pi/L) + O(u^4)$, which falls as L^{-2}; it cannot provide a volume-uniform contour for that single sheet over the entire Brillouin zone. This statement must not be conflated with external isolation of the complete electric shell. Theorem 1 gives the latter a volume-independent free margin $C_F/2$, while Proposition 11 gives an L-independent internal coefficient at every fixed nonzero nested momentum.

Three crossings are named explicitly, and none of them is made here.

Proposition 72 (Exact reduction of the `G17` free-energy target)**.** *For the Euclidean $SU(3)$ Wilson plaquette density $W_p = 1 - \frac{1}{3}\operatorname{Re}\operatorname{Tr} U_p$, let*

$$Z_\Lambda(b) = \int \exp\Big(-\sum_p b_p W_p\Big) d\mu, \qquad b_p(s) = \begin{cases} s, & p \in \Gamma, \\ \beta, & p \notin \Gamma, \end{cases}$$

and write $Z_{\beta,\alpha,\Gamma} = Z_\Lambda(b(\alpha\beta))$ and $Z_\beta = Z_\Lambda(b(\beta))$. Then

$$\log \frac{Z_{\beta,\alpha,\Gamma}}{Z_\beta} = \int_{\alpha\beta}^{\beta} \sum_{p\in\Gamma} \langle W_p \rangle_{s,\Gamma}\, ds. \tag{109}$$

Consequently the single local estimate

$$\sup_{\Lambda,\Gamma,\, p\in\Gamma,\, s\in[\alpha\beta,\beta]} s\,\langle W_p\rangle_{s,\Gamma} \;\le\; C \tag{110}$$

would imply the desired animal-uniform bound with

$$\frac{Z_{\beta,\alpha,\Gamma}}{Z_\beta} \le K_\alpha^{|\Gamma|}, \qquad \log K_\alpha \le C \log(1/\alpha). \tag{111}$$

Proof. Differentiating $\log Z_\Lambda(b(s))$ gives $-\sum_{p\in\Gamma}\langle W_p\rangle_{s,\Gamma}$ and integration from β down to $\alpha\beta$ proves (109). Under (110), each summand is at most C/s, which proves (111). $\square$

The elementary pointwise estimate $0 \le W_p \le 3/2$ gives only $\log K_\alpha \le \frac{3}{2}(1-\alpha)\beta$; its growth in β is precisely why the uniform C/s estimate remains a substantive open problem.

- **Finite lattice to infinite volume** requires an inhomogeneous Wilson free-energy bound with a useful $\log K_\alpha$ and a source-radius reduction. Proposition 72 turns the first item into the explicit one-plaquette estimate (110), but does not prove it; the source-radius theorem is also still unstated. Both are carried in the accompanying gap register as `G17`. The Hamiltonian Riesz island of Theorem 3 belongs to small u and does not prove this Euclidean weak-coupling bound.

- **Protected level to particle** requires a volume-uniform overlap of a smeared, dressed T_1^{+-} operator built on the carrier with the transfer-matrix spectrum, and multi-plaquette survival of the protected level (`G18`). The published overlap measurement of Section 12 is why this theorem must live in the smeared basis. This edition now settles the free external gap, the complete finite-volume interacting Riesz island at existentially small $|u|$, the fixed-k carrier, and the exact price of combining locality with sharp momentum. It does not prove a volume-uniform Wilson-source frame, infinite-volume totality, or the transfer map.
- **Lattice to continuum** (`G19`) is untouched. Small u is strong coupling; the continuum is the opposite end of the axis.

All three crossings are required and none is completed here. The new window and Riesz-island theorems remove the free-spectrum obstruction and its small-u finite-volume persistence subproblem from `G18`, but they do not promote the protected lattice cluster to an infinite-volume physical particle.

The two former charge-odd completeness premises are now theorems. Theorem 1 proves more than shell completeness: below $\frac{5}{2}C_F$ the full trivial-flux electric spectrum is exactly $\{0, 2C_F\}$ for every $N \geq 3$ and $L \geq 3$, and the range of P_- is exactly the charge-odd one-plaquette span. Lemma 17 proves that the only off-diagonal second-order routes are the four adjacent shared-link channels: disjoint exchange routes are orthogonal and the same-face route cancels. The explicit projector matrix element (36) then fixes their signs and normalisations. These statements are analytic results; the finite-volume $L = 3, 4, 5$ enumerations in the verifier are regressions, not substitutes for their proofs. They are new in this edition and have not yet faced external referees; a reader auditing this paper should start there.

One earlier physical identification remains conditional.

That the complete linked charge-even Bloch symbol is the unsigned incidence $\mathcal{N}(k)$ — the signed incidence with the boundary orientations dropped — is not derived. The cubic and range in Section 6 are exact algebraic properties of the defined comparison operator; only their physical charge-even interpretation, bandwidth and Γ-splitting rest on this additional hypothesis.

14 Evidence, authority, and reproducibility

The accompanying dependency-free program `verify_publication_core.py` uses exact integers and `Fraction` arithmetic. Its thirty-eight checks are the coefficient ledger and Weingarten sums, signed incidence and torus homology, closed-cube shell, two-cube connected geometry and reported channel arithmetic, the unsigned cubic, the saved historical Hodge pencil, the fourth-order obstruction, the coupling convention, the planar closed form (38), and the exhaustive shared-link geometry, cycle censuses and torus ranks through $L = 5$ — every value a `Fraction` and no float constructed anywhere in the file.

It does *not* cover the arithmetic new in this edition: the all-rank Casimir shelf of Theorem 1, the $SU(3)$ additive-spectrum and Riesz-contour arithmetic of Theorem 3, the cycle censuses those proofs delegate to, the fixed-momentum carrier, the harmonic representatives and the sharp zone minimum. Those are checked in the accompanying repository instead, and each carries its own printed marker; the boundary is stated rather than blurred, because a standalone verifier that is described as covering more than it runs is worse than one with a stated edge. A second program, `verify_radius2_report.py`, holds everything the sealed radius-two artifact of Section 5.4 can support: it is also standard-library only (the payload is read with `zipfile` and `struct`, and the 11×11 block is diagonalised by cyclic Jacobi), but it works in floats at stated tolerances, and the two files are kept separate precisely so that the exact one stays float-free. Neither reconstructs the finite-precision Clebsch–Gordan inputs, replays the third-order microscopic contractions, or chooses C_{shp}.

Every sentence of this paper that a machine check certifies carries an inline `\chk` marker in the source; the markers are extracted verbatim into Appendix G, so the coverage claim is itself auditable rather than asserted. In the accompanying repository `workhouse verify -only '<name>'` reruns any one of them alone, with its numbers, in about a second.

That device is itself checked. A registered invariant reads every pinned edition in `paper/` — this one included — resolves every printed label against the live registry, and fails if one is renamed, deleted, or left red. The guard is the reason this edition's labels can be trusted at all: the same invariant read only the earliest pinned edition until this one landed, and nineteen labels that had drifted in the intervening editions passed unnoticed until it was made to read them all.

A few displayed objects deliberately carry no marker, because none is a result of this program: the dimension and Casimir table (25), which is standard $SU(N)$ representation theory; the boundary formulas of Appendix E, which are definitions — what is checked there is that they compose to zero; and Table 1, which is provenance about a commit rather than a claim about the physics.

Layer	Count
T0 — Lean 4 theorems, no `sorry`	37
T1 — exact re-derivations	191
T2 — numerical, within a stated tolerance	48
Repository tests	484

Table 1: Counters at the pinned commit. The T0 row counts theorem declarations with no `sorry`, which is a scrape of the tree; `make lean` is what compiles them, and the standard the tree is held to is that every axiom footprint lies in {`propext`, `Classical.choice`, `Quot.sound`}. That compilation was last run elsewhere: the environment this edition was typeset in carries no Lean toolchain, and the row is not asserted as remeasured here. The T1 and T2 rows are `make verify` and the test row is `make check`. They are provenance rather than independent evidence; the argument is above, and the checks are the checks.

A smaller layer is proof-checked rather than merely re-derived. The Lean development compiles the rank law with denominators cleared, $t_3 = 5/612$ and $t_2 = 0$, the third-order ledger identity, the axial law at every exceptional rank, the cycle count $(L^3 - 1) + 3 = L^3 + 2$, the four checkpoint-extraction formulas and the deltas they invert, and four statements about the charge-even cubic including the identity that its middle coefficient is p^2. It is exact-rational and polynomial algebra, plus the trigonometric checkpoint evaluations over $\mathbb{R}$ that supply those deltas — the $\mathbb{R}$ section exists because the extraction theorems formalised only half a statement without it. There is no perturbative derivation and no operator theory. An earlier edition added that "nothing that bears on C_{shp} appears there". That was false, and false about the one genuinely open item here. The tree carries $C_{\text{shp}}^{\text{old}}$ as a rational constant, proves it from the pencil coefficients ($C = (\beta - 2\alpha_3)/16$), and proof-checks the four checkpoint-extraction formulas of Section 9 — which is precisely the C_{shp} algebra. What remains true is the weaker and duller statement: nothing in the Lean tree *decides* between the two recorded values, and by Theorem 41 nothing stated on Γ or an axis could.

Claim	Status	Evidence in this edition
Signed incidence spectrum and L^3+2 carrier	Proven	Analytic integer topology; exact replay
Adjacent all-rank coefficient t_N	Proven locally	Weingarten derivation; exact replay
Global order-u^2 torus assembly	Proven	Retained-shell theorem; process-completeness lemma; explicit projector matrix element
Uniform first electric spectral window	Proven	Spin-network support census; all-rank Casimir shelf; trivial one-form flux
$SU(3)$ all-orders finite-volume Riesz island	Proven, existential domain	Yarotsky enclosure; exact additive spectrum; periodic bounded-degree transport
Two-cube six-channel census	Reported numerical	Rational reconstructions; shared amplitude lineage; assembly test
Unsigned-incidence cubic and range	Proven algebraically	Exact characteristic polynomial and finite-grid replay
Physical charge-even identification	Conditional	Complete linked symbol, including vacuum route, not derived
$SU(3)$ factorisation through u^3	Upstream cold-exact	251/251 exact gates reported by the frozen authority; raw replay not bundled here
Historical fourth-order pencil	Exact for saved kernel	Hash-pinned payload and cold fixed-kernel checks
Complete physical fourth-order kernel	Disputed/open	Historical and August candidates differ off axis
Radius-two rational sub-block and C_4/C_5 increments	Reported numerical	Sealed hash-pinned artifact; stdlib float verifier at stated tolerances; detached replay of pinned parents; no odd Q_3 enclosure
Exact `G17` reduction	Proven reduction; estimate open	Differentiated inhomogeneous partition function; missing uniform one-plaquette bound
Infinite-volume particle or continuum gap	Open	No thermodynamic Wilson frame, transfer identification, or continuum bridge

Table 2: Truth status and evidence level are separate. "Exact for a saved kernel" does not establish that kernel's physical identification.

14.1 Claim-level evidence map

14.2 Upstream corpus and downstream verifier

The organised `ALL THEORY` archive was consulted as the upstream scientific corpus. Its four-document v4.3 authority stack is frozen by byte count and SHA-256 in `export/MAN_GOV_export_manifest.csv`; all four entries were rehashed for this edition and matched. The downstream verifier is WORKHOUSE. Document flow is one-way—frozen corpus to hash manifest to verifier—while verifier findings return to review and status records. The later two-cube results are not in the August 20 corpus and are therefore kept as later WORKHOUSE-local evidence.

The current hostile-scope audit branch is pinned at `ff9a5976a5b03cd763e3e50daf4326138770460d`. On Windows, its registry was rerun locally: 229/230 registered checks passed. The sole failure was the known path-separator comparison for a restored payload; the payload hash and all paper-relevant arithmetic checks passed. This is a platform-specific local audit result, not a claim that the complete test suite or Lean tree was rerun. The source-level `\chk` labels remain in this TeX file but are suppressed typographically; this revised edition is not claimed to be covered by the repository's older manuscript-label guard.

The authority files are identified by the following SHA-256 values:

- master v4.3: `1f8295c3a797aeb854b4b0ca82485bd7f4a149617777d2a43fc147a4a53c26dd`;
- detailed formulas v3.1: `924d5b26f865e33f530d166735e847b89f60df7fe363995e2788ab24e1ff4254`;
- consolidation guide v4.3: `60b8a93c8149049f50ed3ce8e0cfd1b379723b6765f201395a2fda6c4a7e7aa7`;
- canonical source manifest v4.3: `730bc99ed8d749de0dffbabfda85cd4512d8f7748782620b694759a03bacde05`.

14.3 Data and code availability

This release directory contains the TeX source, generated vector figures, figure generator, exact verifier, build log, rendered-page audit, and a SHA-256 manifest. The historical kernel payload used in Reported result 40 has SHA-256 `635d40fa8a5d7da841fd30f36185eb96f14ec4c040678ddd8fb010379afb2900`. The pinned `ymcirc` $d = 3$, $\mathcal{B} = 4$ data blob is identified in Ref. [38] and has SHA-256 `4cad9538f2c052488556362e0cf6e4e63363e7f5801597aeab9013ef945c59dc`. The exact stranded-flux audit source and result have SHA-256 `a42800a91809a77bbd1d339ac604ea80a234eed305377bbabbf890fee98fde55` and `d215fa1e9cd0bde0410eec16c687182c30bc62fa3f28c1445fd81e93888e1d60`. The sealed radius-two spectrum of Section 5.4 and its certificate are bundled, with SHA-256 `52a17f87d3990b234635828a0c92f73f781e9a3abec8bb3c127e990b7a717e6c` and `b9b14d28438585fc4f7d88f96497e9c6e01d0532a8402ffd6db17258a1638c0e`; `verify_radius2_report.py` refuses to run against any other bytes. The upstream release governs the wider supplement through `two_cube_cutoff_free_radius2_finite_u_spectrum_manifest.json`, which pins twelve members — derivation note, builder, hostile tests, six sealed-parent files, result NPZ, certificate and detached verifier — of which only the two named above travelled with this edition. The two-cube build and its finite-precision Clebsch–Gordan payloads are not reproduced by the compact verifier; the exact statements that depend on them are labelled reported. A durable tag or archival DOI is still needed before journal submission.

Machine checks certify algebra relative to their definitions and inputs. They do not replace derivations, promote numerical contractions to symbolic ones, or certify an uncompleted off-axis contraction.

15 Conclusion

The strongest exact coefficient result is a finite-order operator theorem built on a volume-uniform free spectral window: for every $N \geq 3$ and periodic $L \geq 3$, the trivial-flux electric spectrum below $\frac{5}{2}C_F$ is exactly $\{0, 2C_F\}$, the retained-shell theorem identifies the full charge-odd $2C_F$ eigenspace, the process-completeness lemma exhausts its second-order off-diagonal routes, and the four shared-link fusion projectors give the closed coefficient t_N with the oriented cellular boundary fixing its sign and geometry. This assembles *without* a completeness hypothesis into a homological flat carrier of dimension $L^3 + 2$: the flat sector is not a fine-tuned interference band but the cycle space of the lattice two-complex, degenerate because $\partial_2\partial_3 = 0$. Those proofs are the newest and least-reviewed material here, and they deserve the reader's hardest look.

The apparent L^{-2} obstruction is now localised: it is the minimum internal sheet separation near Γ, whereas the free external shell margin is at least $C_F/2$ and the coefficient at every fixed nonzero nested momentum is volume-independent. The $SU(3)$ two-cube calculation supplies a nontrivial operator-level assembly check, including channel-by-channel geometry and orbit localisation, but shares its amplitude lineage and remains numerical at the contraction layer.

For $SU(3)$ the analysis goes one step beyond finite order. Yarotsky's local spectral enclosure, together with the exact additive Casimir arithmetic and a periodic bounded-degree transport, gives an all-orders rank-$3L^3$ Riesz island for the complete neutral charge-odd cluster whenever $27|u| < c_1$ and $27c_2|u| < 1/33$. This is a volume-uniform finite-volume statement with existential constants. It neither isolates one internal sheet nor proves a source-carrying thermodynamic particle. A later detached-replayed radius-two extension adds nested finite-u Ritz spectra and leading incremental order-u^4 and order-u^5 effects; because the odd exterior action is absent, it supplies neither a matched truncation error nor a complete fourth- or fifth-order physical coefficient.

The unsigned-incidence cubic is exact as an algebraic comparison; its physical charge-even identification is not assumed silently. At fourth order the saved historical kernel has an exact positive generalized Hodge pencil and stencil, yet the later candidate differs in an off-axis coefficient that no scalar reanchoring, Γ datum, or axial datum can determine. The decisive next object is therefore one target-blind, hash-sealed calculation that returns the rest scalar and off-axis shape from the same kernel. Independently, promoting the finite-volume Riesz island to a thermodynamic, source-carrying multiplet with a volume-uniform Wilson frame and a controlled transfer map is now the precise fixed-spacing `G18` target, and Proposition 72 isolates `G17`'s first missing estimate without pretending to prove it. Until those steps are supplied, the operator and Riesz-island theorems are substantial, but the physical fourth-order kernel, infinite-volume particle interpretation, and continuum limit remain open.

A Channel weights in closed form

Substituting (25) into (32),

$$w_{\mathbf{1}} = -\frac{2}{N(N^2-1)}, \qquad w_{\Lambda^2 F} = -\frac{N-1}{(N+1)(2N-3)},$$
$$w_{\mathrm{Adj}} = -\frac{2(N^2-1)}{N(2N^2-1)}, \qquad w_{\mathrm{Sym}^2 F} = -\frac{N+1}{(N-1)(2N+3)}.$$

Their family sums are (33) and (34), and the difference has numerator $2N(N^2-4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once. At $N = 3$: $w_{\mathbf{1}} = -1/12$, $w_{\mathrm{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$, $w_{\mathrm{Sym}^2 F} = -2/9$, which are the four values Reported result 22 matches against (42).

B The Weingarten index sums

With $\mathrm{Wg}(e) = 1/(N^2-1)$ and $\mathrm{Wg}((12)) = -1/(N(N^2-1))$, and σ, τ running over S_2,

$$\begin{aligned} &\langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle \\ &\qquad = \sum_{\sigma,\tau} \mathrm{Wg}(\sigma\tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}. \end{aligned}$$

Summing the delta products over all N^4 index quadruples,

$$M_{\mathrm{direct}} = \mathrm{Wg}(e)(N^4 + N^2) + 2\,\mathrm{Wg}((12))N^3 = N^2,$$

$$M_{\mathrm{cross}} = 2\,\mathrm{Wg}(e)N^3 + \mathrm{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$. The related fourth moment $\int |U_{11}|^4\, dU = 2/[N(N+1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2, 2)$ sector.

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

The u^1 row is $SU(3)$-only rather than an $SU(3)$ specialisation: it is $-\langle p, -|V|p, -\rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. Of the rest, the u^0 row and the u^2 $BB^\dagger$ coefficient specialise all-rank statements ($2C_F$ and t_N); the u^2 flat scalar inherits the $SU(3)$ tower term through (46); and the three u^3 rows are $SU(3)$-only and conditional on the supplied domino ledger, exactly as Section 7 says.

D Unsigned comparison and conditional charge-even ledger

Conditional on identifying the unsigned-incidence comparison with the complete linked charge-even symbol, the same convention, with $\lambda \in \{12, -4\}$ and (46), gives:

Order	Quantity	Value
u^2	hopping $t_{2,+} = \ell_3$	$-11/306$
u^2	on-site diagonal	$223/1020$
u^2	band bottom ($\lambda = 12$)	$-217/1020$
u^2	band top ($\lambda = -4$)	$1109/3060$
u^2	bandwidth $16\|t_+\|$	$88/153$
u^3	hopping $t_{3,+}$	$-6335/249696$
u^3	on-site diagonal ($\lambda = 0$)	$52163/260100$
u^3	band bottom ($\lambda = 12$)	$-54049/520200$
u^3	band top ($\lambda = -4$)	$471353/1560600$

Because $t_{r,+} < 0$ at both orders, the $\lambda = -4$ endpoint is the band *maximum* in this sector. A certificate key naming it `bandmin` at both orders is a naming error against its own arithmetic, recorded rather than silently corrected.

E Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$
\begin{aligned}
\partial_3[x;1,2,3] = [x+e_1;2,3] &- [x;2,3] \\
&- [x+e_2;1,3] + [x;1,3] \\
&+ [x+e_3;1,2] - [x;1,2].
\end{aligned}
$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2\partial_3 = 0$ over $\mathbb{Z}$. On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\operatorname{rank} \partial_3 = L^3 - 1$.

F The two-cube geometry

The eleven oriented plaquettes of the $(3,2,2)$ prism, by base vertex and ordered plane axes, are

$$
\begin{aligned}
&(000;12),\ (000;13),\ (000;23),\ (001;12),\\
&(010;13),\ (100;12),\ (100;13),\ (100;23),\\
&(101;12),\ (110;13),\ (200;23),
\end{aligned}
$$

with left, right and shared-face index sets $I_L = (0,1,2,3,4,7)$, $I_R = (5,6,7,8,9,10)$, $I_F = (7)$. Building $G = BB^{\mathsf{T}}$ from the oriented boundaries and applying the geometric Möbius fold (40) leaves exactly four nonzero off-diagonal pairs, $(0,5),(1,6),(3,8),(4,9)$, each equal to -1: four disjoint 2×2 blocks and three isolated directions, which is why the spectra in Section 5 are closed form once the diagonals below are supplied. The two connected diagonals are

$$
\begin{aligned}
D_{\mathcal{B}=6} = \tfrac{1}{612}\,\mathrm{diag}(&-2317,-2317,-2295,-2317,-2317,\\
&-2317,-2317,0,-2317,-2317,-2295),\\
D_{\mathcal{B}=4} = \mathrm{diag}(&-\tfrac{7}{4},-\tfrac{7}{4},-\tfrac{15}{4},-\tfrac{7}{4},-\tfrac{7}{4},\\
&-\tfrac{7}{4},-\tfrac{7}{4},0,-\tfrac{7}{4},-\tfrac{7}{4},-\tfrac{15}{4}),
\end{aligned}
$$

so both spectra are one line of arithmetic: a pair block contributes $D_{ii} \mp c$ twice and each isolated direction its own D_{ii}. The $\mathcal{B}=4$ comparator on the same geometry gives 0 once, $-\frac{15}{4}$ twice, $-\frac{11}{6}$ four times, and $-\frac{5}{3}$ four times: the paired levels sit *above* the isolated $-15/4$ pair, where at $\mathcal{B}=6$ they sit below it. The three isolated levels $\{-\frac{15}{4},-\frac{15}{4},0\}$ are identical in the two truncations, so the entire effect of the cutoff lives in the four cross-cell blocks. That reversal is not the coefficient's sign by itself — within a 2×2 block the pair splits as $D \mp c$, which is invariant under $c \to -c$ — but the truncation's effect on the connected diagonal as well. What the sign of c alone does reverse is the ordering on a *single* cube, where the kernel is $\alpha I + tG$ with one diagonal and the spectrum of G fixed; that is the shell reversal the one-cube reconstructions report.

G Machine-check coverage

The paper's discipline is that a machine check, not a sentence, is the unit of evidence. This appendix makes that auditable: every inline `\chk` marker in the source — each one naming the check that certifies the sentence it sits beside — is extracted verbatim by `make_coverage.py` and printed here. A reader can therefore see exactly which statements are covered, and, by their absence from this list, which are not: the remaining load-bearing hypothesis of Section 13 — the charge-even identification — appears nowhere below, and that is the point.

The source carries 141 inline markers. Each line below is one marker, verbatim, under the section that carries it.

Introduction

- the discrete index theorem gives 0 on the signed face-edge operator, bounding nothing

Hamiltonian and projected sector

- the printed towers are canonical-u: 4*Delta(3u/2) reproduces them verbatim
- the 4**r rescaling breaks the bridge: order 2 off by 16, order 3 by 64
- at N = 2 the C-odd hopping vanishes and the C-even one does not
- every nontrivial irrep clears 5/4 C_F, except the fundamental pair
- the torus graph is simple, and the only short cycles are the L = 3 winding triangles
- every 4-cycle is an elementary face, except exactly the 3L^2 straight winding loops at L = 4
- the SU(3) additive electric spectrum leaves 14, 16, 17 around the shell

Oriented incidence and the exact carrier

- B B^dagger = q I - d conj(d)^T for the curl incidence
- dim Z_2 = L^3 + 2 by rank, not by re-arranging the formula
- cube boundaries and three wrapping sheets SPAN Z_2
- the L^3+2 count is chain-level, not the Bloch convention
- the Bloch and chain routes to the carrier agree
- the sheets average to harmonic representatives, and the Gamma block IS b_2
- the Fourier sum of cube boundaries IS the carrier, and a local seed costs L^-3

The wrapping sheets are cycles, not harmonic — and their harmonic representatives

- FINDING: the wrapping sheets are cycles but NOT harmonic
- the sheets average to harmonic representatives, and the Gamma block IS b_2

Exact adjacent shared-link dynamics and global assembly

- each channel gap is C_F + C_R/2, and the weights sum to one

The channel weights are a theorem

- the shared-link weights are Weingarten, not an isotropy assumption
- the Weingarten route is independent of the corpus
- the published dimension-ratio matrix element is this registry's weight formula
- the fourth moment integral |U_11|^4 = 1/6 at N = 3

The off-diagonal process list is complete

- the plaquette graph is 12-regular and two faces share at most one link
- connected first order vanishes by inclusion-exclusion, not by cancellation of numbers
- the four channel weights follow from dimension and Casimir
- A_N and B_N are the channel sums, not transcriptions

The shared-link law

- t_N = B_N - A_N
- t_N > 0 for N >= 3
- t_2 = 0 and t_3 = 5/612
- large-N expansion of t_N through 1/N^9
- deficit identity 1/4 - N^3 t_N
- N^3 t_N increases monotonically to 1/4

The vacuum-mediated route

- ell_N = A_N + B_N + 1/C_F, the vacuum-mediated route at every rank

Finite-volume width

- the zone maximum of q is 12 only at even L
- q_min on the L-torus grid is 4 sin^2(pi/L)

The orientation signs are essential

- the two band spans ARE the two incidence spectra

The second-order coefficient on a two-cube space

- connected first order vanishes by inclusion-exclusion, not by cancellation of numbers
- the connected two-cube geometry has exactly four cross-cell pairs, each -1
- the B=6 connected kernel is (5/612) G_conn + diag, with the certified spectrum
- the certificate's own gates pass, target-blind, with the wrong-sign control rejected
- the B=6 six-channel census sums to the registry's own t_3 = 5/612
- the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel
- every shared-link channel is separately proportional to the geometry, on all 56 pairs

One truncation, two constructions

- B=6 retains every adjacent shared-link channel; B=4 provably cannot
- the retention rule is C2(rho) + 2 C2(3) <= B, and both retentions it decides are equalities
- the B=6 six-channel census IS the Weingarten four-channel ledger, channel by channel
- FINDING: the T1 link cutoff reverses the sign of t_3, and 14/153 is what it omits
- FINDING: a full T1 = B = 4 cube Hamiltonian reproduces -1/12 and the reversed shell
- FINDING: the B = 6 cube flips the sign back to +5/612, closing the decisive test
- the B = 6 scalar misses the bridge's by exactly the same-face sextet route

What the cutoff moves, and where

- the connected diagonal is orbit-constant, and only the transporting orbit moves

The one-cube shell, and why its flat level cannot move

- the one-cube shell is A1 + T1 + E, and its flat level is the S^2 fundamental class
- the certificate's finite-volume fingerprints, and 29 = L^3 + 2 is the Lean cycle count

A detached-replayed radius-two Ritz extension

- the sealed radius-two report passes its six float checks at stated tolerances
- the sealed radius-two report passes its six float checks at stated tolerances
- the sealed radius-two report passes its six float checks at stated tolerances

The unsigned-incidence comparison operator

- the C-even characteristic polynomial is mu(mu - p)^2 = 4 a_1 a_2 a_3
- p + q = 12: one zone function runs both sectors
- the Bloch cubic IS the finite L = 3 and L = 4 plaquette spectrum, exactly
- the C-even range [-4, 12] is exact, and each edge is attained at one point only
- the C-even band touches its floor exactly on the three planes k_j = pi
- the C-even band touches its floor exactly on the three planes k_j = pi
- the C-even spectra at the four high-symmetry momenta

Conditional charge-even assembly

- one assembly formula gives every registered band value
- one assembly formula gives every C-even value at both orders
- the plaquette graph is 12-regular and two faces share at most one link

- both declared coincidences, checked: one is ell_N at all ranks, the other is bare

Where the rest frame is not blind

- the C-even Gamma point pins t_+, exactly where no C-odd Gamma datum can
- the C-even curvature is (4/3)|t_+| at both orders, and isotropic
- the C-even bandwidth is 16|t_+| at every order; the C-odd manifold width is 12|t_-|
- at N = 2 the C-odd hopping vanishes and the C-even one does not

$SU(3)$ **through third order**

- d_3 = 7/32 + 12*leak_3 - 4*b_3
- leak_3 is assembled from the domino diagonal and the vacuum piece
- E_flat and t(u) carry the ledger coefficients
- d_- = 1/2 + 12*leak_2, and leak_2 = -11/306
- FINDING: no Gamma-point datum can constrain the hopping

What the homology protects

- a boundary-factorised correction shifts the carrier without dispersing it, for generic M
- the carrier projection is where the 1/q comes from
- clearing the denominator reproduces the five-element numerator basis

An exact theorem for the saved historical kernel

- v10a.26 A, B, D match the sealed rationals within 2.3e-13
- FINDING: alpha_new falls outside the corpus's own 2.3e-13 bound

The obstruction is polynomial

- FINDING: the retained Gamma/axis data cannot identify C_shp
- Phi_C vanishes at Gamma along every direction
- the crosswalk is exactly scalar on the momentum axes
- on an axial cut the mixed invariants vanish and the norm divides

A second witness

- FINDING: an explicit second witness C_alt exhibits the C2 non-identifiability
- the C_alt witness IS the balanced eps-free continuation, and Delta_C/(A/2) = 15/32

Where the coefficient lives

- checkpoint values at X, M, P, R
- the four extraction formulas invert the ansatz
- X is blind to B, C, D — it fixes A alone
- the shape fit's C row sums to zero, so no Gamma-anchor error can move C_shp at all
- a translation-local scalar shift changes nothing observable
- FINDING: C_shp is carried by 6 of 189 records, not spread across the kernel
- FINDING: A = 5/48 pins the normal sector's whole C4 contribution
- C_normal = -A_normal/2: the agreed axial coefficient pins the normal channel
- the two 189-record kernels agree everywhere except three amplitudes, and the on-site anchor swap moves C by exactly zero

What cannot decide it

- FINDING: the marked-cluster engine emits the Gamma scalar only — a completed 609-sweep cannot decide C_shp

The kernel is six amplitudes

- the shape coefficients are solved, not fitted: A = 5/48, B = D = 0 exactly
- the 189 records carry exactly six weight magnitudes
- every orbit's carrier projection is closed form in e1, e2, e3
- A = 5/48 forces the normal amplitude: nu = -(5/48 + 4u)
- C_shp = -5/96 - u - (rho + pi)/2, exactly

The tier collapse, in the orbit basis

- RETRACTED: the vertex count does NOT forbid B_shp and D_shp
- the vanishing coefficients are exactly the degree-3 ones
- FINDING: the tier collapse is two integer cancellations at record level
- B = 0 is unpopulated, not cancelled
- the tier collapse is one identity: D = -6u + 3*u2 with u2 = 2u
- u2 = 2u is the cross term of a perfect square

Both branches in one basis

- RETRACTED: B_3 - beta_historical = 25/64 is a forbidden substitution, not an exact branch
- the SU(5) stage-1 scan is shipped: 895,524 pairs, zero determinant sectors
- the SU(6) determinant correction is exactly 6/343 and momentum-independent
- the N=3 continuation shift is exactly 25/64, with two exact corollaries
- the on-site orbit IS the momentum-independent channel
- PREDICTION: the eps-sector at N=3 is Delta(rho + pi) = -25/512 and nothing else
- FINDING: the cold kernel is the same six orbits with rho and pi sign-flipped

A near-Γ statement that does not adjudicate

- Jordan bound q(k) >= (4/pi^2)|k|^2 holds on the whole zone
- the exact zone minimum of q on |k| >= r is 4 sin^2(r/2), and Jordan overstates it by pi/2
- the criterion survives C2: K depends on the kernel only through sqrt(W_4)
- the statement is non-vacuous only below an explicit coupling

External comparisons

- Hamer's 1+- series matches the C-odd Gamma-point coefficients through x^3
- Hamer's 0++ series matches the C-even Gamma-point coefficients through x^3
- the m_n = 2^(n-1) a_n bridge is the x = 2u conversion
- the Hamer table is pinned, and the a_4 agreement is primary-source
- the KPS 1980 string-tension table equals the certified sigma series EXACTLY
- sigma_n^phys = (-1)^n sigma_n^raw, and C5 is the n = 3 case
- the ratio and sigma series reproduce E_flat exactly
- the errata-resolved Euclidean series is doubly sourced, transcription for transcription
- FINDING: Munster's 1985 table shifts his 1982 erratum at eighth order
- the overlap obstruction was published in 1988, and it scales
- a cross-regime paper never supplies a value

Scope

- Delta_L = 4 tau(u) sin^2(pi/L) is positive and falls as L^-2

Channel weights in closed form

- the four channel weights follow from dimension and Casimir

The Weingarten index sums

- the shared-link weights are Weingarten, not an isotropy assumption

- SU(3) Weingarten values follow from the general formula
- the fourth moment integral |U_11|^4 = 1/6 at N = 3

The $SU(3)$ coefficient ledger

- the manuscript's SU(3) ledger is this registry, value by value

Unsigned comparison and conditional charge-even ledger

- one assembly formula gives every C-even value at both orders
- FINDING: the certificate key 'bandmin' holds the band MAXIMUM, at both orders

Finite-volume chain calculation

- d_2 d_3 = 0 on the built complex
- rank d_3 = L^3 - 1 on the built complex

The two-cube geometry

- the B=4 comparator on the same geometry gives -1/12 with the reversed spectrum

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